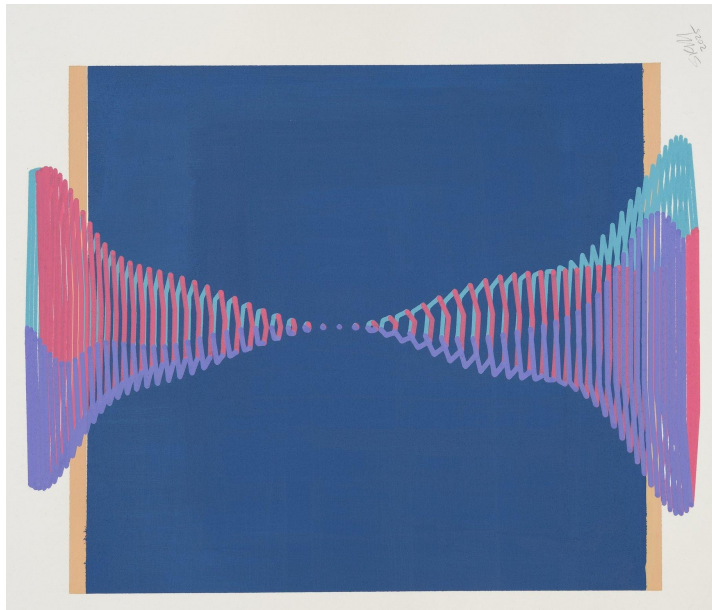


Optimizing the extraction of entanglement from quantum fields

Marcos Morote Balboa

T. Rick Perche

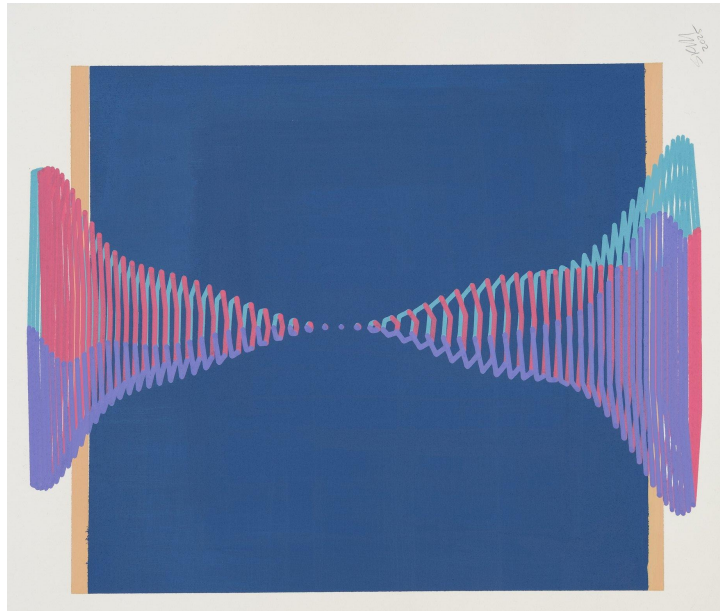


Credit: Sohan Murthy

$\langle \text{IS} | \text{RQI} \rangle$



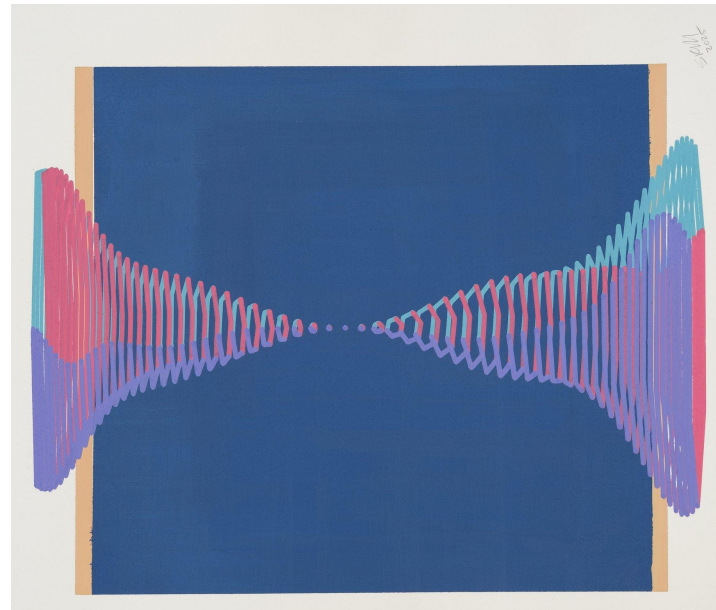
Outline



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Outline

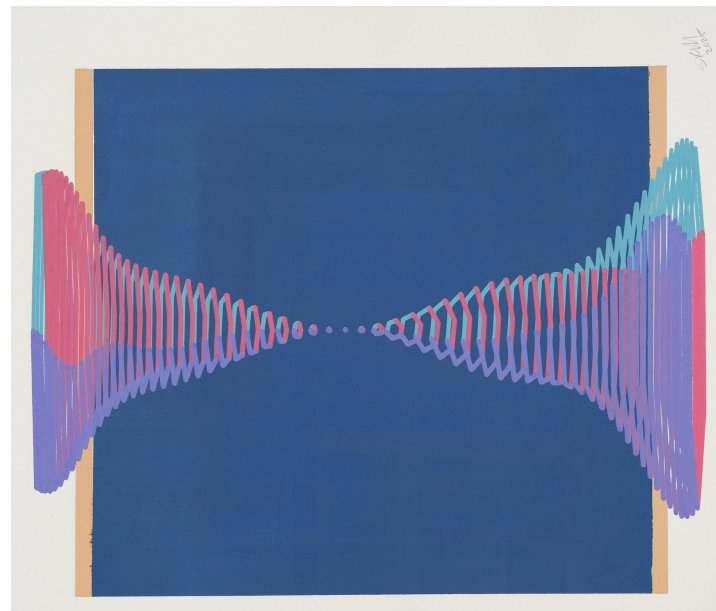
1. Entanglement Harvesting (EH)



Credit: Sohan Murthy

Outline

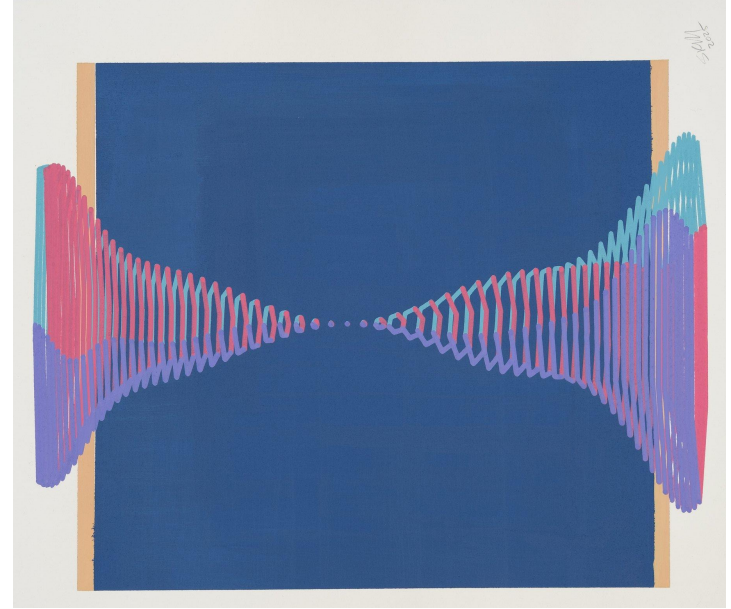
1. Entanglement Harvesting (EH)
2. EH in Bose-Einstein condensates



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Outline

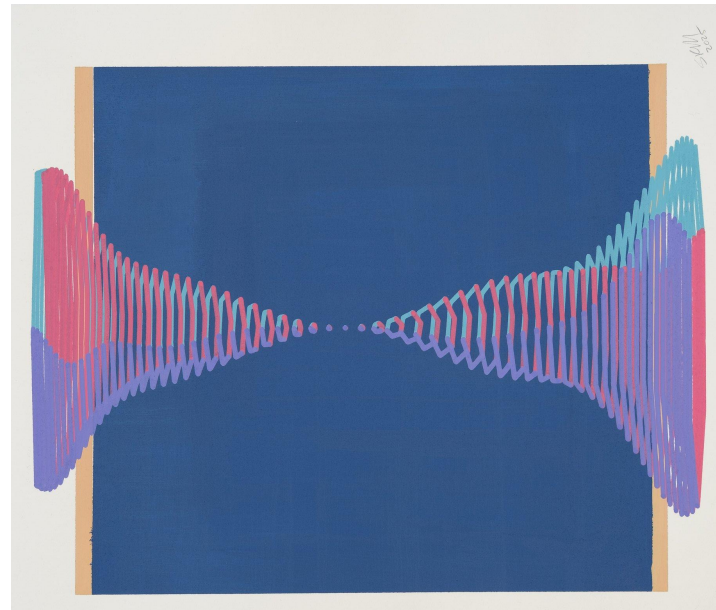
1. Entanglement Harvesting (EH)
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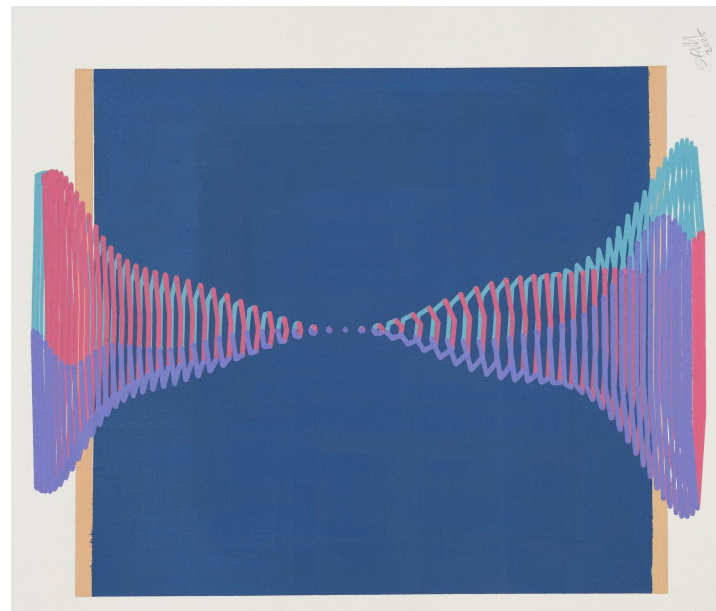
1. Entanglement Harvesting (EH)
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 - 3.1. EH with arbitrary spacetime smearing



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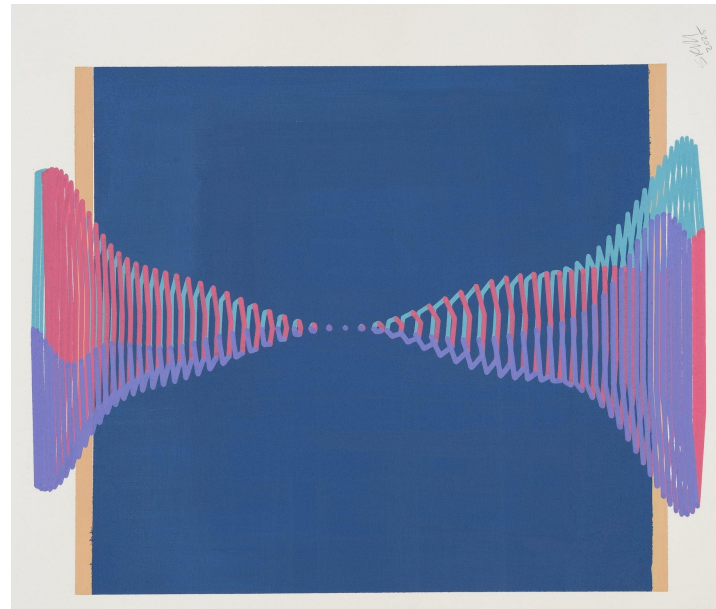
1. Entanglement Harvesting (EH)
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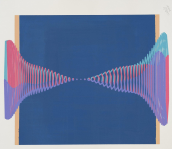
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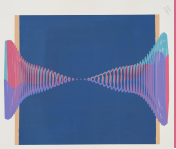
1. Entanglement Harvesting (EH)
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 - 3.2. Optimization over compactly supported functions
 - 3.3. Further work



Credit: Sohan Murthy

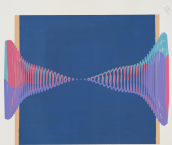


Entanglement harvesting (EH)



Entanglement harvesting (EH)

→ How much entanglement between regions of a quantum field?



EH: Quantifying entanglement

————→ **Pure states**



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$$S(|\psi\rangle\langle\psi|) = -\text{Tr}(\hat{\rho}_i \log(\hat{\rho}_i))$$

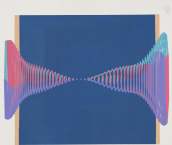


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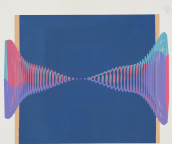
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→ not enough if there is classical uncertainty



EH: Pere's criterion

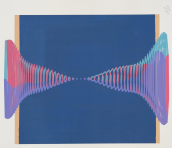
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$$\longrightarrow \textbf{Negativity}^* \quad \mathcal{N}(\hat{\rho}_{AB}) = \sum_{\lambda_i < 0} |\lambda_i|$$



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$$\longrightarrow \textbf{Negativity}^* \quad \mathcal{N}(\hat{\rho}_{AB}) = \sum_{\lambda_i < 0} |\lambda_i|$$

*it is not a faithful quantifier in general, but works for two qubits



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→ How much entanglement between regions of a quantum field?

→ Negativity



Entanglement harvesting (EH)

- How much entanglement between regions of a quantum field?
- Negativity
 - Still very complicated for quantum fields



Entanglement harvesting (EH)

————→ How much entanglement between regions of a quantum field?

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————→ Still very complicated for quantum fields

————→ Probes



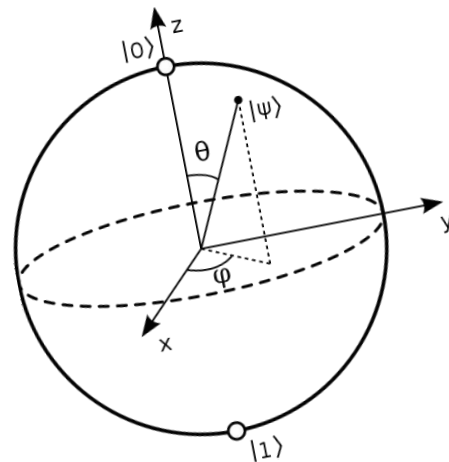
Entanglement harvesting (EH)

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 - Probes
- Couple qubits to quantum field in different regions



Entanglement harvesting (EH)

- How much entanglement between regions of a quantum field?
- Negativity
 - Still very complicated for quantum fields
 - Probes
- Couple qubits to quantum field in different regions
 - **Entanglement harvesting**



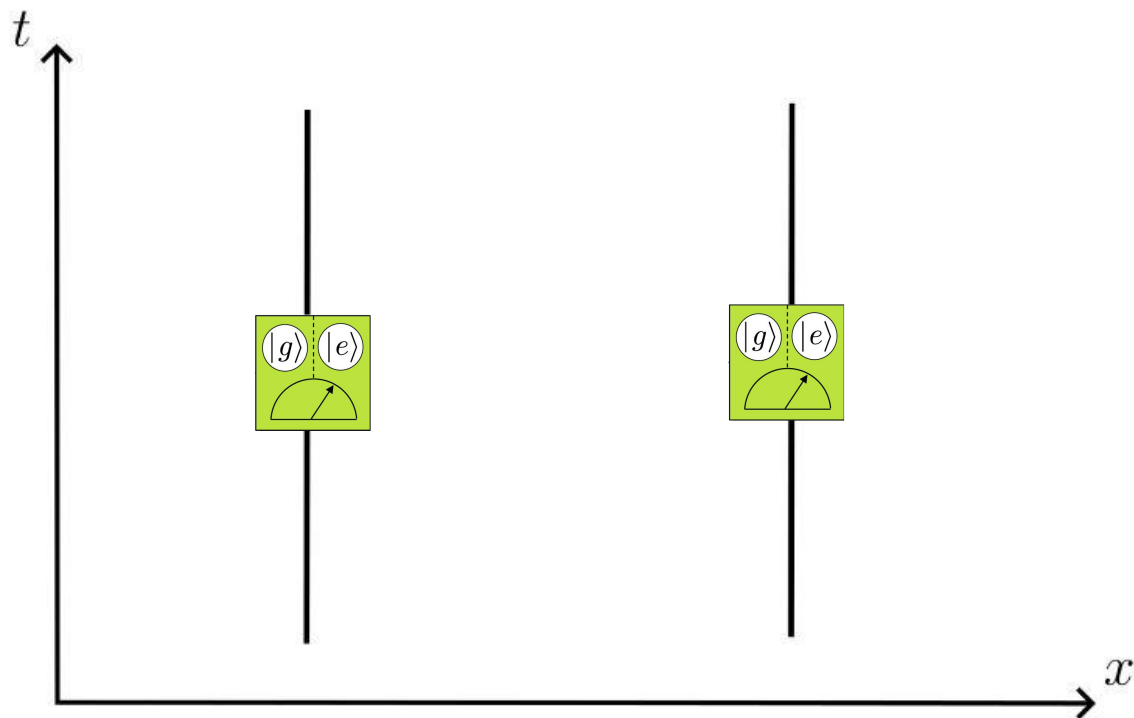
A. Valentini, Non-local correlations in quantum electrodynamics, Phys. Lett. A 153, 321 (1991).

B. Reznik, Entanglement from the vacuum, Foundations of Physics 33, 167 (2003).

A. Pozas-Kerstjens and E. Martín-Martínez, Harvesting correlations from the quantum vacuum, Phys. Rev. D 92, 064042 (2015).



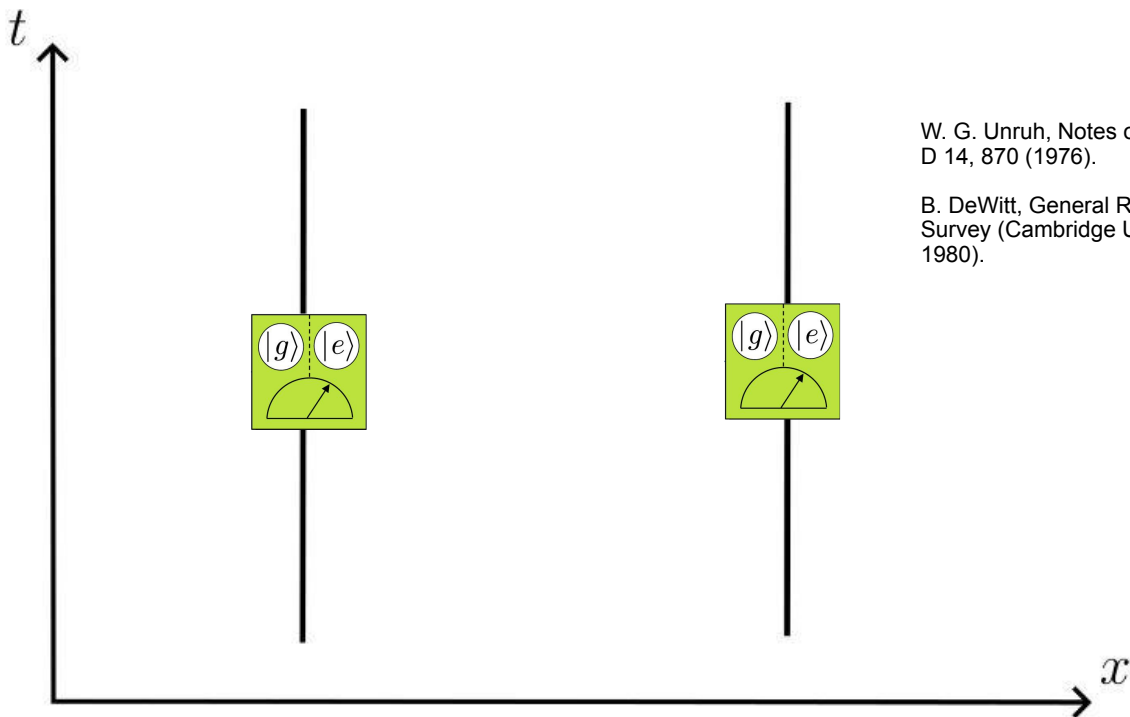
EH: Unruh-DeWitt (UDW) detectors





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→ Unruh-DeWitt (UDW)



W. G. Unruh, Notes on black-hole evaporation, Phys. Rev. D 14, 870 (1976).

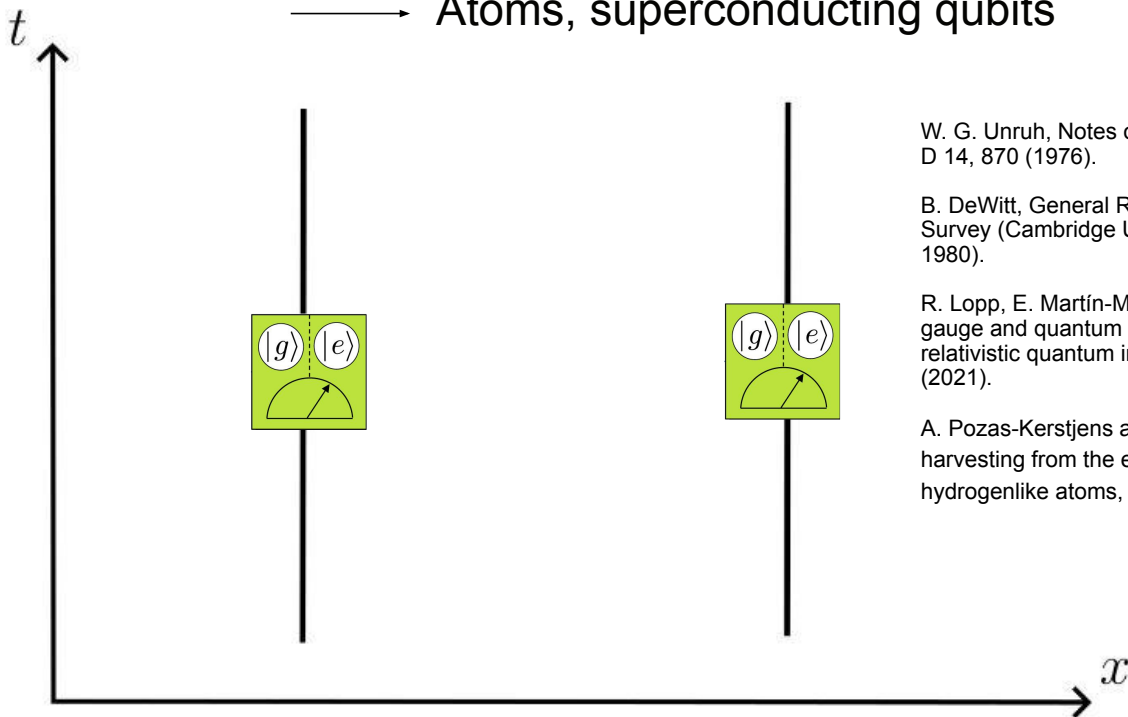
B. DeWitt, General Relativity; an Einstein Centenary Survey (Cambridge University Press, Cambridge, UK, 1980).



EH: Unruh-DeWitt (UDW) detectors

→ Unruh-DeWitt (UDW)

→ Atoms, superconducting qubits



W. G. Unruh, Notes on black-hole evaporation, Phys. Rev. D 14, 870 (1976).

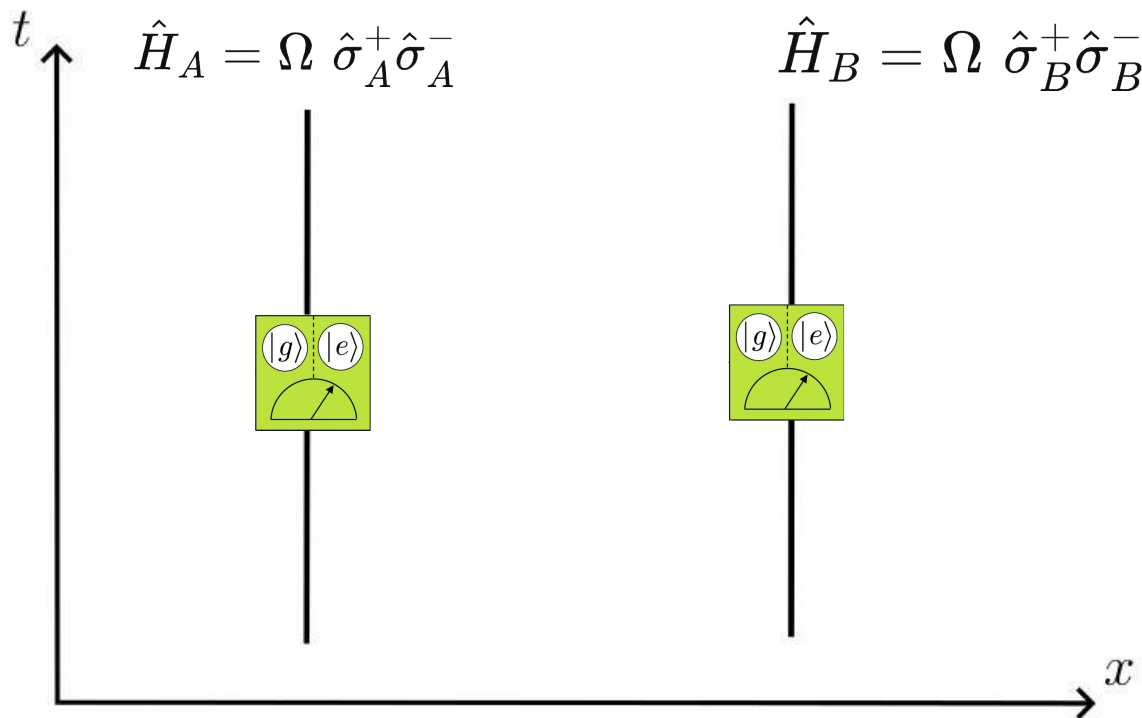
B. DeWitt, General Relativity; an Einstein Centenary Survey (Cambridge University Press, Cambridge, UK, 1980).

R. Lopp, E. Martín-Martínez, Quantum delocalization, gauge and quantum optics: The light-matter interaction in relativistic quantum information, Phys. Rev. A 103, 013703 (2021).

A. Pozas-Kerstjens and E. Martín-Martínez, Entanglement harvesting from the electromagnetic vacuum with hydrogenlike atoms, Phys. Rev. D 94, 064074 (2016)

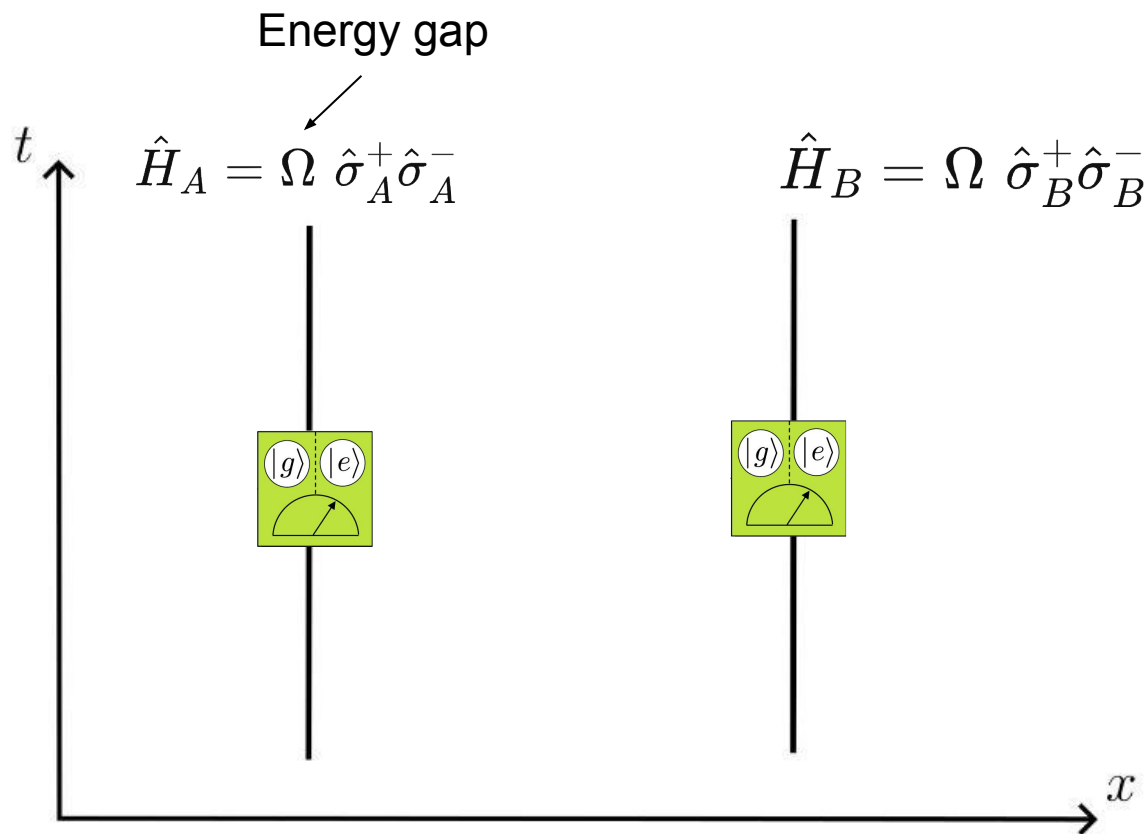


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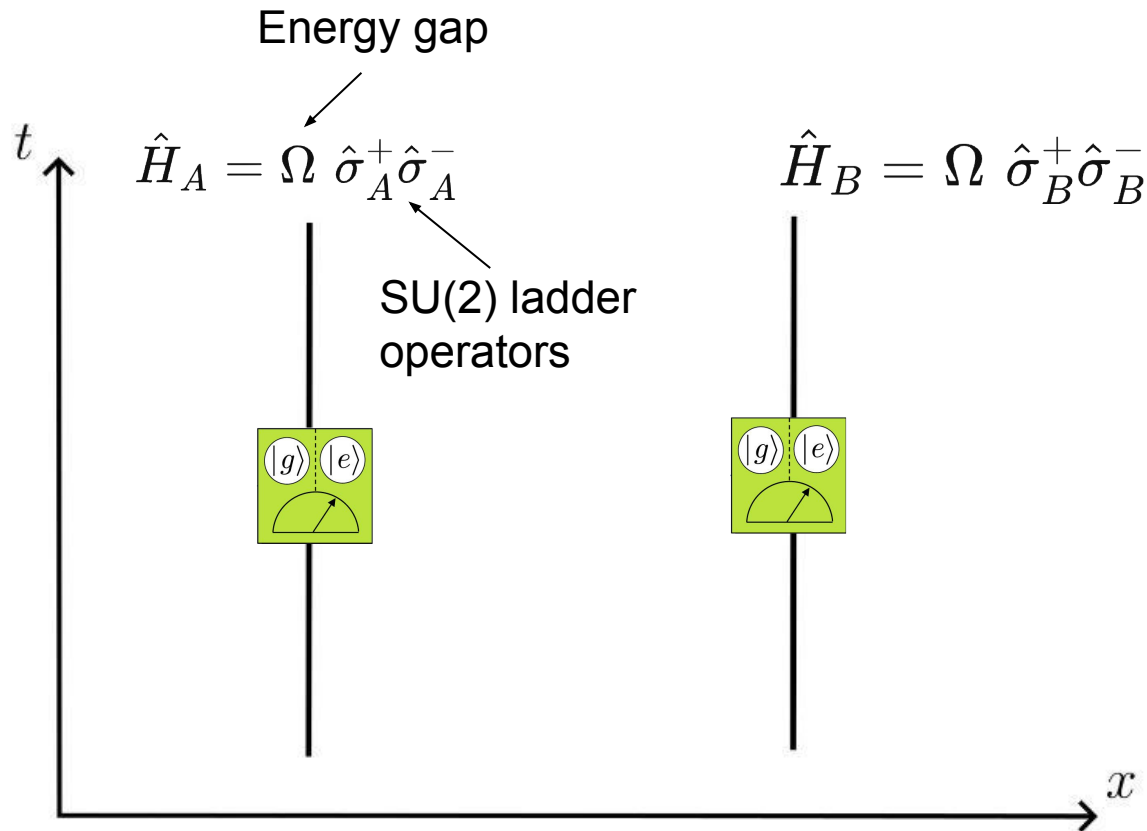


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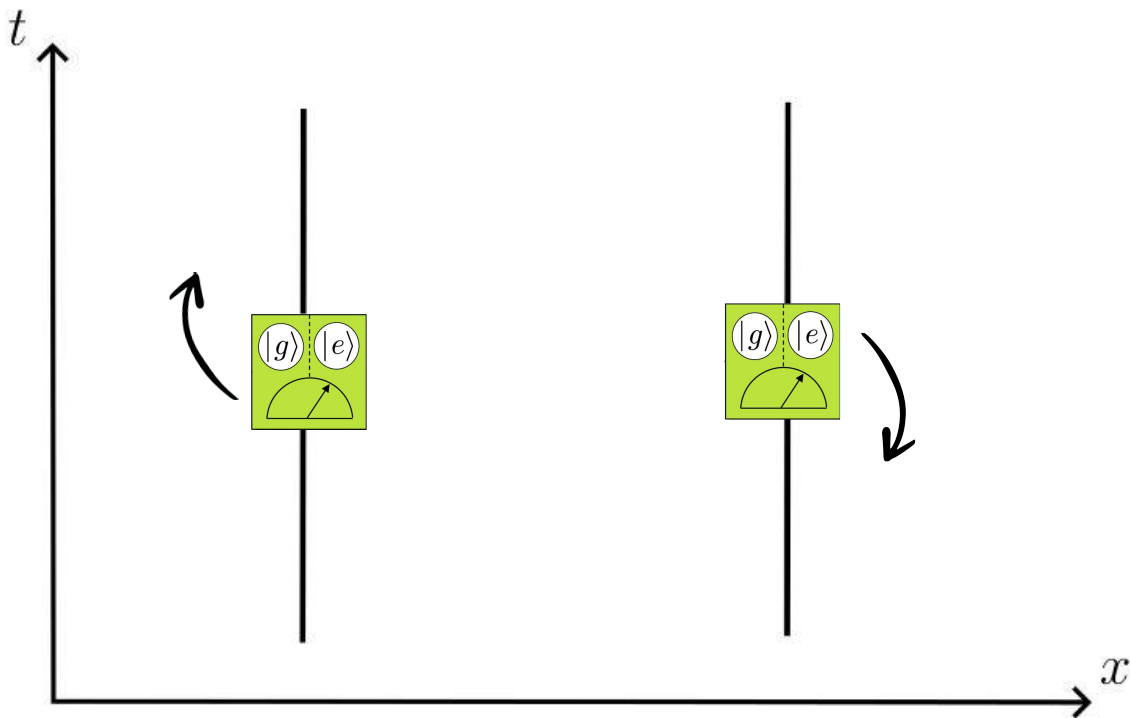


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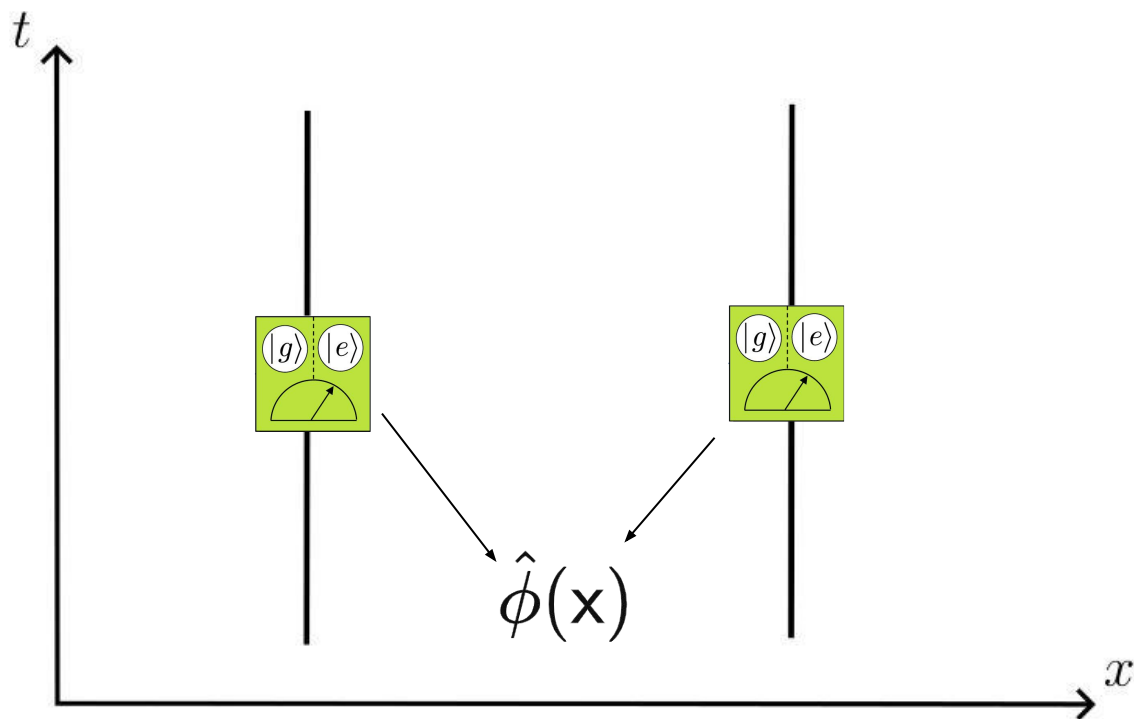


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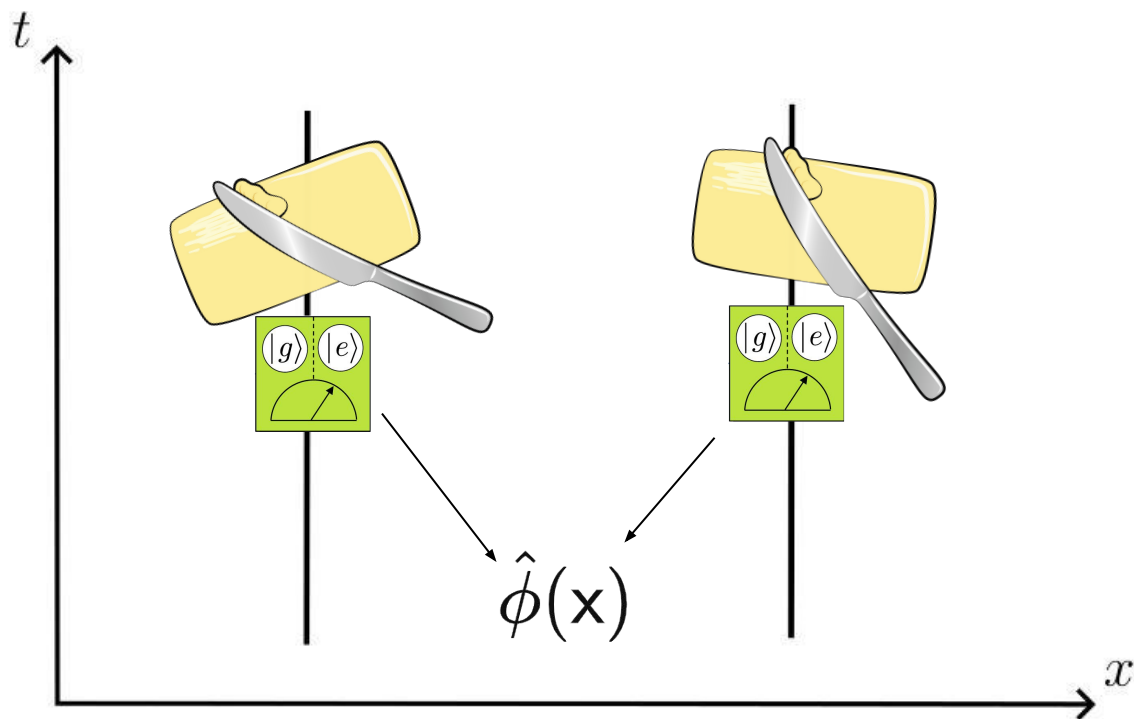


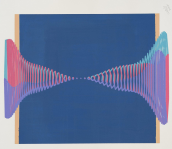
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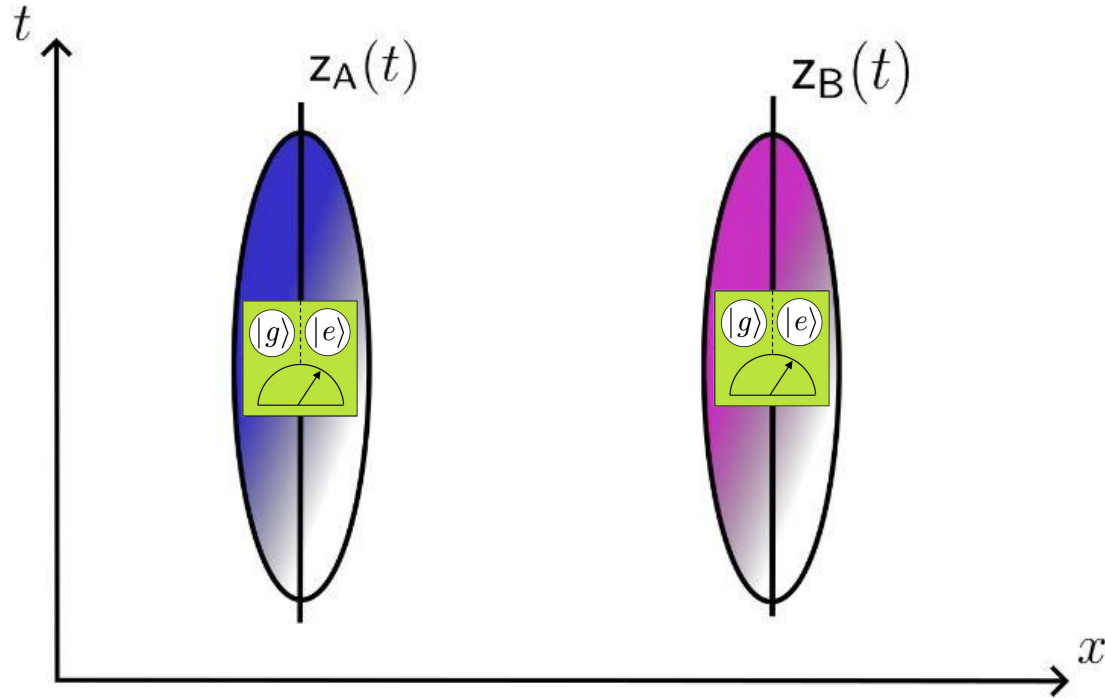
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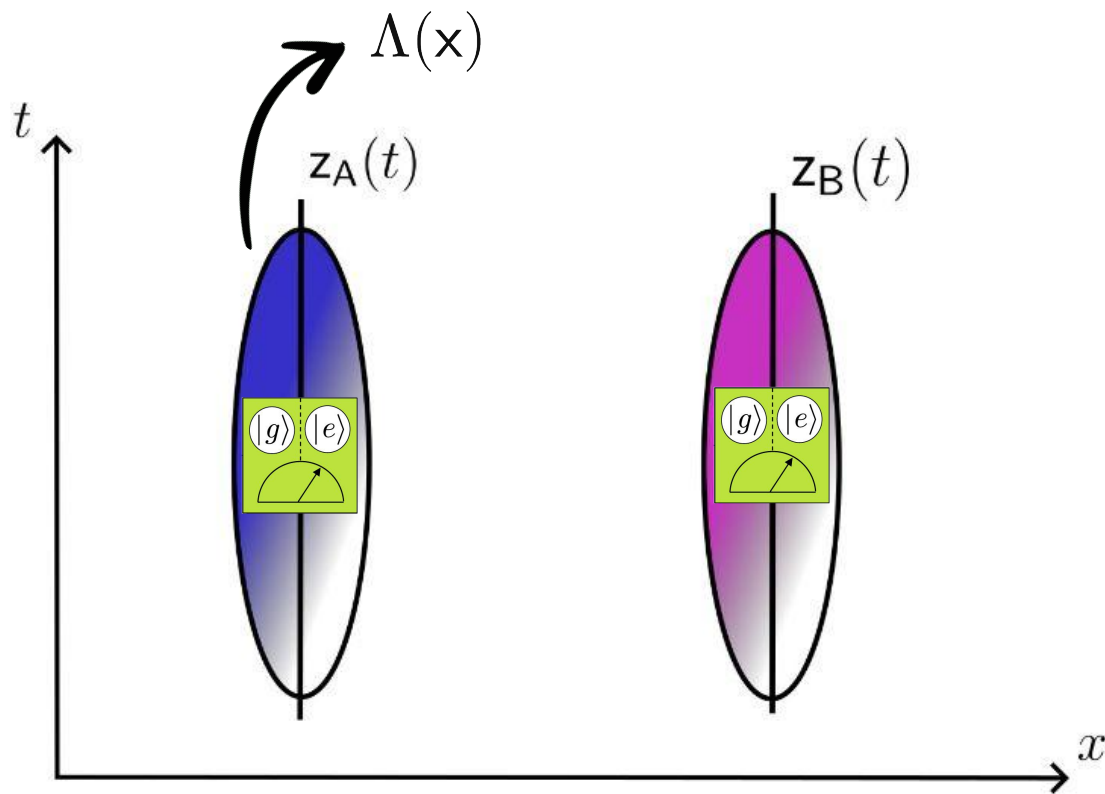
EH: Unruh-DeWitt (UDW) detectors

→ Physically, the detectors have finite size



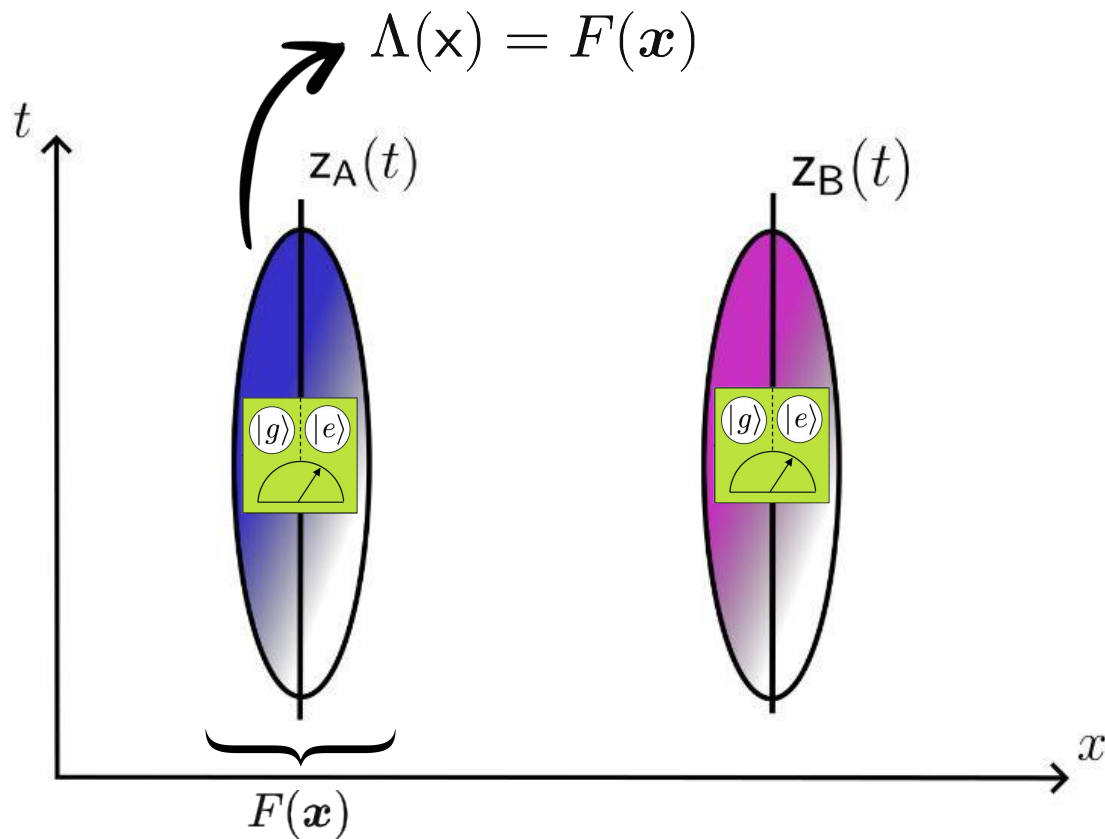


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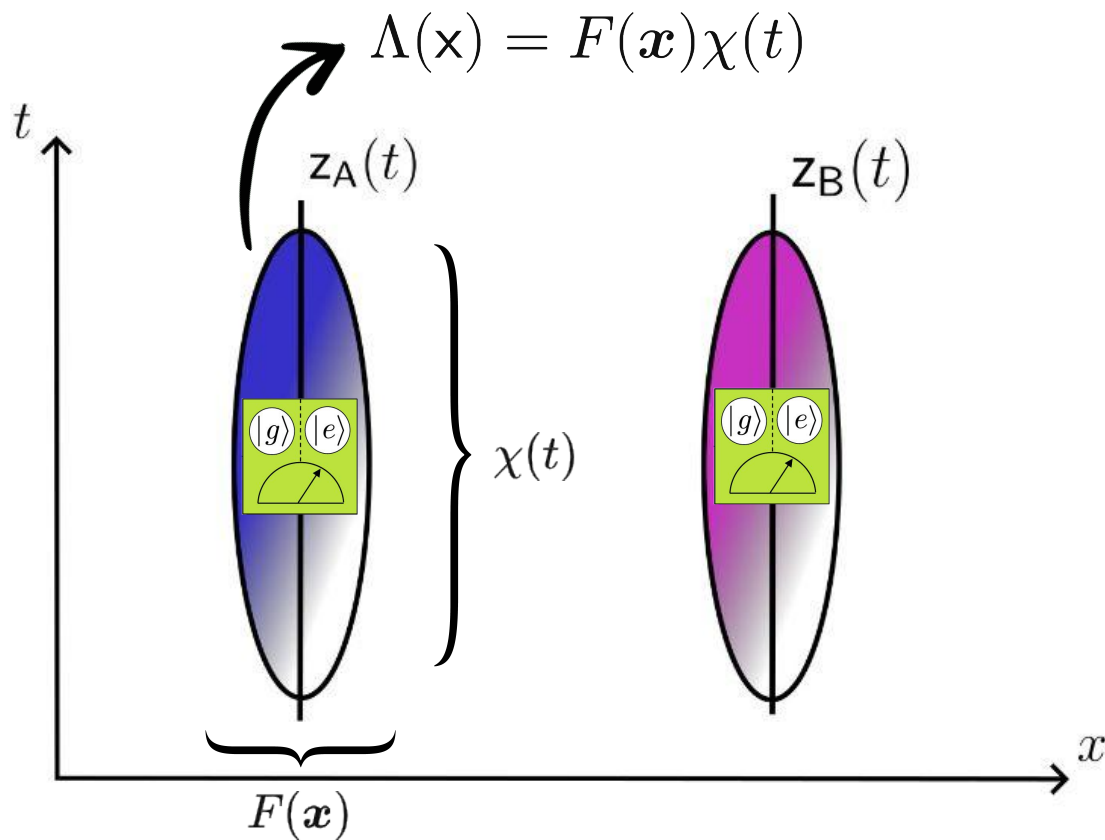


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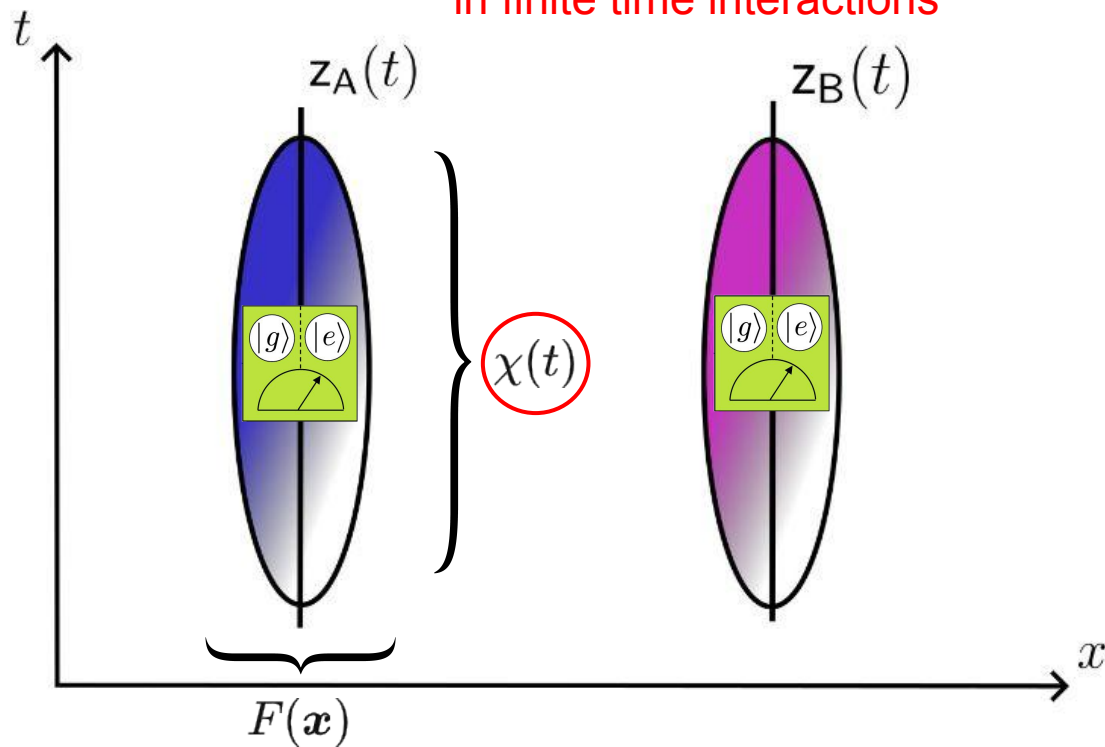
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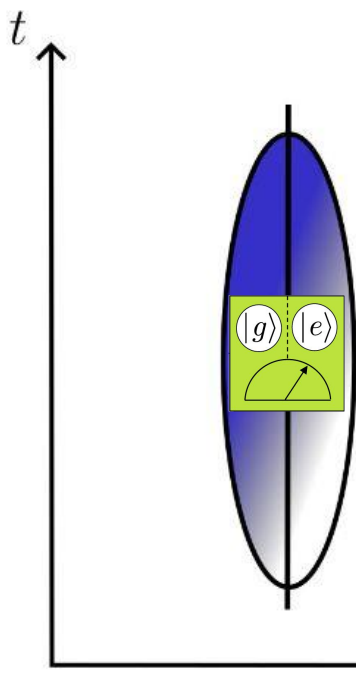
EH: Unruh-DeWitt (UDW) detectors

→ We can only see vacuum excitations
in finite time interactions





EH: Unruh-DeWitt (UDW) detectors



$$\hat{h}_I(x) = \lambda \Lambda(x) (e^{i\Omega t} \hat{\sigma}^+ + e^{-i\Omega t} \hat{\sigma}^-) \hat{\phi}(x)$$



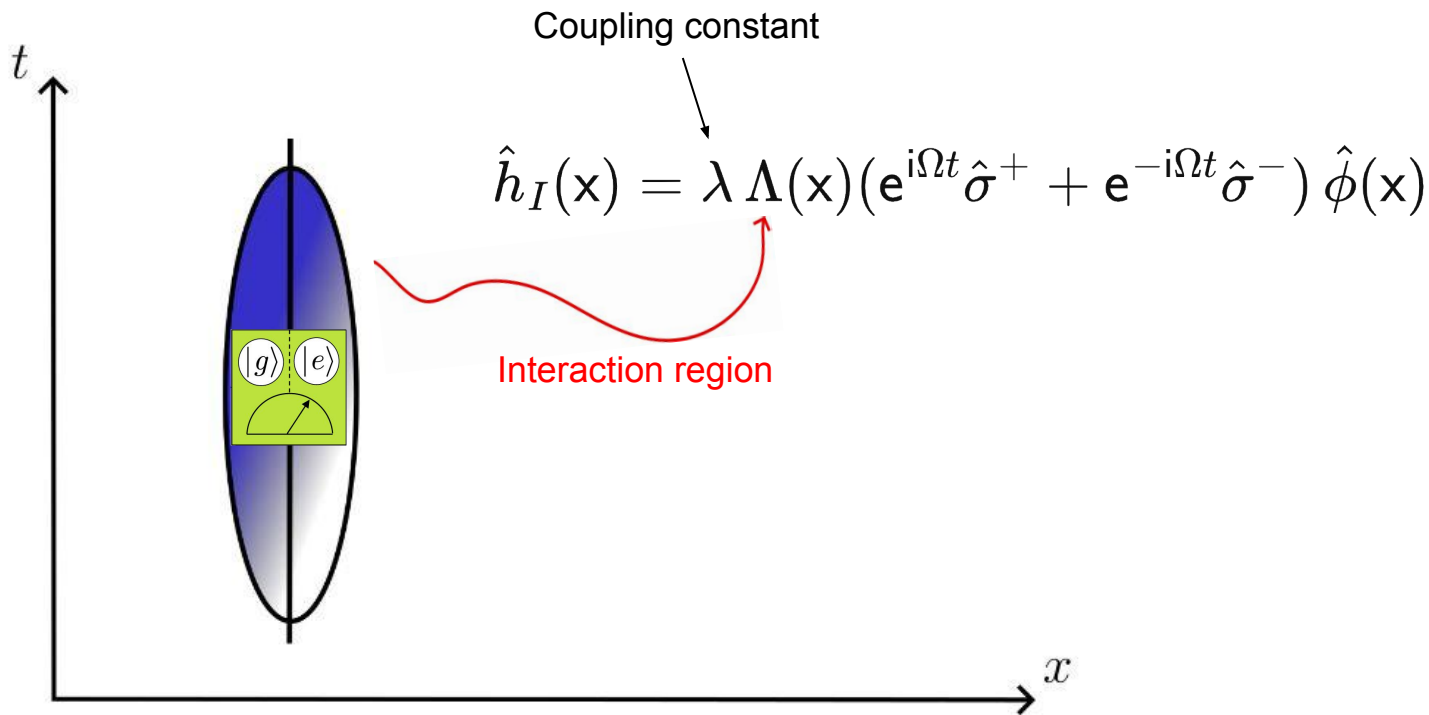
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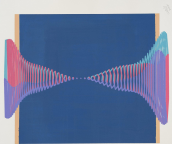
Coupling constant

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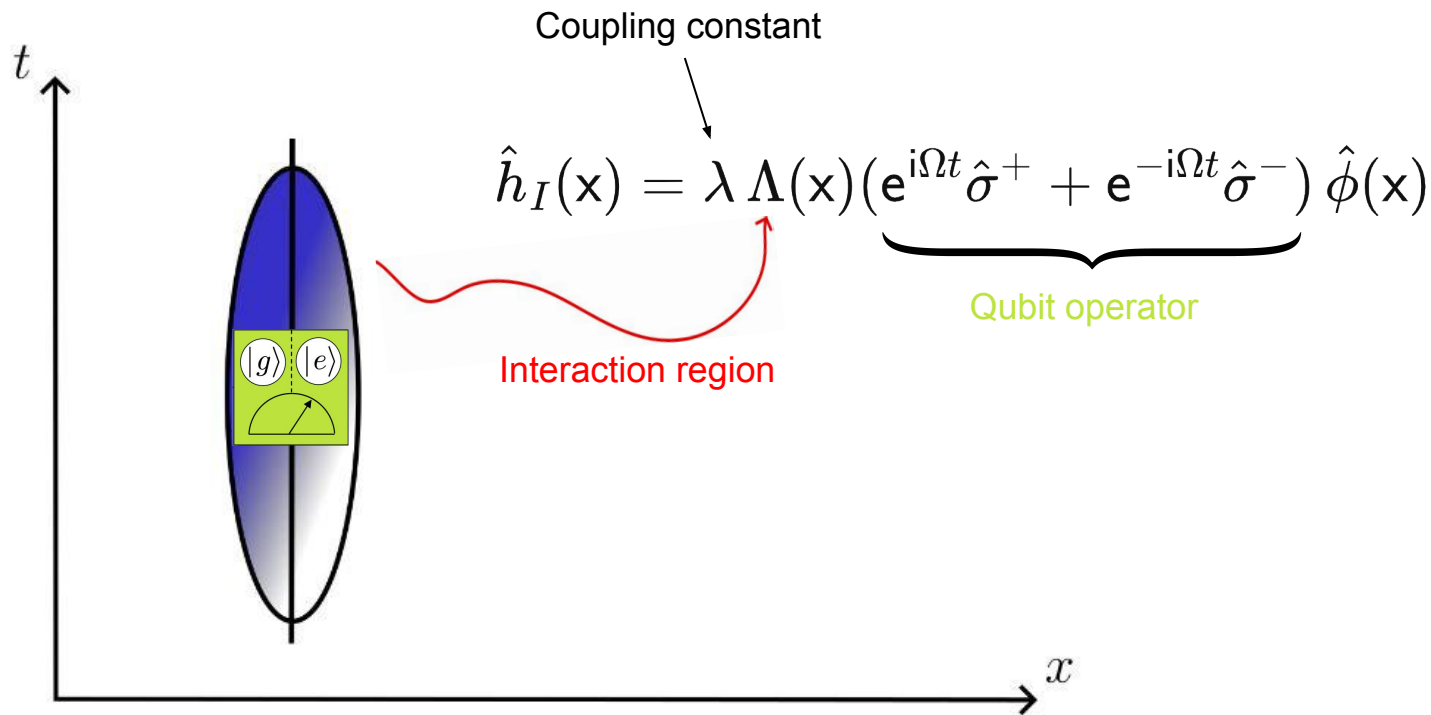


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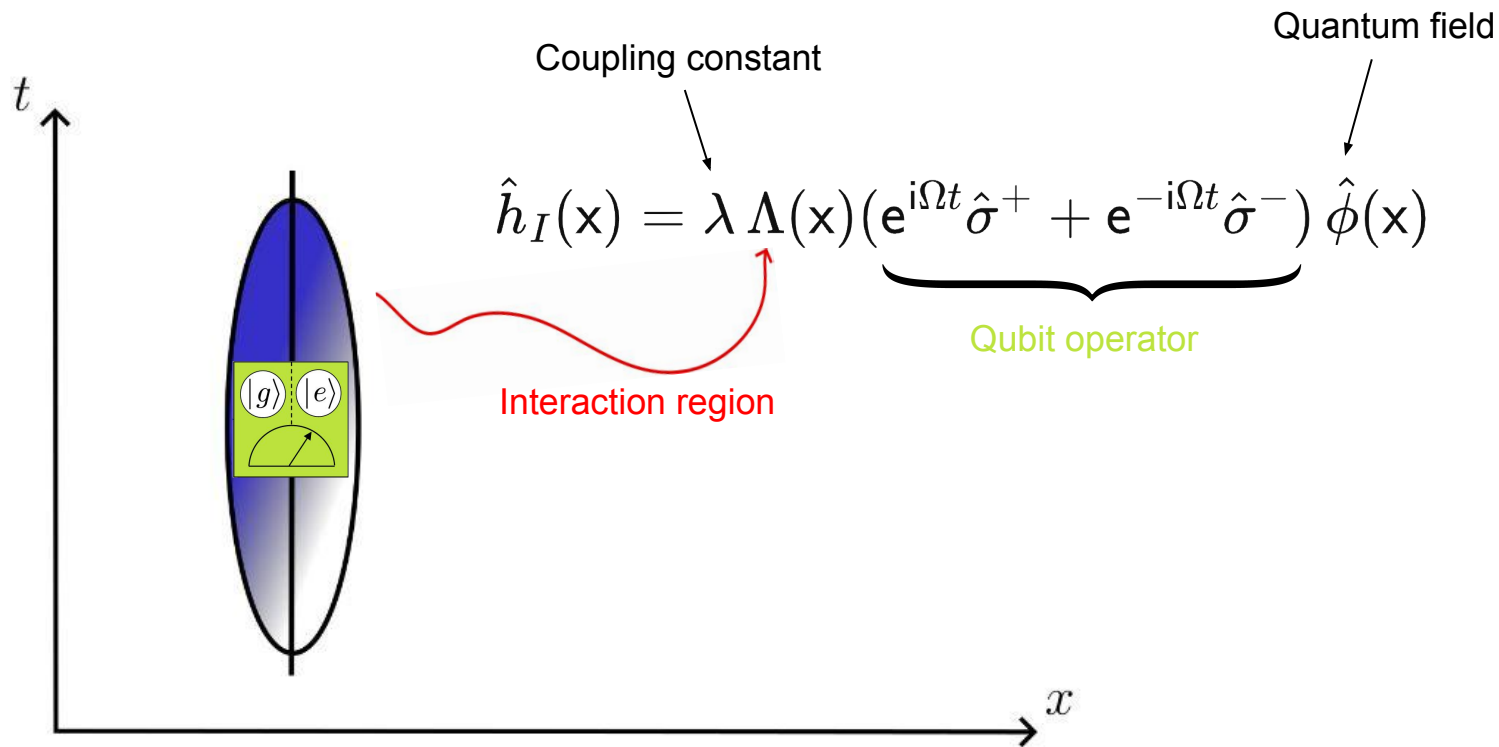


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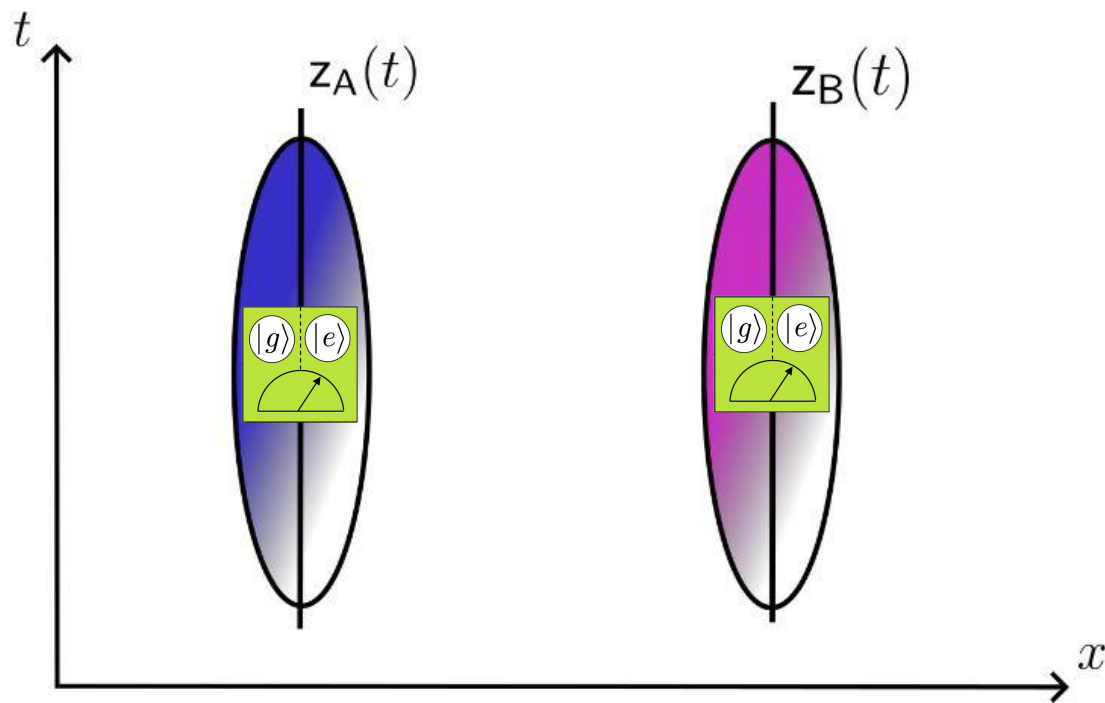


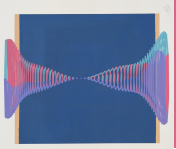
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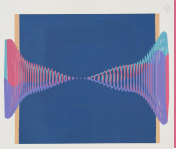


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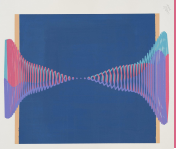


EH: Propagators



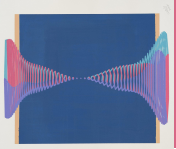
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EH: Propagators

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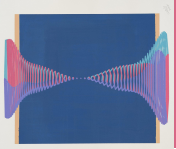


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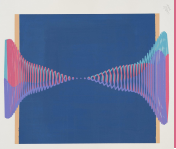


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$$W(\mathbf{x}, \mathbf{x}') \equiv \langle \hat{\phi}(\mathbf{x}) \hat{\phi}(\mathbf{x}') \rangle$$



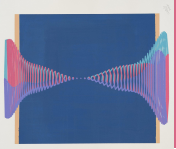
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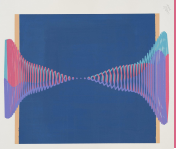
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→ Feynman propagator (time-ordered 2-point function)

$$G_F(\mathbf{x}, \mathbf{x}') \equiv \langle T \hat{\phi}(\mathbf{x}) \hat{\phi}(\mathbf{x}') \rangle$$



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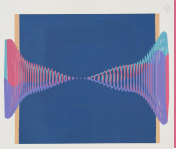
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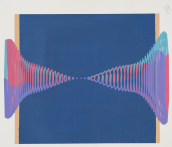
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EH: Negativity



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→ Eigenvalues of $\hat{\rho}_{AB}^T$?



EH: Negativity

→ Eigenvalues of $\hat{\rho}_{AB}^{\text{T}_B}$?

$$\lambda_{\text{neg}} = -|G_{AB}^{++}| + W^{-+}$$



EH: Negativity

→ Eigenvalues of $\hat{\rho}_{AB}^{\top_B}$?

$$\lambda_{\text{neg}} = -|G_{AB}^{++}| + W^{-+}$$

$$\longrightarrow G_{AB}^{++} = \lambda^2 G_F(\Lambda_A^+, \Lambda_B^+)$$



EH: Negativity

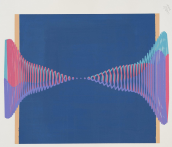
→ Eigenvalues of $\hat{\rho}_{AB}^{\top_B}$?

$$\lambda_{\text{neg}} = -|G_{AB}^{++}| + W^{-+}$$

$$\longrightarrow G_{AB}^{++} = \lambda^2 G_F(\Lambda_A^+, \Lambda_B^+)$$

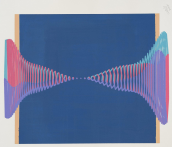
$$\longrightarrow W^{-+} = \lambda^2 W(\Lambda_A^-, \Lambda_A^+) = \lambda^2 W(\Lambda_B^-, \Lambda_B^+)^*$$

(*identical detectors)



EH: Negativity

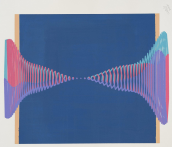
$$\longrightarrow G_F(\Lambda_A^+, \Lambda_B^+)$$



EH: Negativity

$$\longrightarrow G_F(\Lambda_A^+, \Lambda_B^+)$$

$\longrightarrow$ Feynman propagator

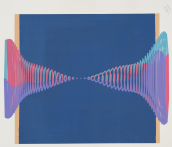


EH: Negativity

————→ $G_F(\Lambda_A^+, \Lambda_B^+)$

————→ Feynman propagator

————→ Non local



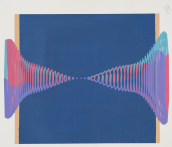
EH: Negativity

————→ $G_F(\Lambda_A^+, \Lambda_B^+)$

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————→ Encodes quantum correlations



EH: Negativity

→ $G_F(\Lambda_A^+, \Lambda_B^+)$

→ Feynman propagator

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→ Encodes quantum correlations → Entanglement



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→ Related to **probability amplitude** that **both** detectors become excited mutually **due** to the quantum correlations



EH: Negativity

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→ $W(\Lambda_A^-, \Lambda_A^+) = W(\Lambda_B^-, \Lambda_B^+)$



EH: Negativity

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→ Non local

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→ Related to **probability amplitude** that **both** detectors become excited mutually **due** to the quantum correlations

→ $W(\Lambda_A^-, \Lambda_A^+) = W(\Lambda_B^-, \Lambda_B^+)$

→ Wightman function



EH: Negativity

→ $G_F(\Lambda_A^+, \Lambda_B^+)$

→ Feynman propagator

→ Non local

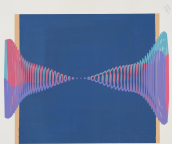
→ Encodes quantum correlations → Entanglement

→ Related to **probability amplitude** that **both** detectors become excited mutually **due** to the quantum correlations

→ $W(\Lambda_A^-, \Lambda_B^+) = W(\Lambda_B^-, \Lambda_A^+)$

→ Wightman function

→ Local



EH: Negativity

→ $G_F(\Lambda_A^+, \Lambda_B^+)$

→ Feynman propagator

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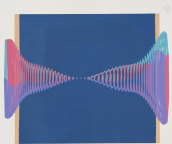
→ Related to **probability amplitude** that **both** detectors become excited mutually **due** to the quantum correlations

→ $W(\Lambda_A^-, \Lambda_B^+) = W(\Lambda_B^-, \Lambda_A^+)$

→ Wightman function

→ Local

→ Encodes quantum noise



EH: Negativity

→ $G_F(\Lambda_A^+, \Lambda_B^+)$

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→ Non local

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→ Related to **probability amplitude** that **both** detectors become excited mutually **due** to the quantum correlations

→ $W(\Lambda_A^-, \Lambda_A^+) = W(\Lambda_B^-, \Lambda_B^+)$

→ Wightman function

→ Local

→ Encodes quantum noise

→ **Probability** that **a** detector become excited



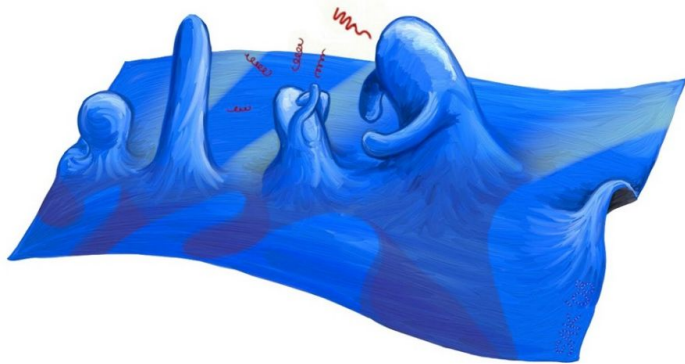
EH: Negativity

$$\mathcal{N} = \max\{0, |G_{AB}^{++}| - W^{-+}\}$$



EH: Negativity

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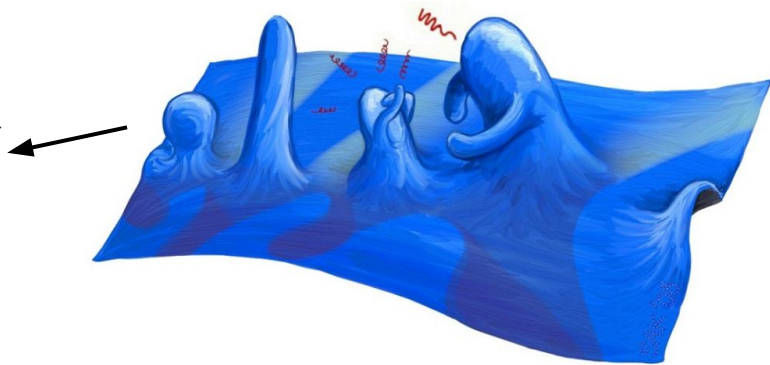




EH: Negativity

$$\mathcal{N} = \max\{0, |G_{AB}^{+++}| - W^{-+}\}$$

$$|G_{AB}^{+++}| > W^{-+}$$

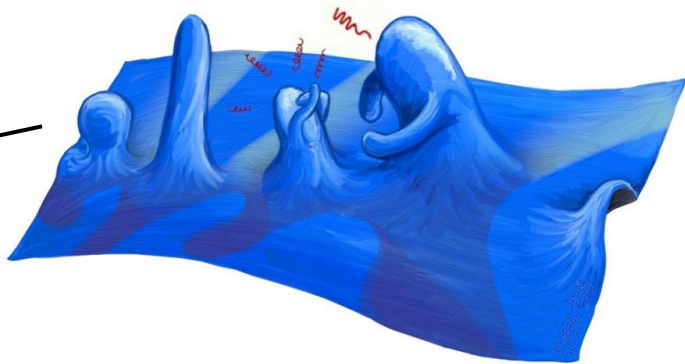




EH: Negativity

$$\mathcal{N} = \max\{0, |G_{AB}^{+++}| - W^{-+}\}$$

$$|G_{AB}^{+++}| > W^{-+}$$



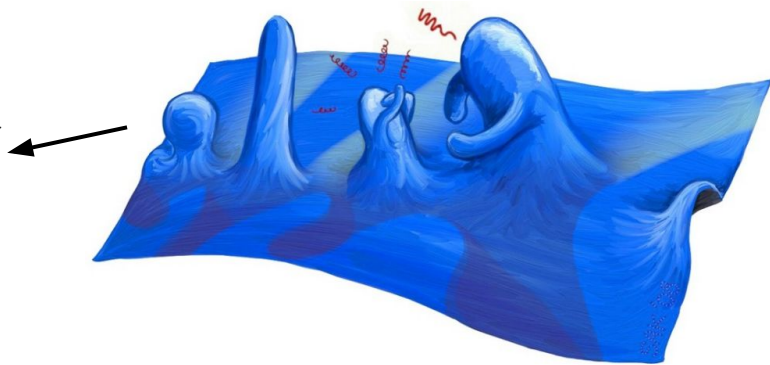
$$\mathcal{N} > 0$$



EH: Negativity

$$\mathcal{N} = \max\{0, |G_{AB}^{++}| - W^{-+}\}$$

$$|G_{AB}^{++}| > W^{-+}$$



$$\mathcal{N} > 0$$

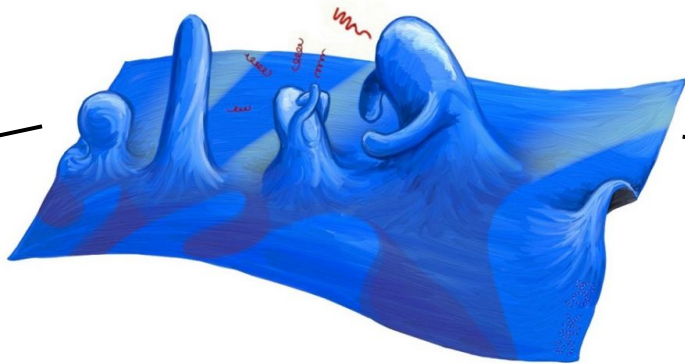




EH: Negativity

$$\mathcal{N} = \max\{0, |G_{AB}^{+++}| - W^{-+}\}$$

$$|G_{AB}^{+++}| > W^{-+}$$



$$|G_{AB}^{+++}| < W^{-+}$$

$$\mathcal{N} > 0$$



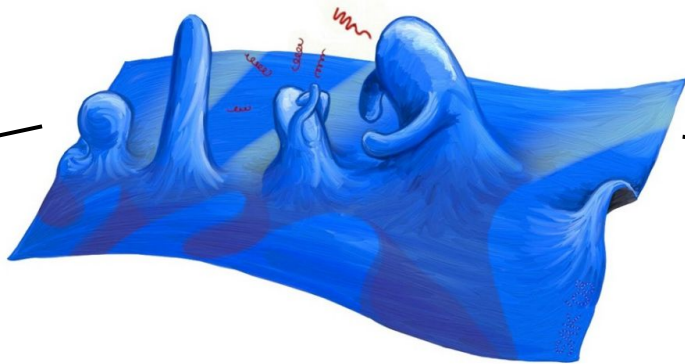


EH: Negativity

$$\mathcal{N} = \max\{0, |G_{AB}^{+++}| - W^{-+}\}$$

$$|G_{AB}^{+++}| > W^{-+}$$

$$\mathcal{N} > 0$$



$$|G_{AB}^{+++}| < W^{-+}$$

$$\mathcal{N} = 0$$

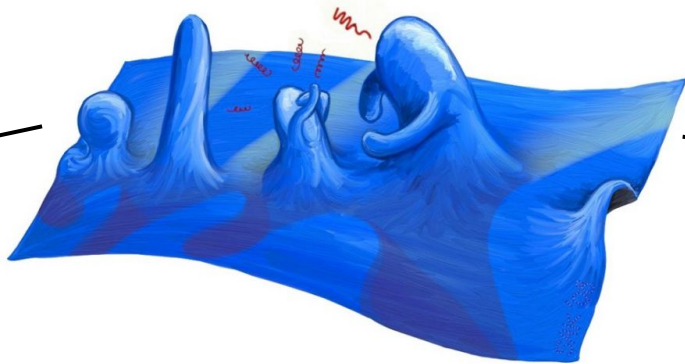


EH: Negativity

$$\mathcal{N} = \max\{0, |G_{AB}^{++}| - W^{-+}\}$$

$$|G_{AB}^{++}| > W^{-+}$$

$$\mathcal{N} > 0$$



$$|G_{AB}^{++}| < W^{-+}$$

$$\mathcal{N} = 0$$



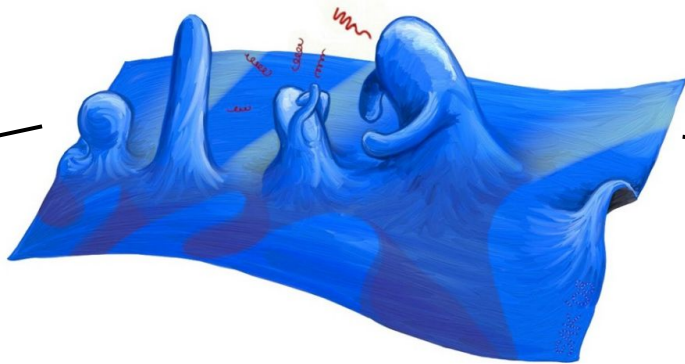


EH: Negativity

$$\mathcal{N} = \max\{0, |G_{AB}^{++}| - W^{-+}\}$$

$$|G_{AB}^{++}| > W^{-+}$$

$$\mathcal{N} > 0$$



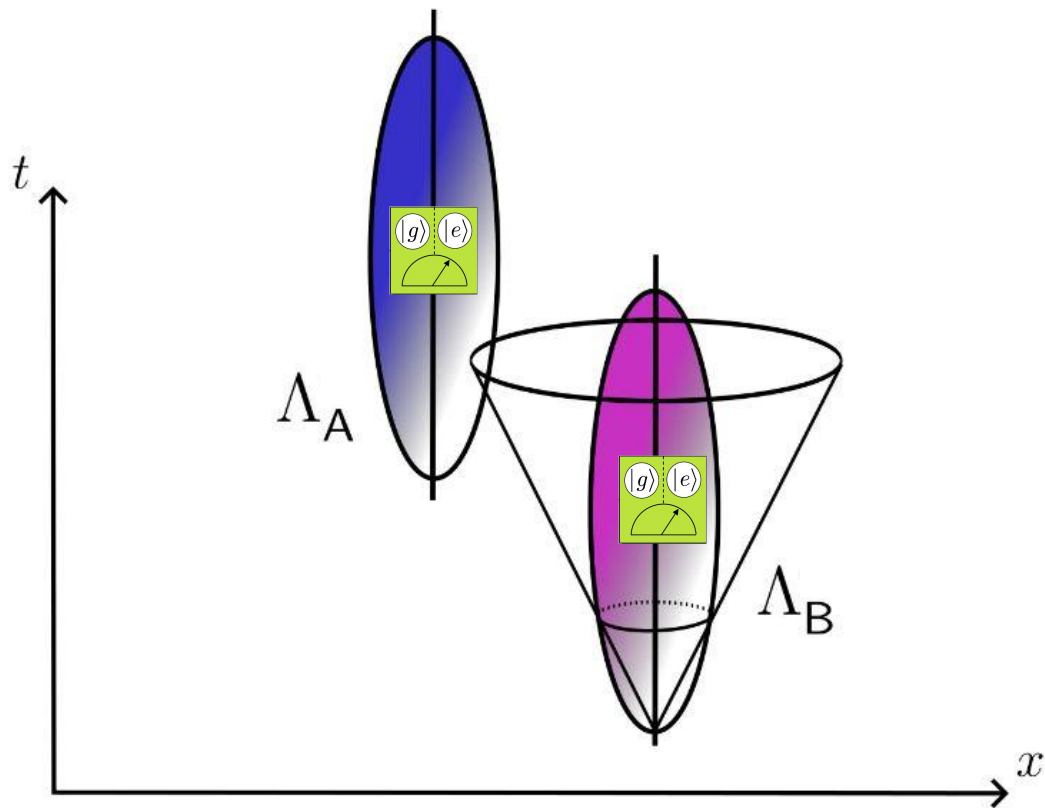
$$|G_{AB}^{++}| < W^{-+}$$

$$\mathcal{N} = 0$$



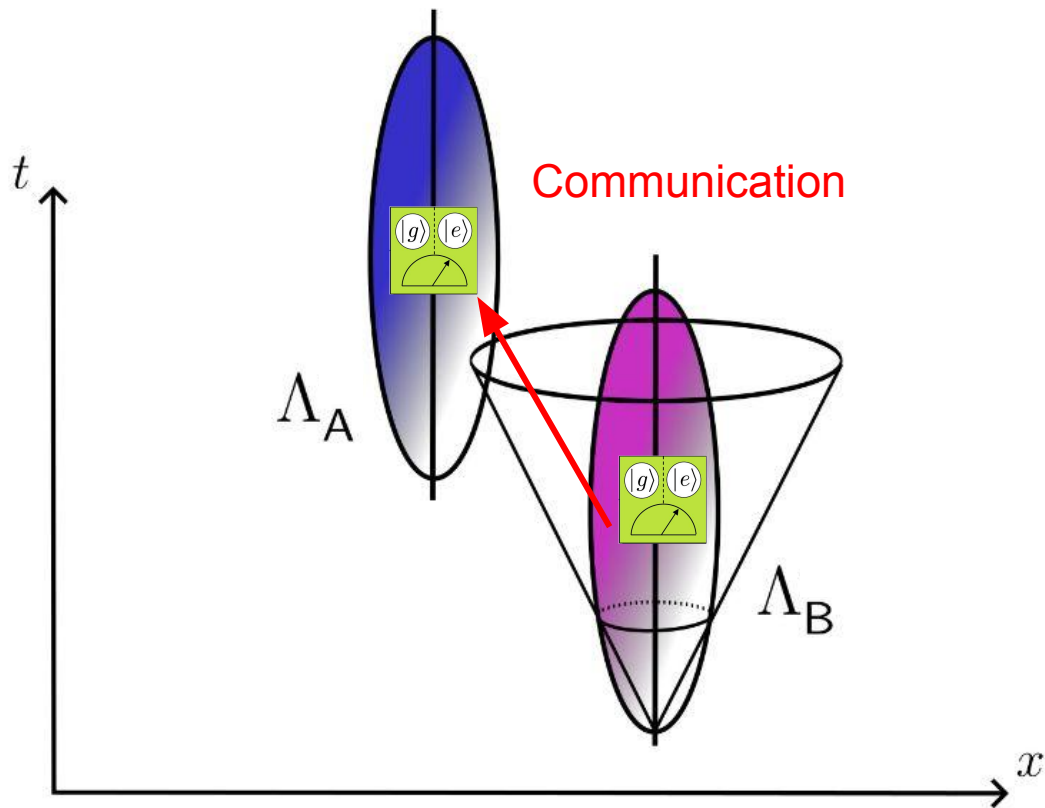


EH: Quantifying signalling



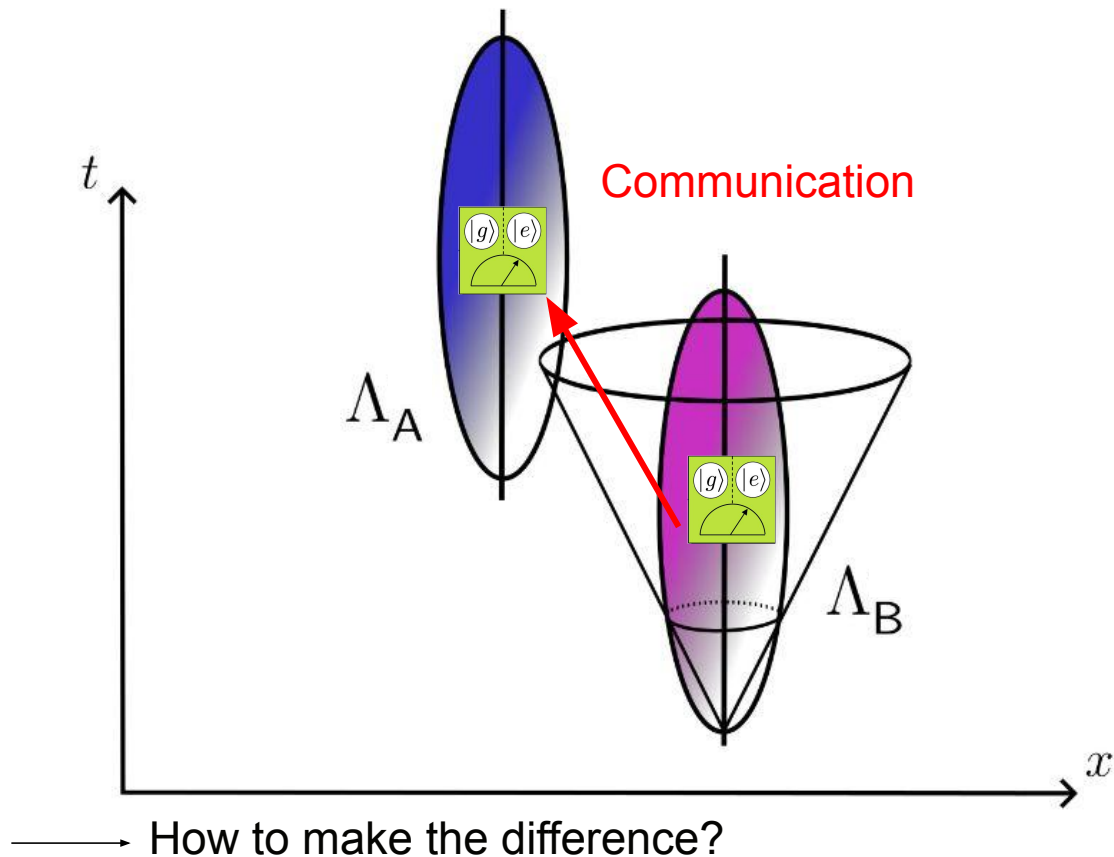


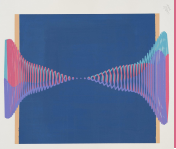
EH: Quantifying signalling



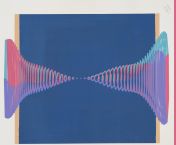


EH: Quantifying signalling



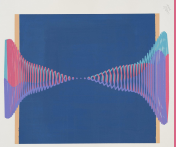


EH: Quantifying signalling



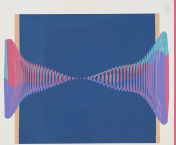
EH: Quantifying signalling

$$\longrightarrow G_F(\Lambda_A^+, \Lambda_B^+)$$



EH: Quantifying signalling

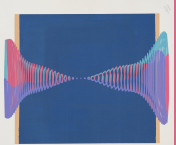
$$\longrightarrow G_F(\Lambda_A^+, \Lambda_B^+) = \frac{1}{2} \left(H(\Lambda_A^+, \Lambda_B^+) + i\Delta(\Lambda_A^+, \Lambda_B^+) \right)$$



EH: Quantifying signalling

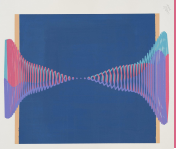
$$\longrightarrow G_F(\Lambda_A^+, \Lambda_B^+) = \frac{1}{2} \left(H(\Lambda_A^+, \Lambda_B^+) + i\Delta(\Lambda_A^+, \Lambda_B^+) \right)$$

state dependence

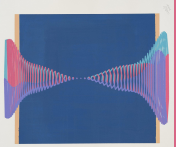


EH: Quantifying signalling

$$\longrightarrow G_F(\Lambda_A^+, \Lambda_B^+) = \frac{1}{2} \left(\underset{\substack{\nearrow \\ \text{state dependence}}}{H(\Lambda_A^+, \Lambda_B^+)} + i \underset{\substack{\nearrow \\ \text{exchange of information}}}{\Delta(\Lambda_A^+, \Lambda_B^+)} \right)$$



EH: Quantifying signalling



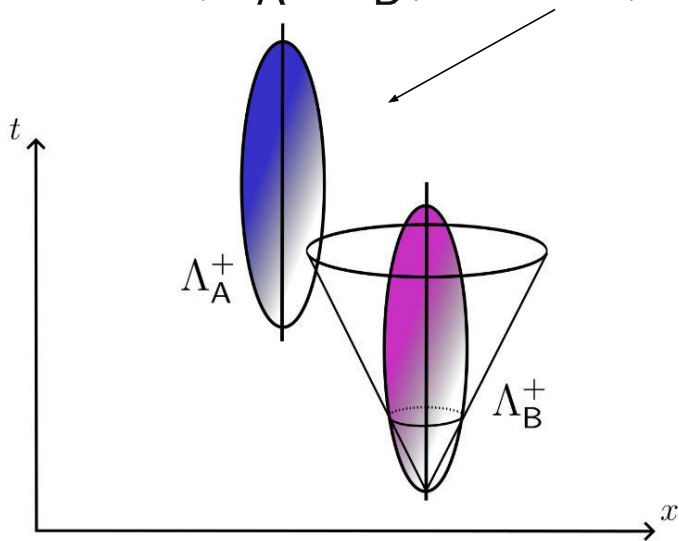
EH: Quantifying signalling

$$\Delta(\Lambda_A^+, \Lambda_B^+) = G_R(\Lambda_A^+, \Lambda_B^+) + G_R(\Lambda_B^+, \Lambda_A^+)$$



EH: Quantifying signalling

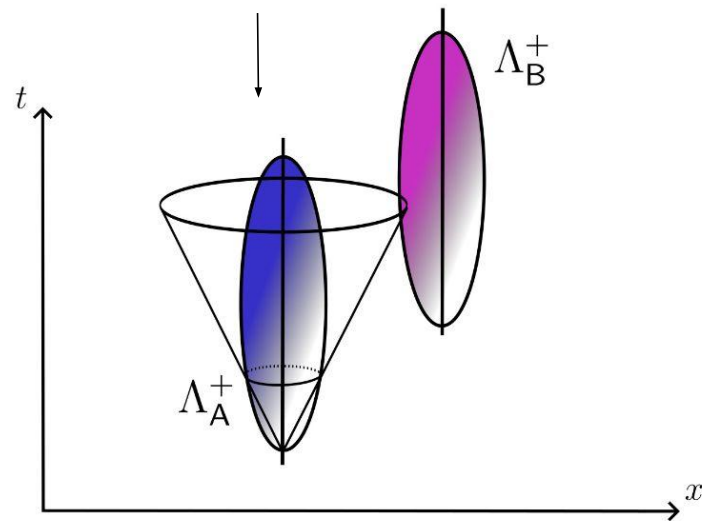
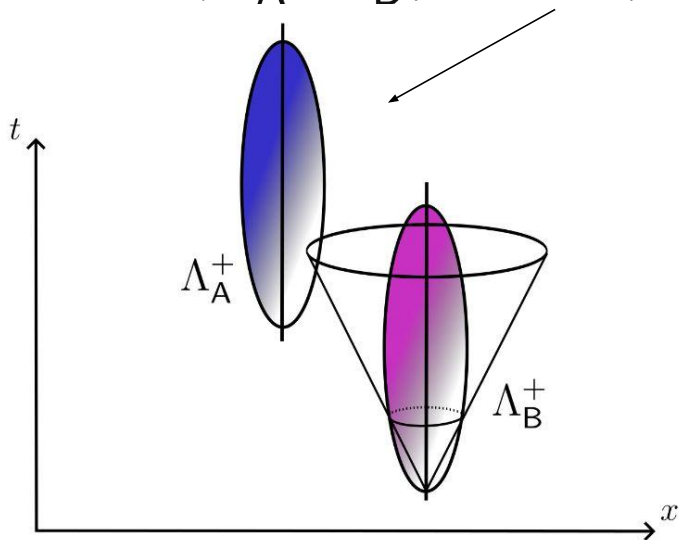
$$\Delta(\Lambda_A^+, \Lambda_B^+) = G_R(\Lambda_A^+, \Lambda_B^+) + G_R(\Lambda_B^+, \Lambda_A^+)$$





EH: Quantifying signalling

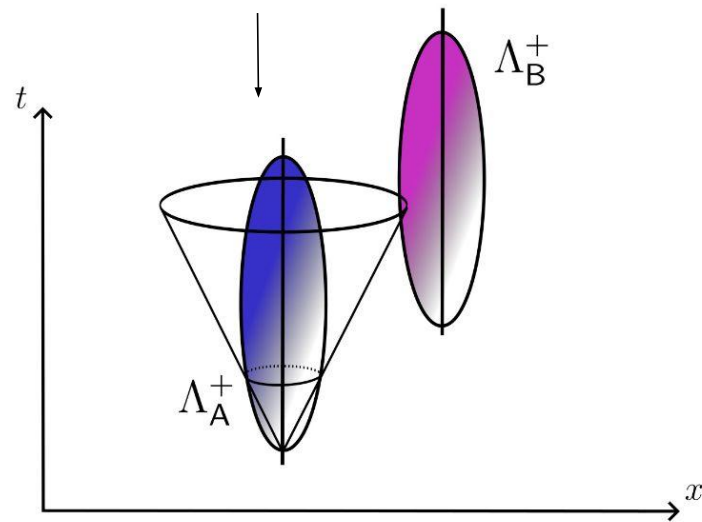
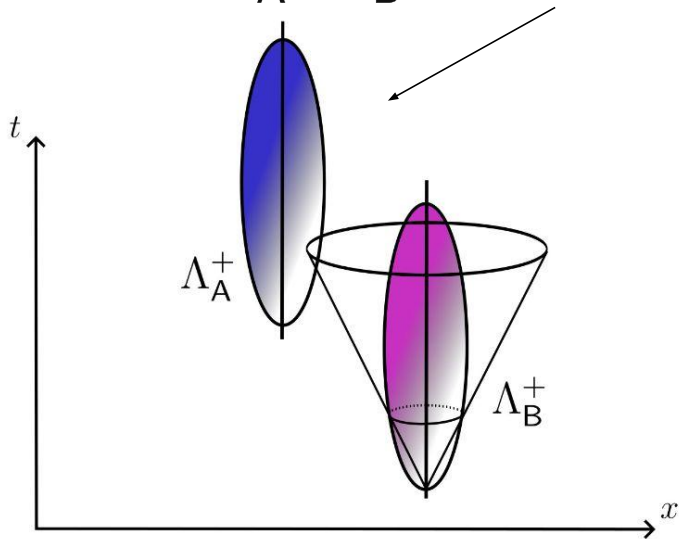
$$\Delta(\Lambda_A^+, \Lambda_B^+) = G_R(\Lambda_A^+, \Lambda_B^+) + G_R(\Lambda_B^+, \Lambda_A^+)$$





EH: Quantifying signalling

$$\Delta(\Lambda_A^+, \Lambda_B^+) = G_R(\Lambda_A^+, \Lambda_B^+) + G_R(\Lambda_B^+, \Lambda_A^+)$$

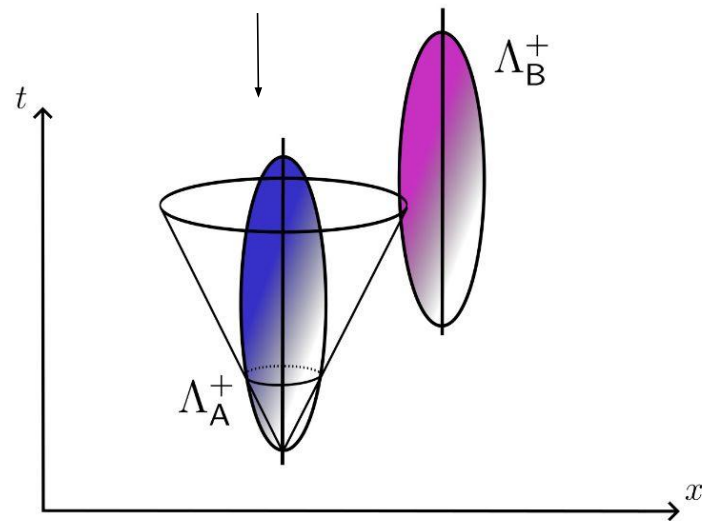
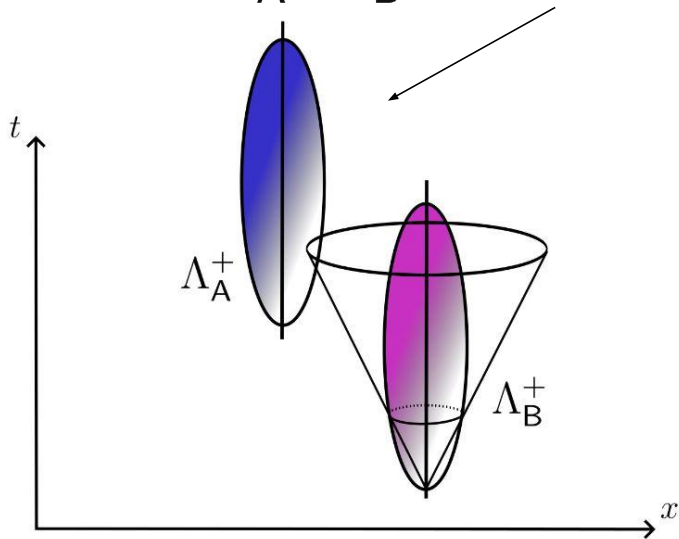


→ If $\Delta(\Lambda_A^+, \Lambda_B^+) \neq 0$, there is exchange of information



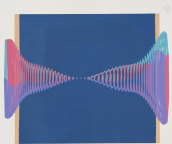
EH: Quantifying signalling

$$\Delta(\Lambda_A^+, \Lambda_B^+) = G_R(\Lambda_A^+, \Lambda_B^+) + G_R(\Lambda_B^+, \Lambda_A^+)$$



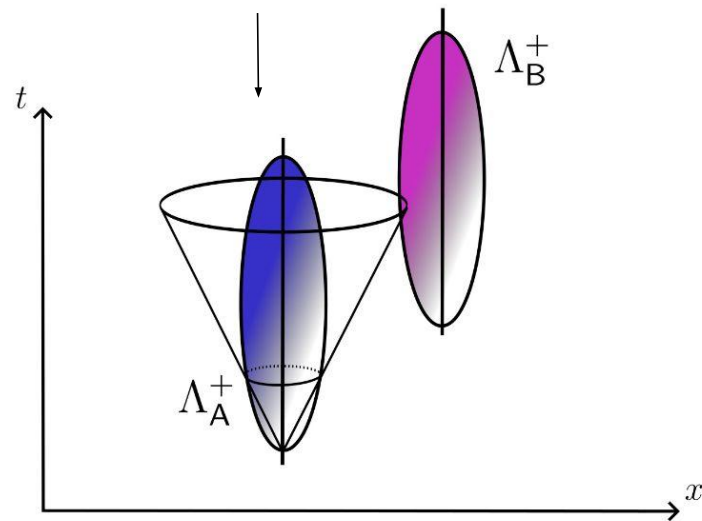
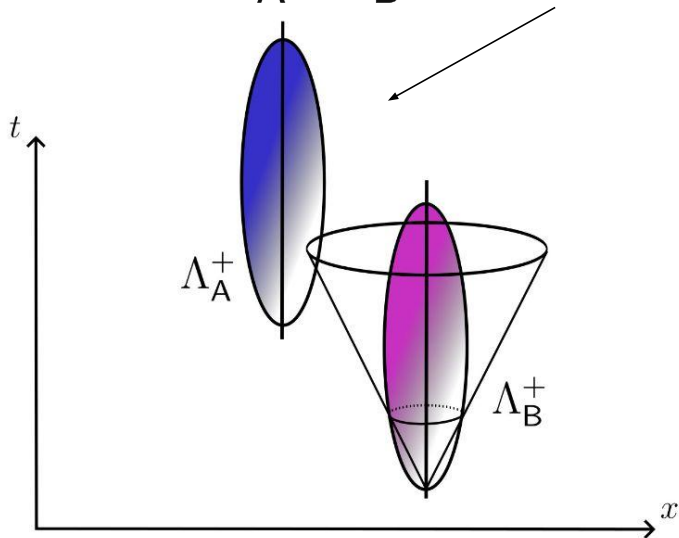
→ If $\Delta(\Lambda_A^+, \Lambda_B^+) \neq 0$, there is exchange of information

→ entanglement due to communication



EH: Quantifying signalling

$$\Delta(\Lambda_A^+, \Lambda_B^+) = G_R(\Lambda_A^+, \Lambda_B^+) + G_R(\Lambda_B^+, \Lambda_A^+)$$



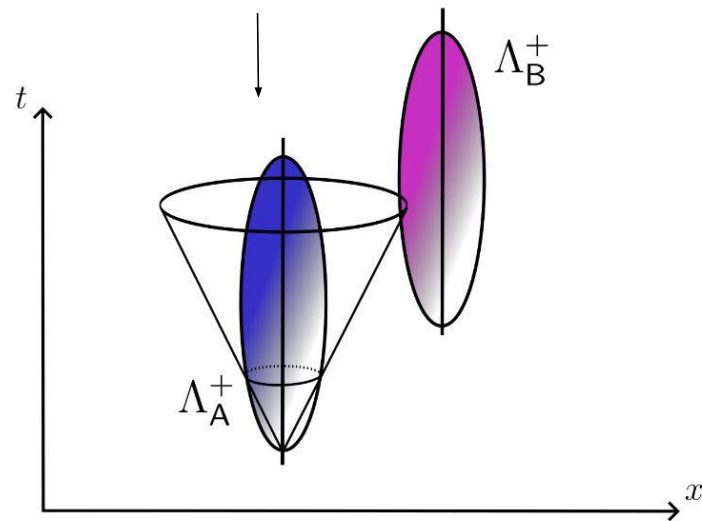
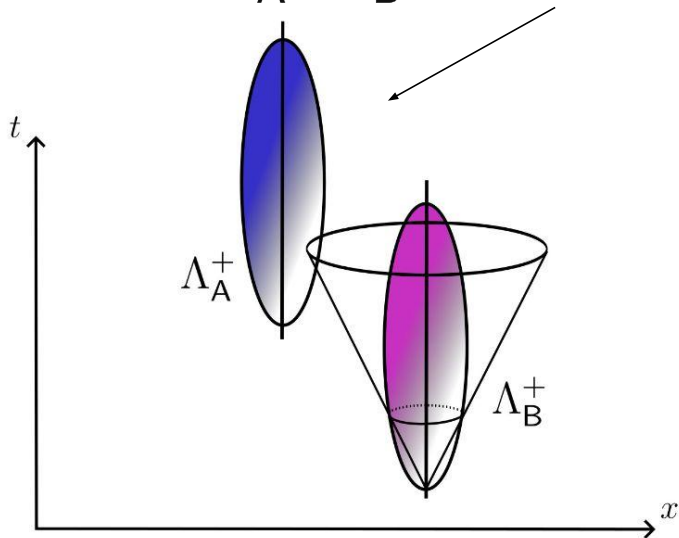
————→ If $\Delta(\Lambda_A^+, \Lambda_B^+) \neq 0$, there is exchange of information —————→ entanglement due to communication

————→ If $\Delta(\Lambda_A^+, \Lambda_B^+) = 0$, there is **no** exchange of information



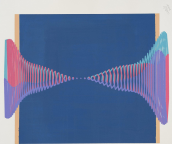
EH: Quantifying signalling

$$\Delta(\Lambda_A^+, \Lambda_B^+) = G_R(\Lambda_A^+, \Lambda_B^+) + G_R(\Lambda_B^+, \Lambda_A^+)$$



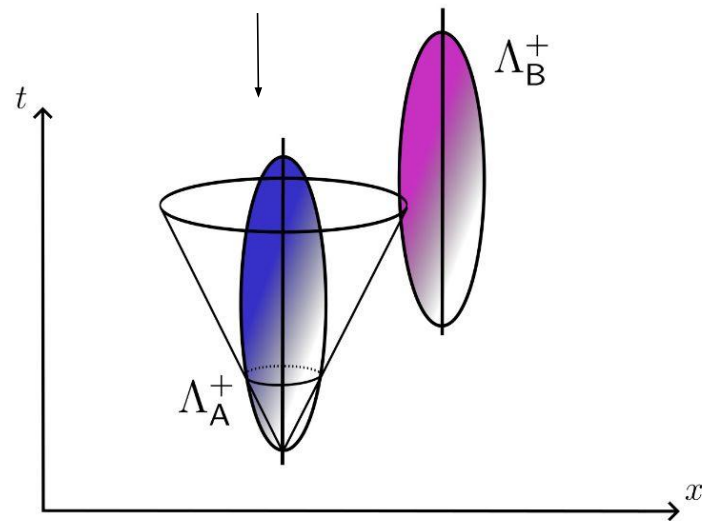
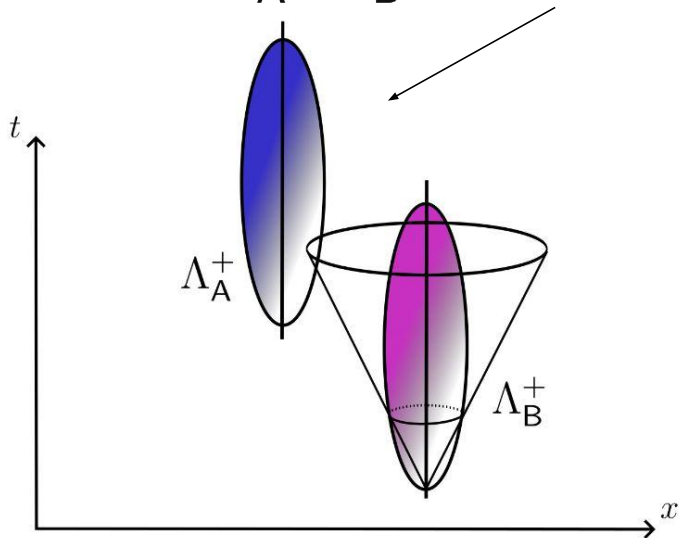
————→ If $\Delta(\Lambda_A^+, \Lambda_B^+) \neq 0$, there is exchange of information —————→ entanglement due to communication

————→ If $\Delta(\Lambda_A^+, \Lambda_B^+) = 0$, there is **no** exchange of information —————→ genuine entanglement from vacuum



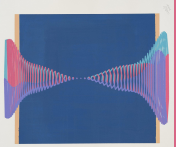
EH: Quantifying signalling

$$\Delta(\Lambda_A^+, \Lambda_B^+) = G_R(\Lambda_A^+, \Lambda_B^+) + G_R(\Lambda_B^+, \Lambda_A^+)$$



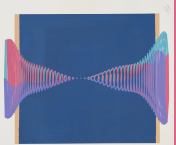
—————> If $\Delta(\Lambda_A^+, \Lambda_B^+) \neq 0$, there is exchange of information —————> entanglement due to communication

—————> If $\Delta(\Lambda_A^+, \Lambda_B^+)^* = 0$, there is **no** exchange of information —————> genuine entanglement from vacuum



EH: Quantifying signalling

$$\longrightarrow \Delta(\Lambda_A^+, \Lambda_B^+) = 0$$



EH: Quantifying signalling

$$\longrightarrow \Delta(\Lambda_A^+, \Lambda_B^+) = 0$$

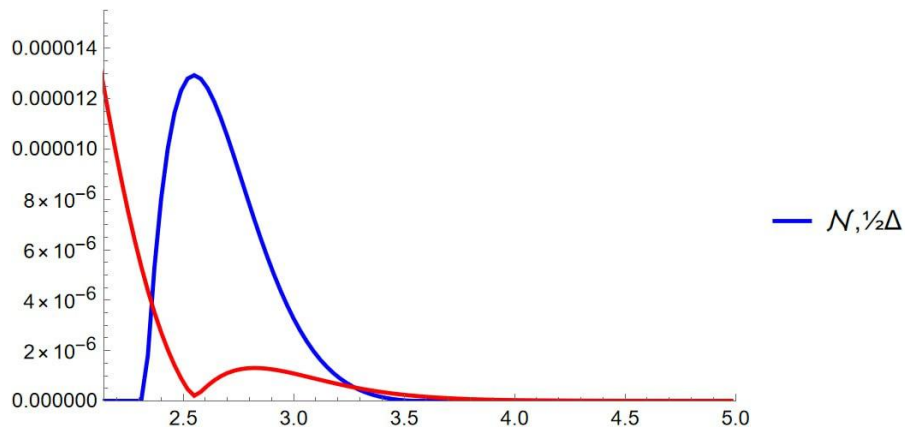
$$\longrightarrow \Delta(\Lambda_A^+, \Lambda_B^+) = G_R(\Lambda_A^+, \Lambda_B^+) + G_R(\Lambda_B^+, \Lambda_A^+)$$

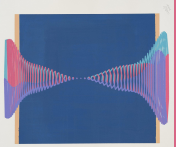


EH: Quantifying signalling

$$\longrightarrow \Delta(\Lambda_A^+, \Lambda_B^+) = 0$$

$$\longrightarrow \Delta(\Lambda_A^+, \Lambda_B^+) = G_R(\Lambda_A^+, \Lambda_B^+) + G_R(\Lambda_B^+, \Lambda_A^+)$$





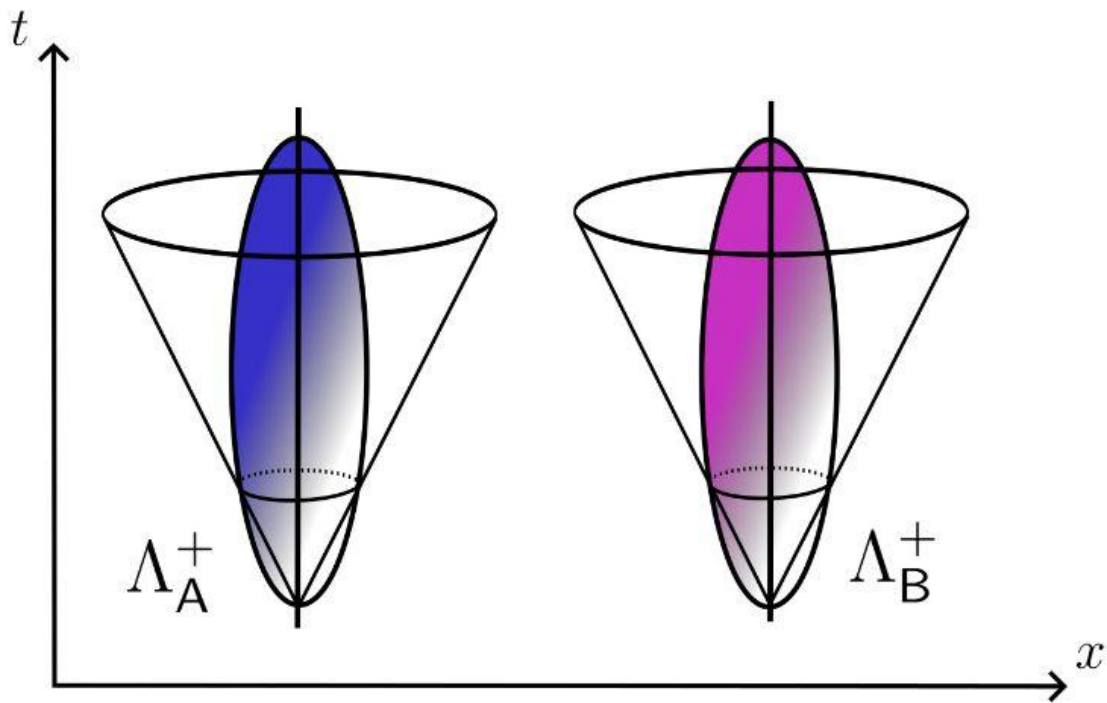
EH: Quantifying signalling

$$\Delta(\Lambda_A^+, \Lambda_B^+) = 0$$



EH: Quantifying signalling

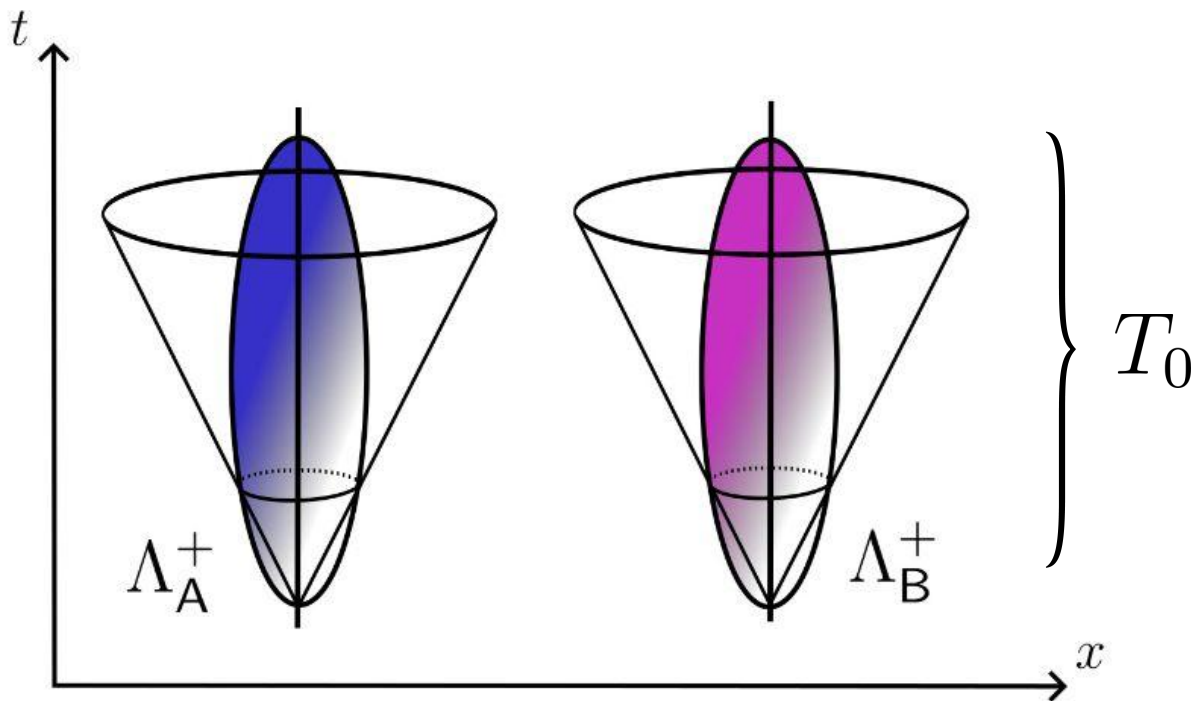
$$\Delta(\Lambda_A^+, \Lambda_B^+) = 0$$





EH: Quantifying signalling

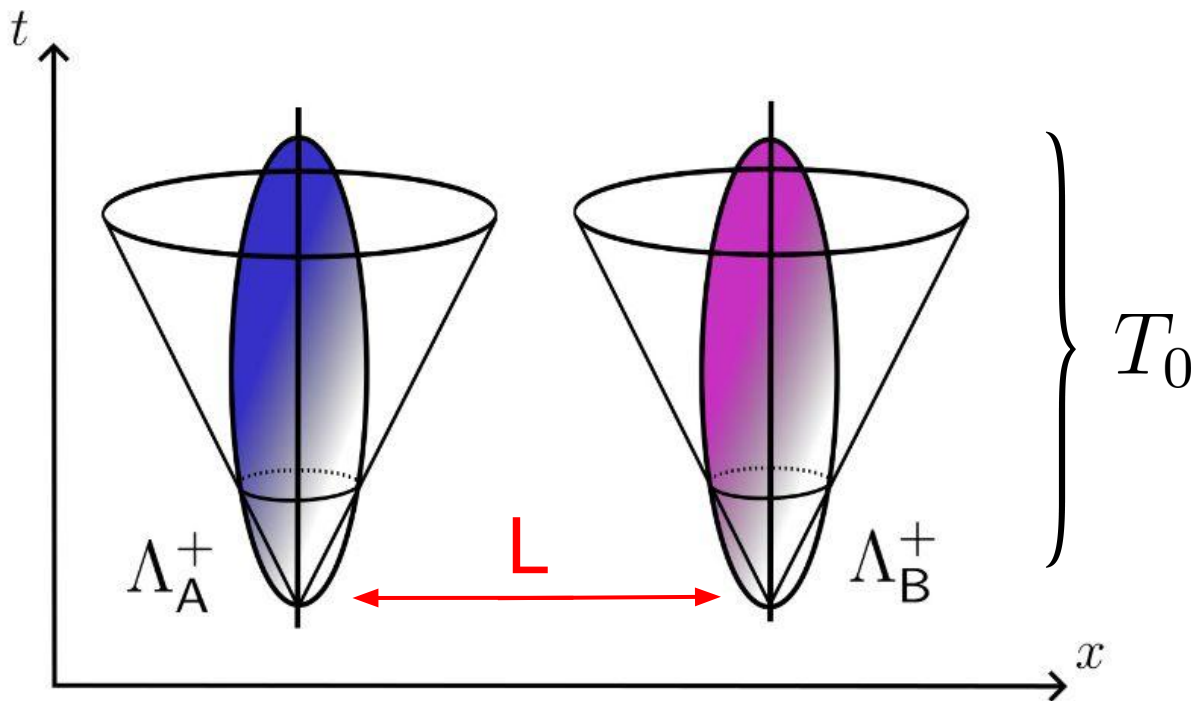
$$\Delta(\Lambda_A^+, \Lambda_B^+) = 0$$

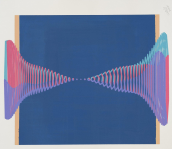




EH: Quantifying signalling

$$\Delta(\Lambda_A^+, \Lambda_B^+) = 0$$

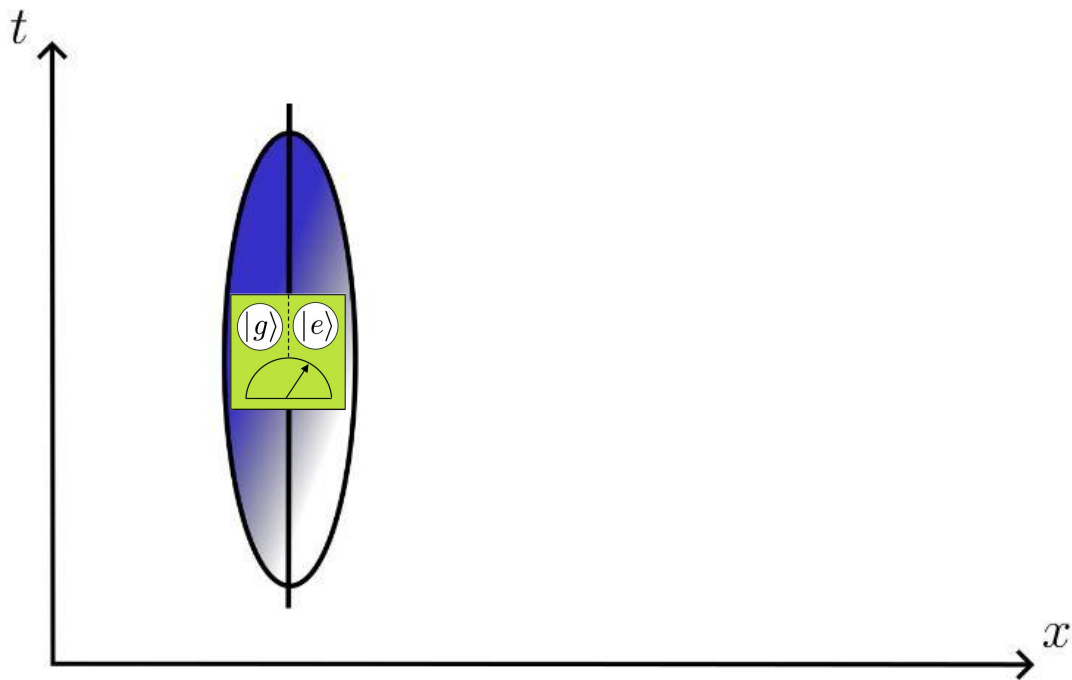




EH: Negativity in practice

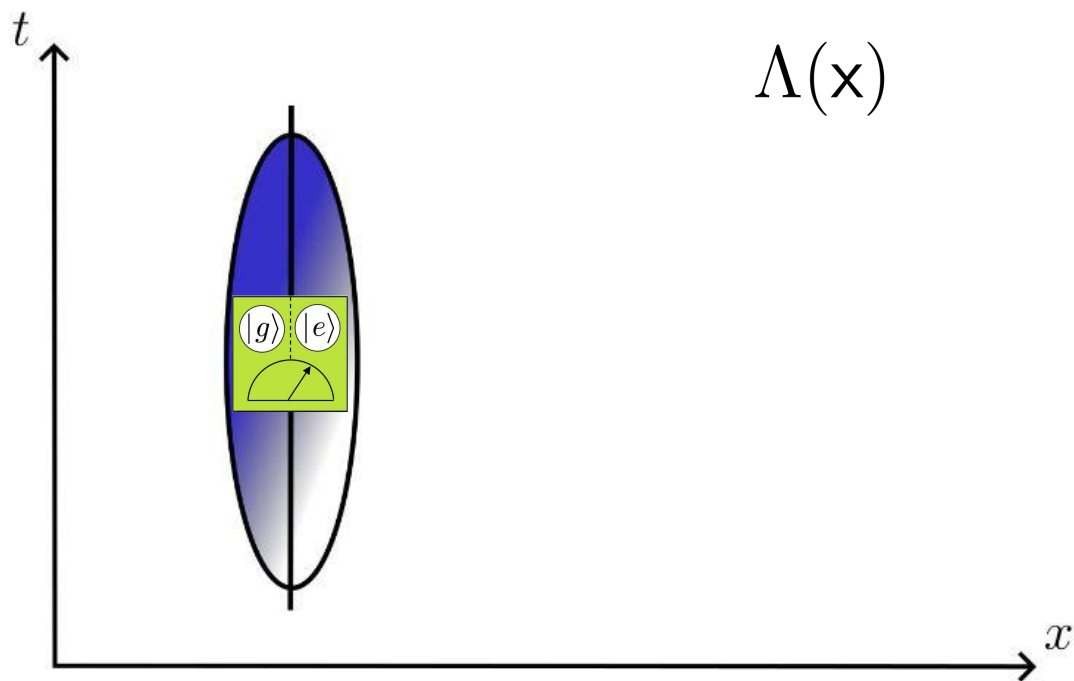


EH: Negativity in practice



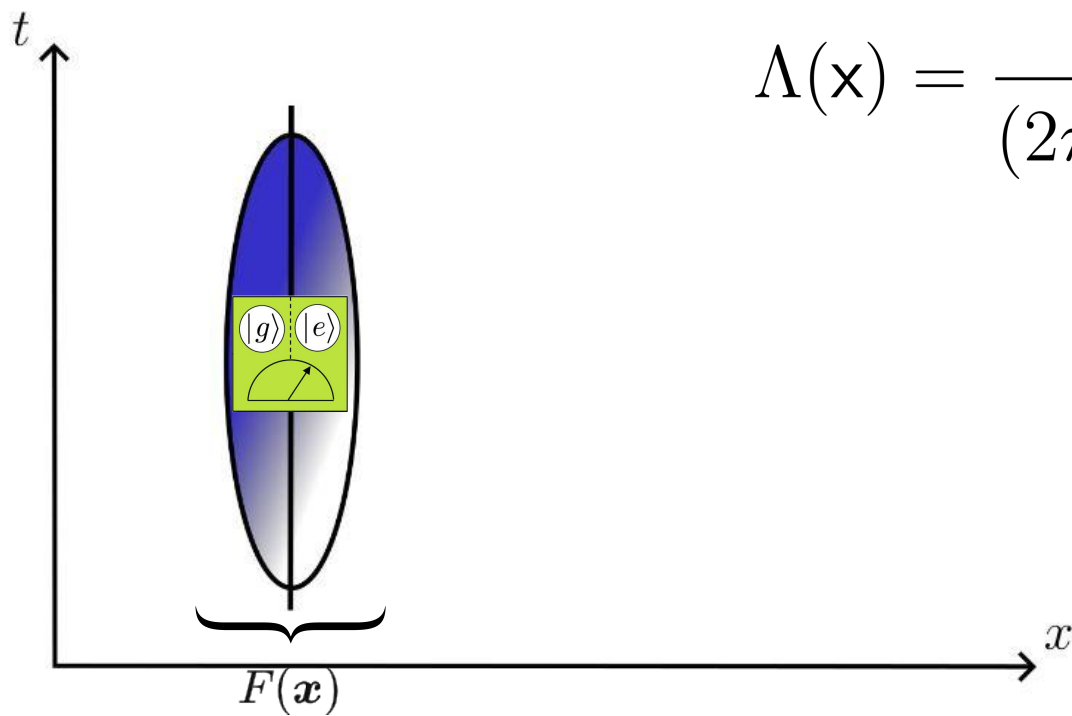


EH: Negativity in practice





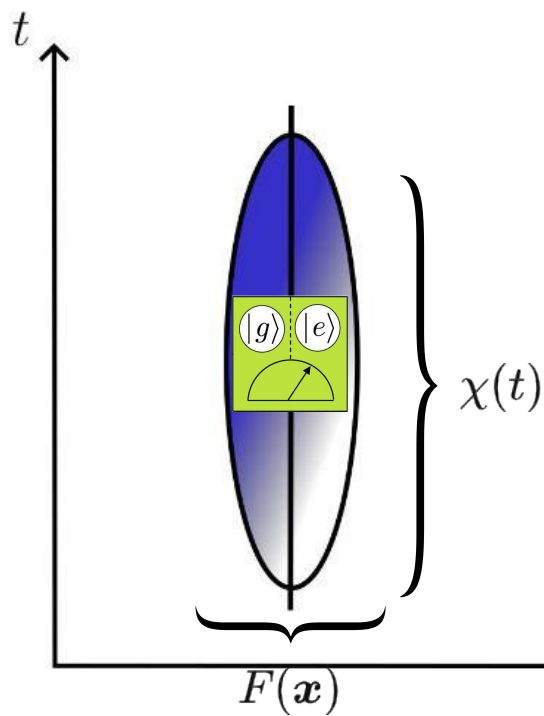
EH: Negativity in practice



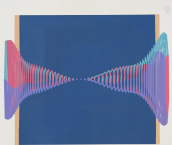
$$\Lambda(\mathbf{x}) = \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}} e^{-\frac{|\mathbf{x}|^2}{2\sigma^2}}$$



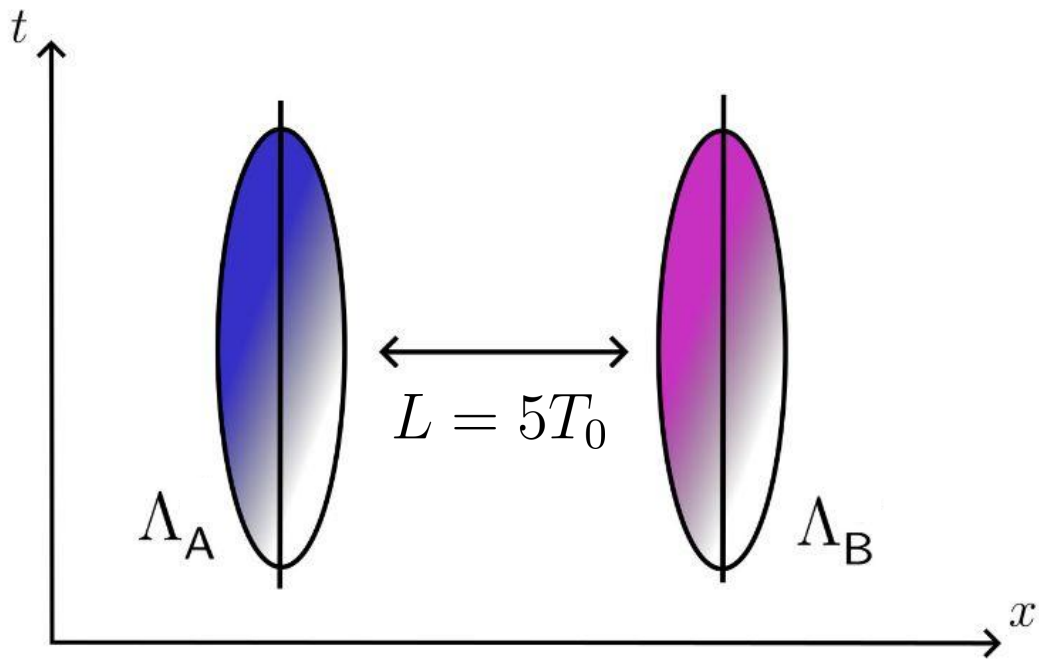
EH: Negativity in practice

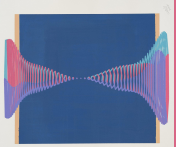


$$\Lambda(\mathbf{x}) = \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}} e^{-\frac{|\mathbf{x}|^2}{2\sigma^2}} e^{-\frac{t^2}{2T^2}}$$



EH: Negativity in practice





EH: Quantifying signalling

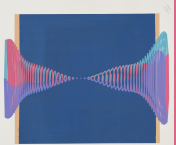
$$\longrightarrow \mathcal{N}^{\pm}[\hat{\rho}_{AB}] := \max\{0, |\mathcal{M}^{\pm}| - \mathcal{L}_{jj}\} + \mathcal{O}(\lambda^4)$$



EH: Quantifying signalling

$$\longrightarrow \mathcal{N}^{\pm}[\hat{\rho}_{AB}] := \max\{0, |\mathcal{M}^{\pm}| - \mathcal{L}_{jj}\} + \mathcal{O}(\lambda^4)$$

$$\mathcal{I}[\hat{\rho}_{AB}] := \begin{cases} \frac{\mathcal{N}^{-}[\hat{\rho}_{AB}]}{\mathcal{N}[\hat{\rho}_{AB}]}, & \mathcal{N}[\hat{\rho}_{AB}] > 0 \\ 0 & \mathcal{N}[\hat{\rho}_{AB}] = 0 \end{cases}$$

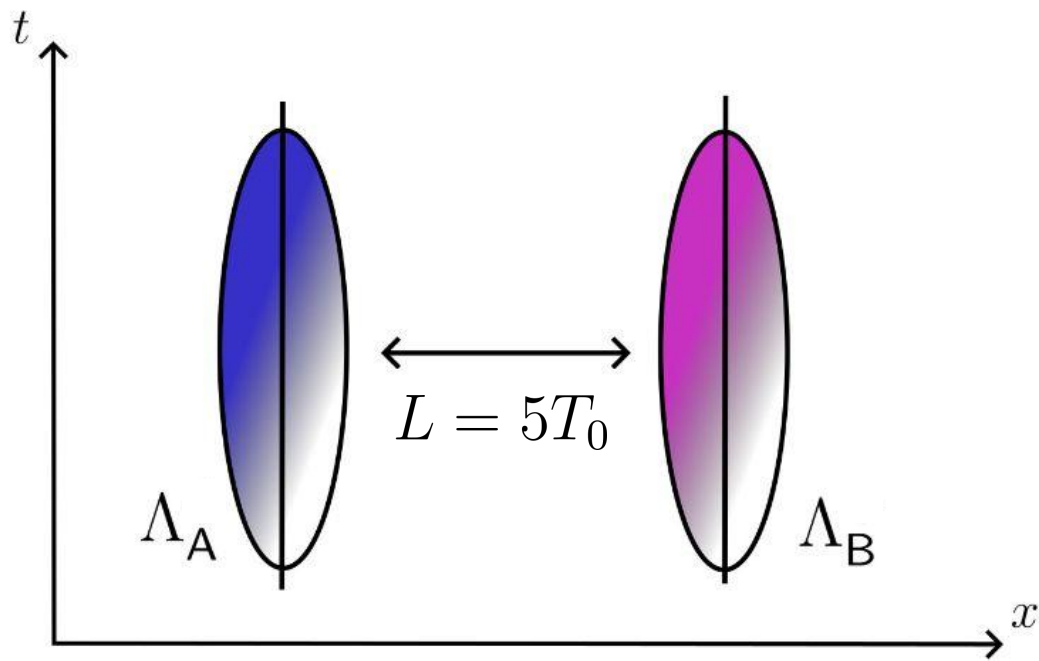


EH: Quantifying signalling

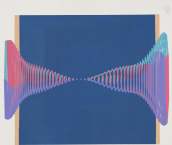
$$\longrightarrow \frac{\Delta}{\mathcal{N}} \leq \varepsilon$$



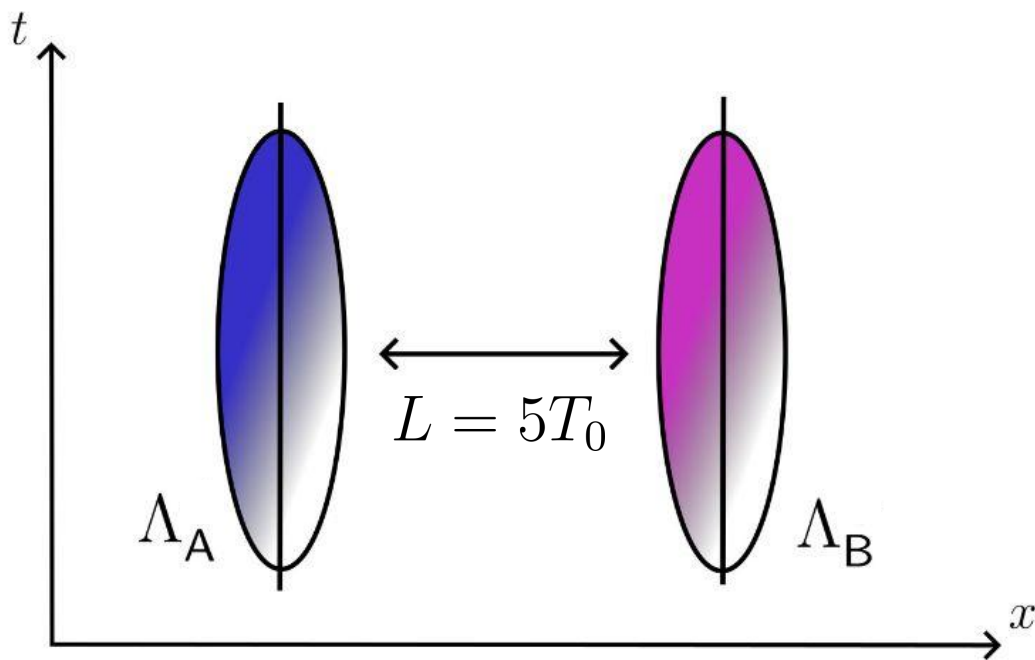
EH: Negativity in practice



$$\Lambda_A = \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}} e^{\frac{-t^2}{2T^2}} e^{\frac{-|x|^2}{2\sigma^2}}$$



EH: Negativity in practice

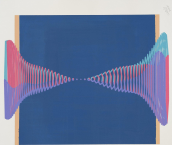


$$\Lambda_A = \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}} e^{\frac{-t^2}{2T^2}} e^{\frac{-|x|^2}{2\sigma^2}}$$

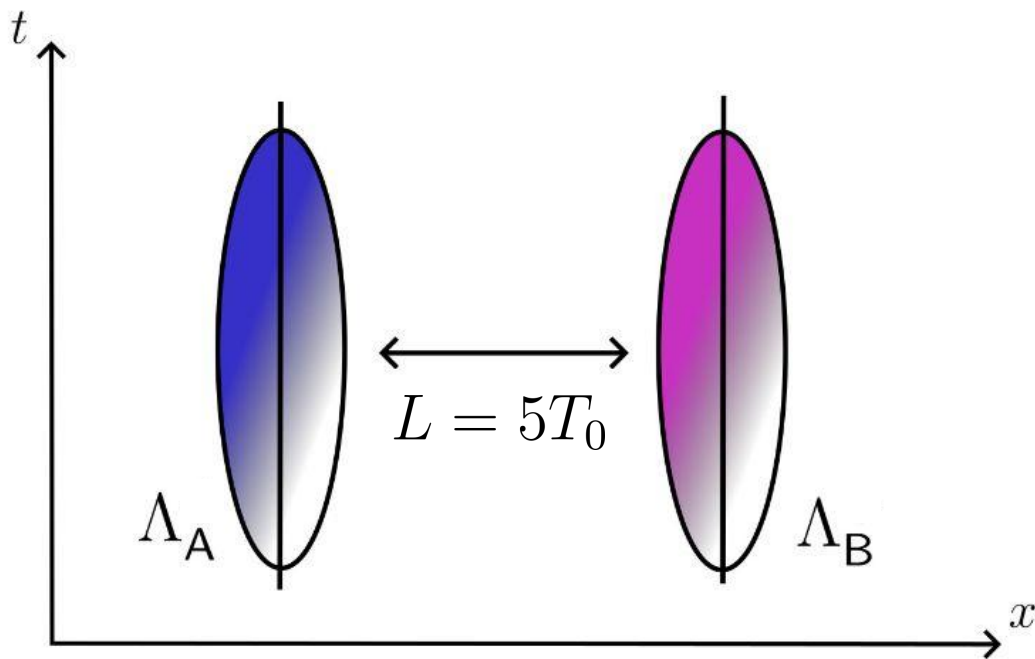
$$\Lambda_B = \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}} e^{\frac{-t^2}{2T^2}} e^{\frac{-|x-L|^2}{2\sigma^2}}$$

A. Pozas-Kerstjens and E. Martín-Martínez, Harvesting correlations from the quantum vacuum, Phys. Rev. D 92, 064042 (2015).

R. T. Perche, "Closed-form expressions for smeared bi-distributions of a massless scalar field: non-perturbative and asymptotic results in relativistic quantum information", 20 Dec 2023, <https://doi.org/10.48550/arXiv.2312.13343>



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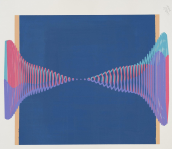


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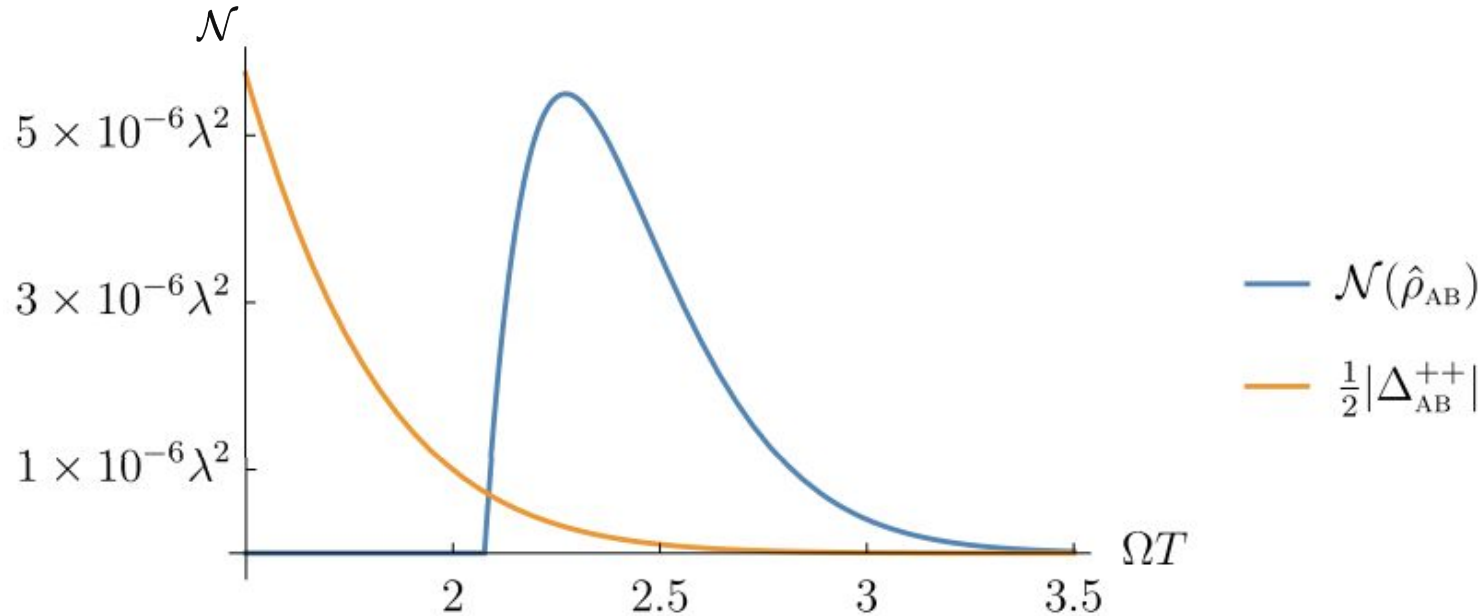
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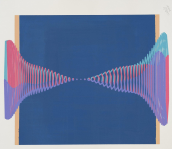
————— Closed forms!

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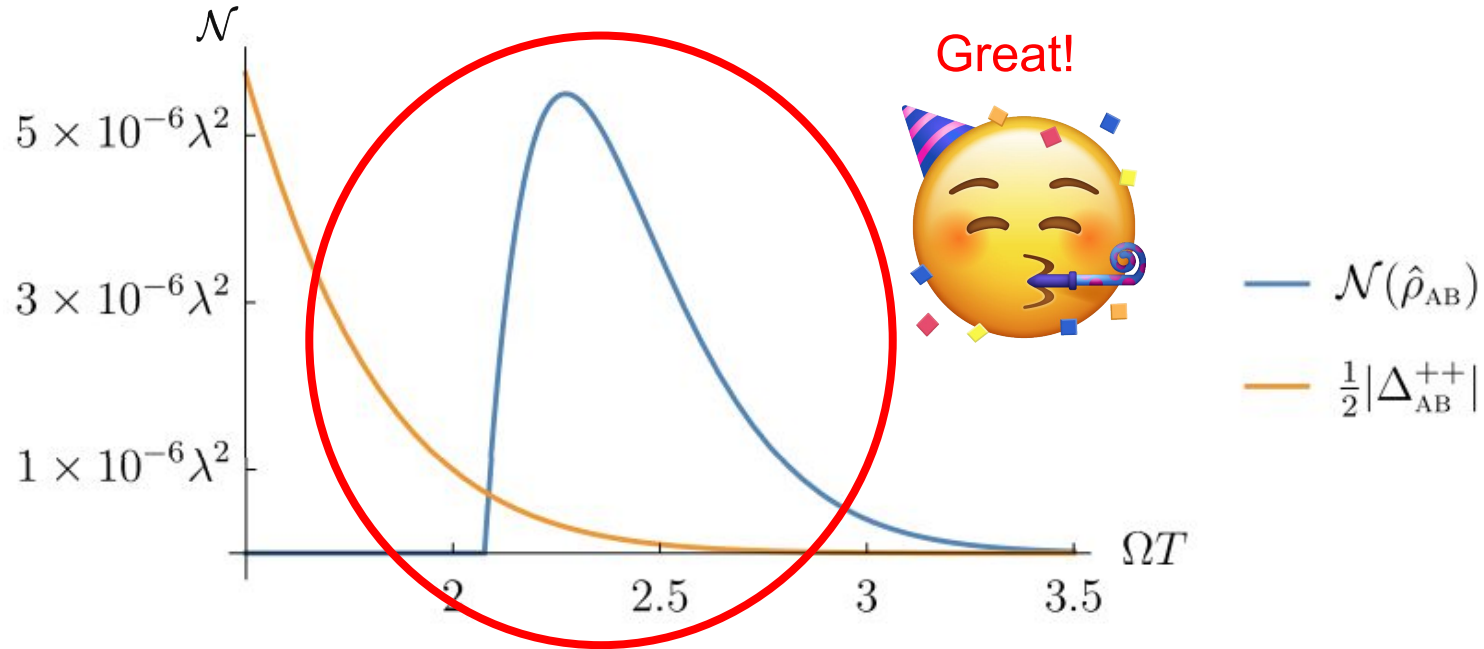


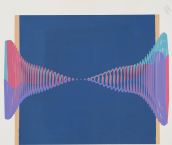
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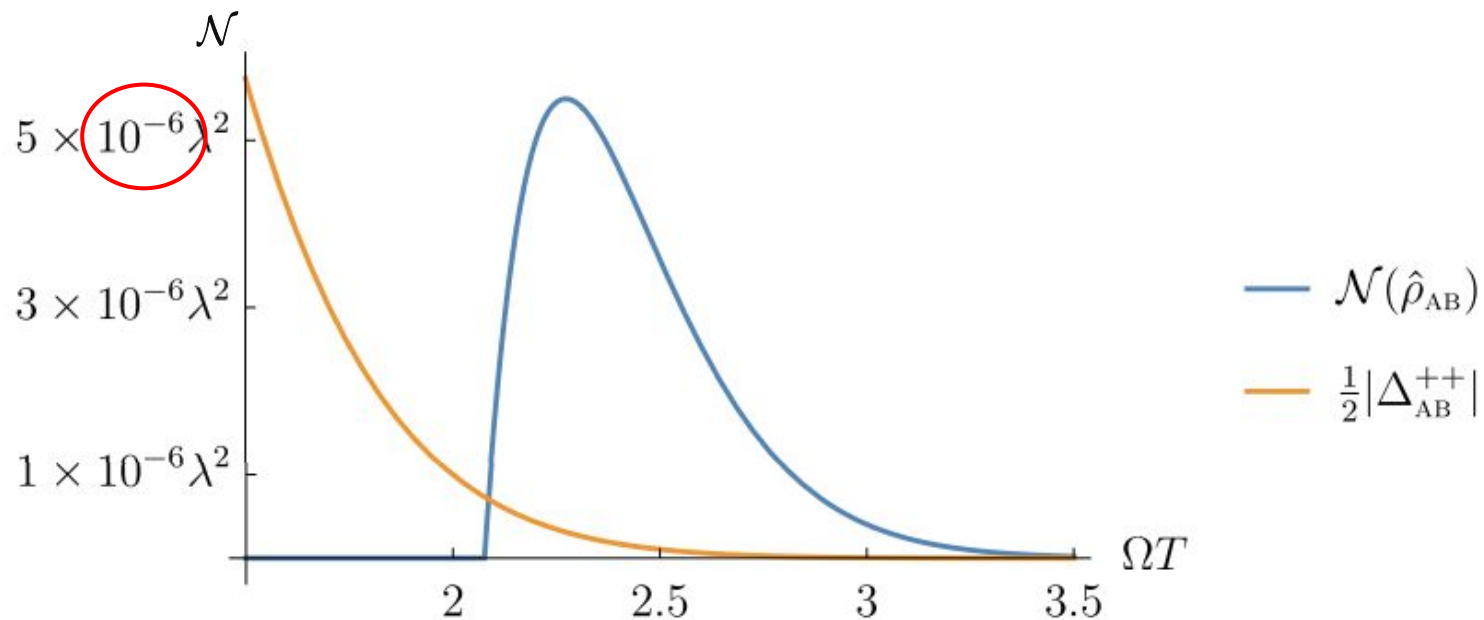


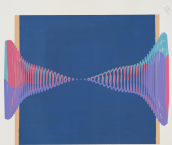
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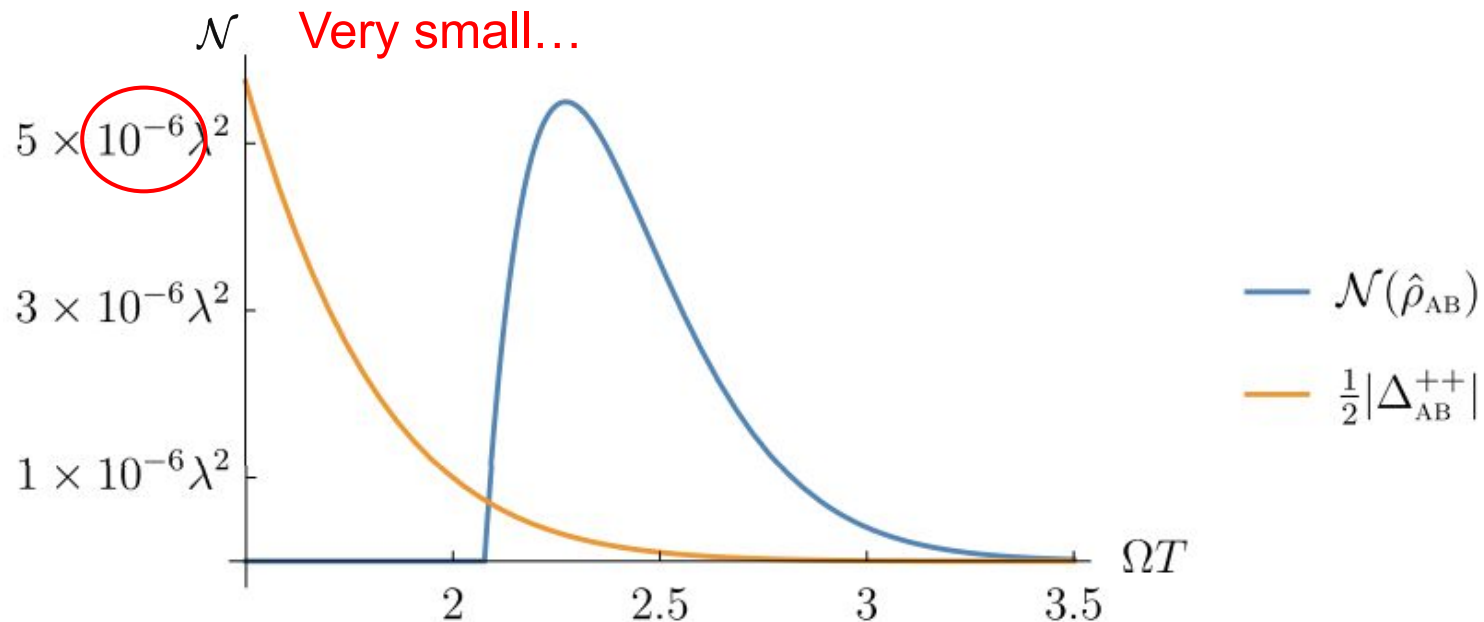


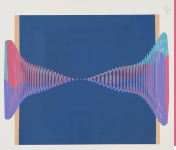
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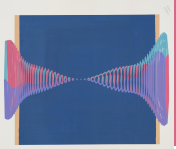


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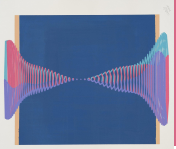


EH: Summary



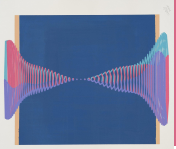
EH: Summary

————→ We place 2 UDW detectors in two regions of spacetime



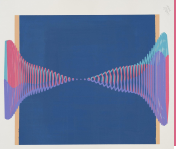
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- We place 2 UDW detectors in two regions of spacetime
- we consider causally disconnected regions to avoid signaling



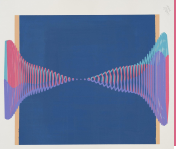
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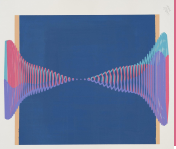
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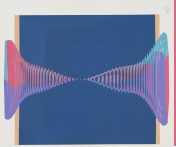
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→ Feynman propagator	}	→ Negativity (entanglement)
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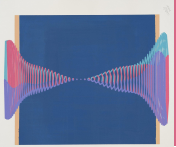
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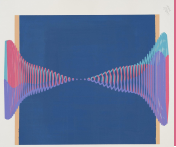
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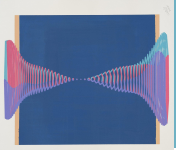
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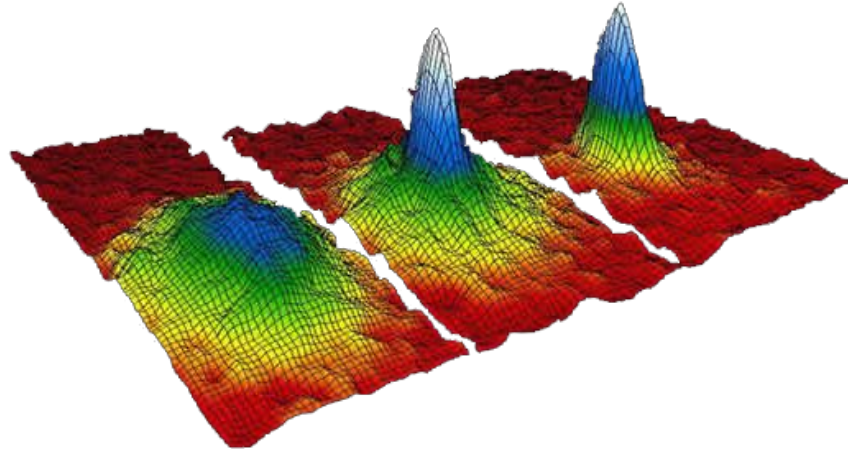


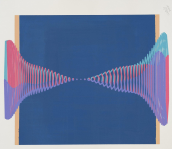
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- The observed negativity is of the order of $\mathcal{O}(10^{-6})$

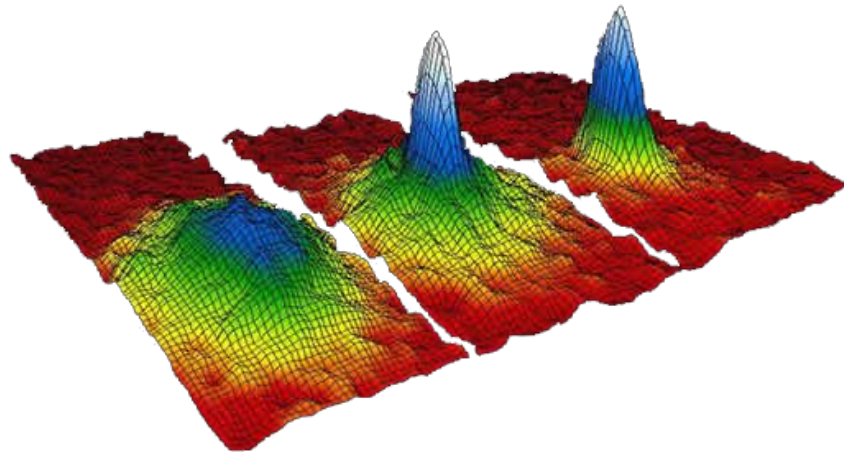


Analogous Quantum Field Theory





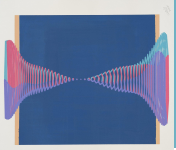
Analogous Quantum Field Theory



Markus Oberthaler

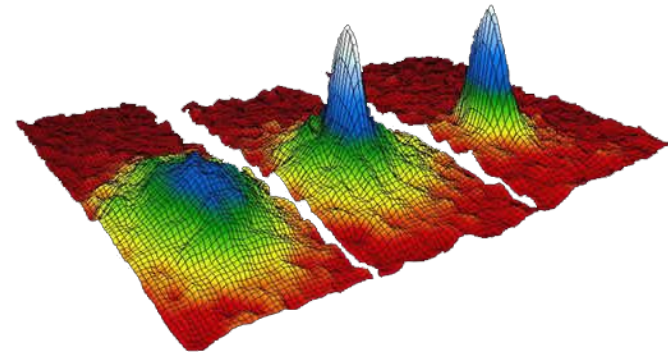


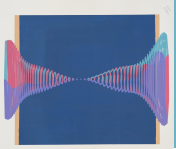
T. Rick Perche



Analogous Quantum Field Theory

————→ Bose-Einstein condensate





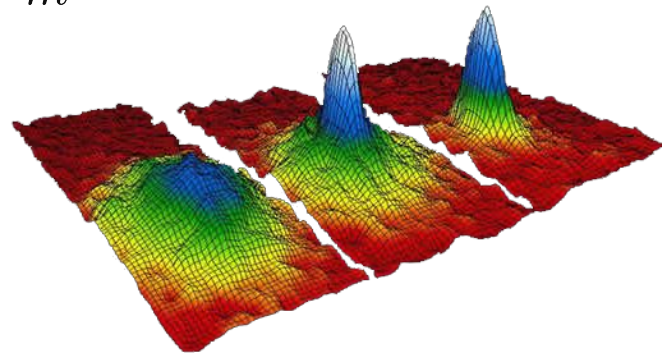
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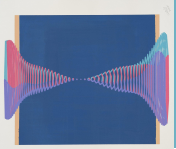
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$$i\hbar \frac{\partial}{\partial t} \psi = \left(-\frac{\hbar^2}{2m} \nabla^2 + g|\psi|^2 \right) \psi$$

“Gross-Pitaevskii”

$$g = \frac{4\pi\hbar^2 a_s}{m}$$





Analogous Quantum Field Theory

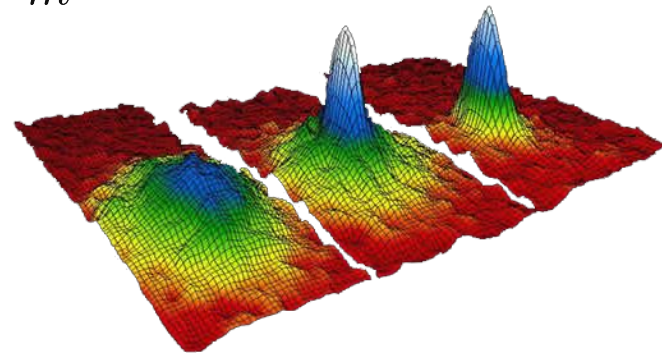
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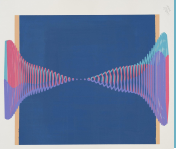
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scattering length





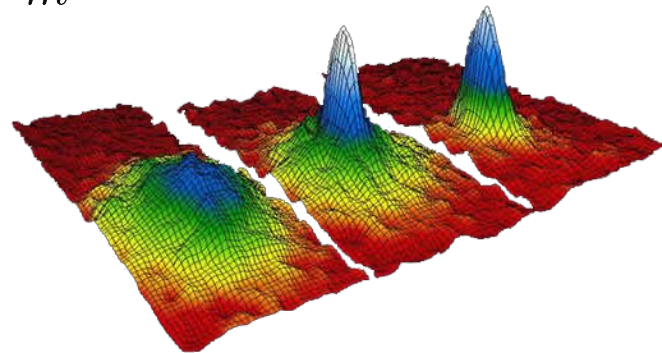
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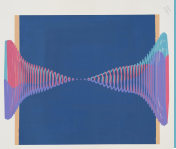
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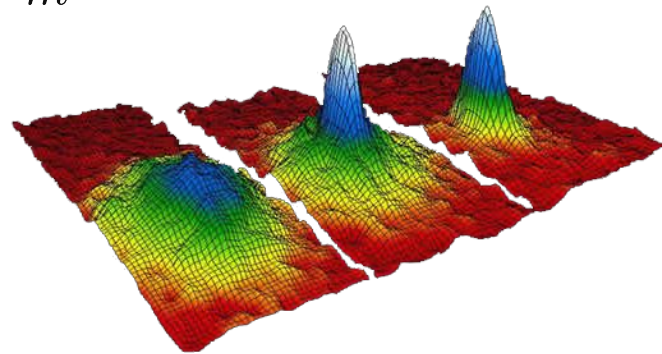
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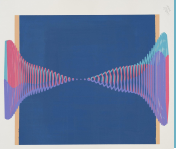
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Analogous Quantum Field Theory

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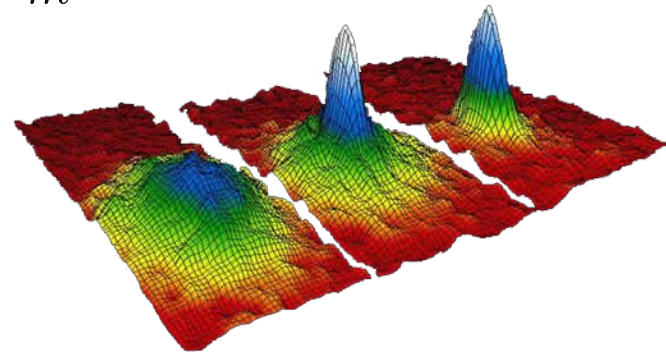
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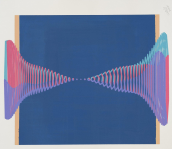
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$$E(p) = \sqrt{\left(\frac{p^2}{2m} \right)^2 + c_s^2 p^2}$$





Analogous Quantum Field Theory

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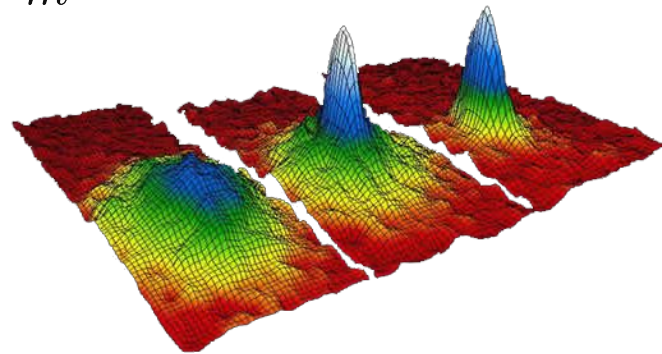
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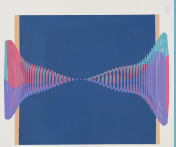
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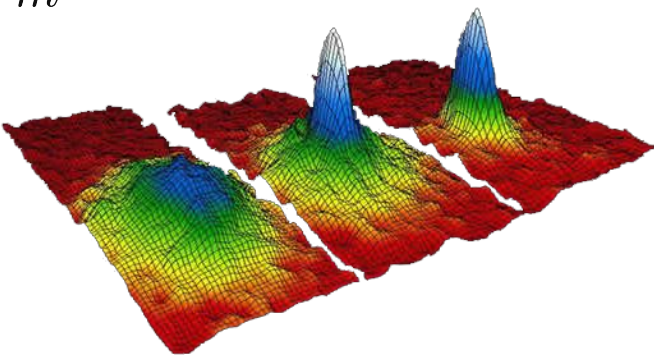
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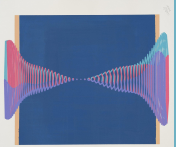
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relativistic, real scalar quantum field



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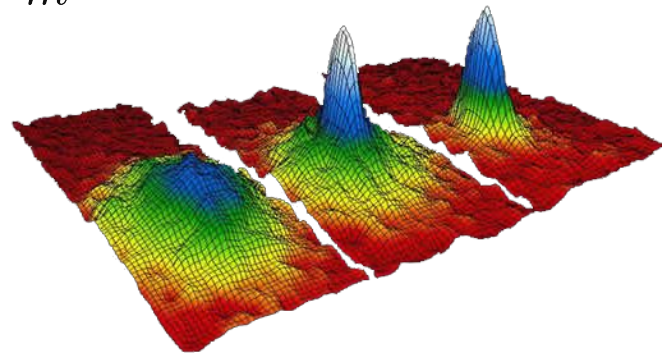
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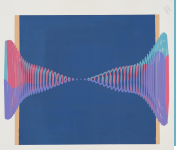
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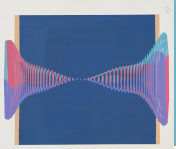
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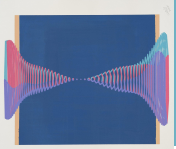


Entanglement Harvesting (BEC)



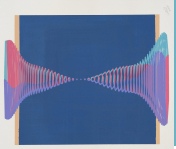
Entanglement Harvesting (BEC)

————→ Polaron (single atom), trapped by dipole interaction



Entanglement Harvesting (BEC)

- Polaron (single atom), trapped by dipole interaction
- when a phonon passes, it interacts with the atom

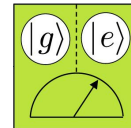


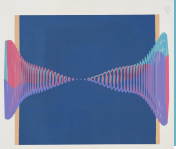
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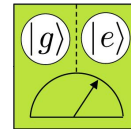


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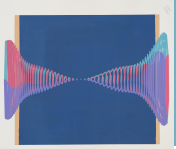
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————→ $F(x)$

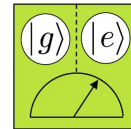


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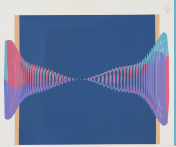
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————→ $F(\boldsymbol{x})$ ———→ wavefunction

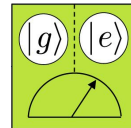


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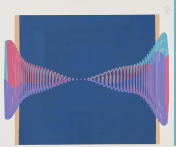
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————→ $\chi(t)$

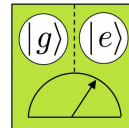


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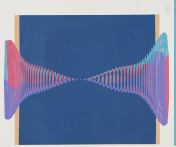
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————→ $F(\mathbf{x})$ ———→ wavefunction

————→ $\chi(t)$ ———→ $g = \frac{2\pi\hbar^2 a_{ab}}{m}$

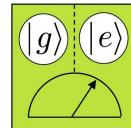


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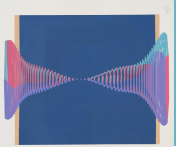
————→ “the detector clicks” ———→



————→ $F(\mathbf{x})$ ———→ wavefunction

————→ $\chi(t)$ ———→ $g = \frac{2\pi\hbar^2 a_{ab}}{m}$

————→ We make the coupling time dependent

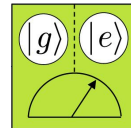


Entanglement Harvesting (BEC)

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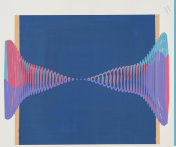


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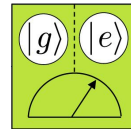


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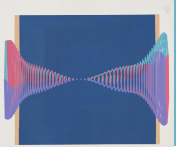
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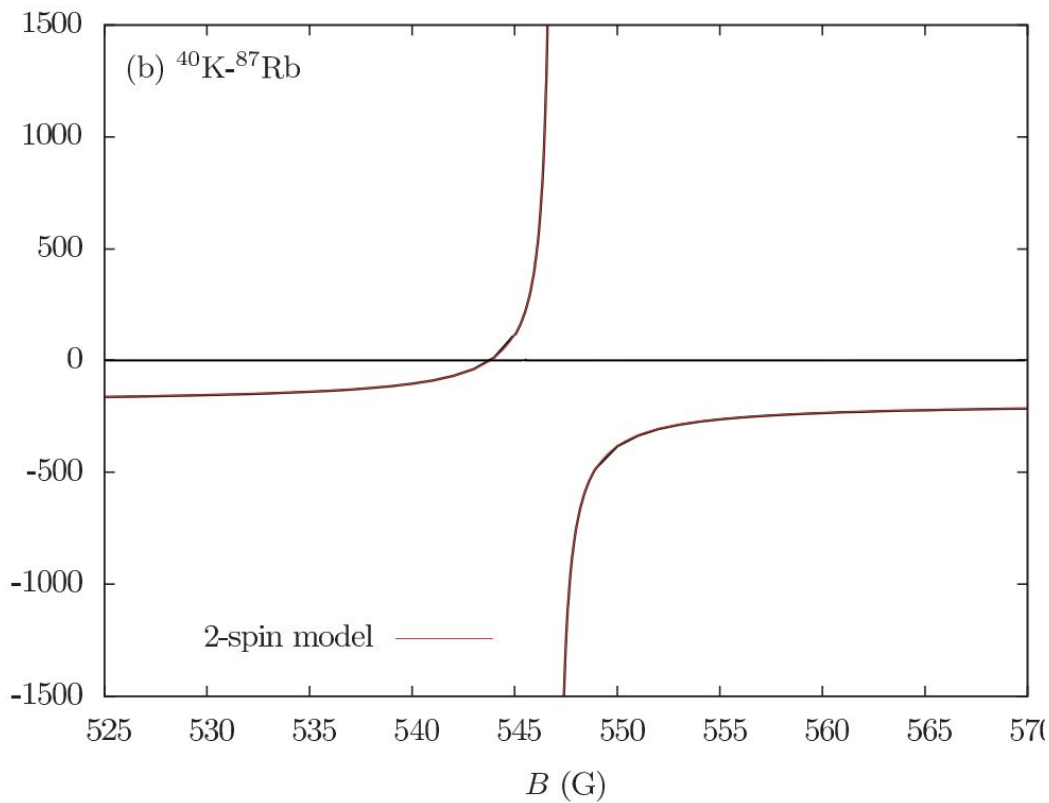
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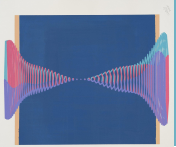
—————→ We make the coupling time dependent

—————→ external magnetic field

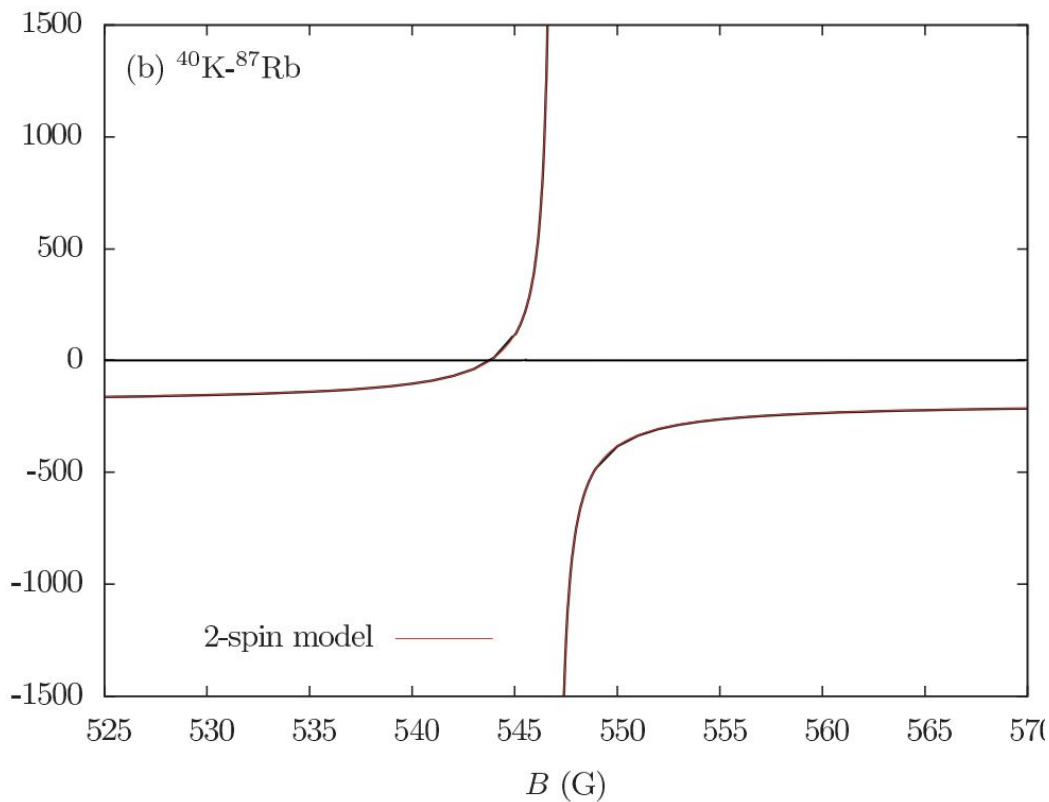
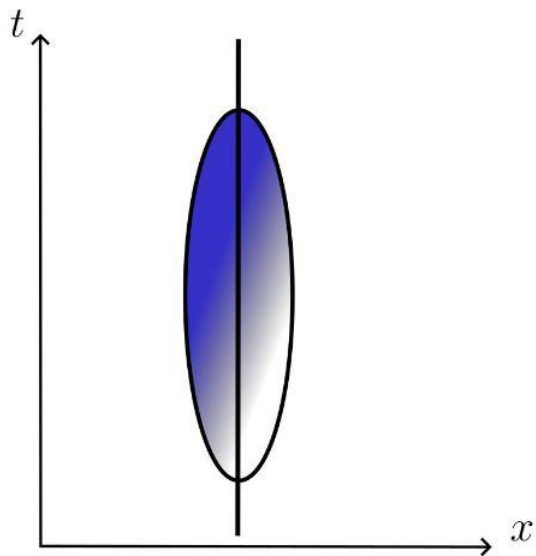


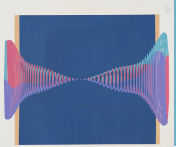
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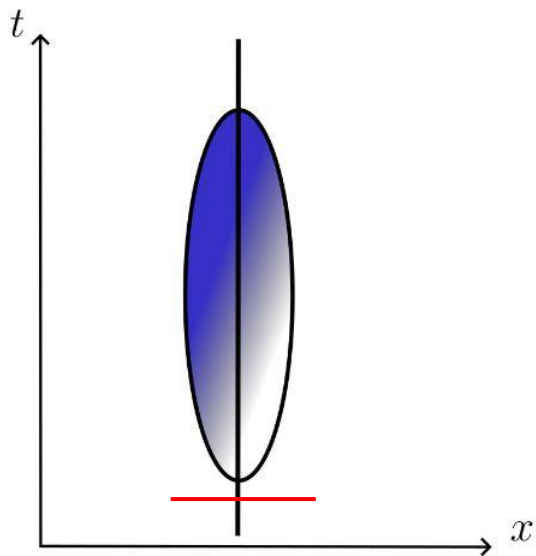


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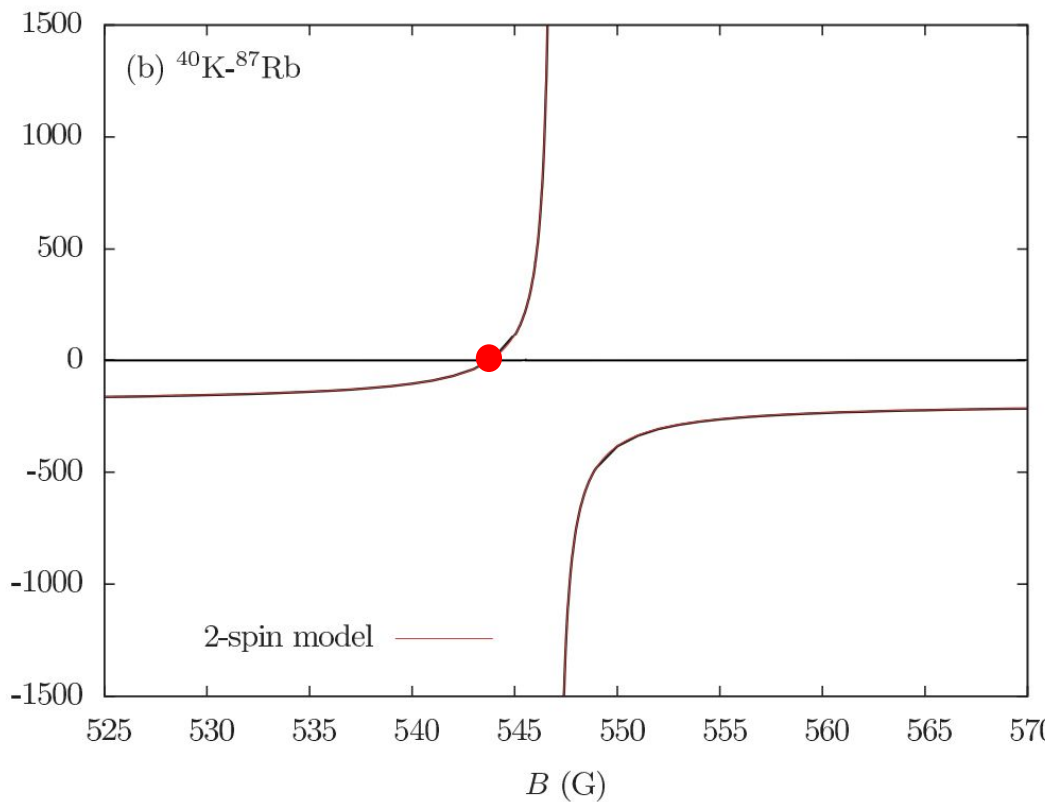


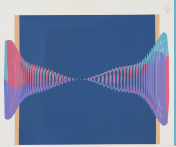


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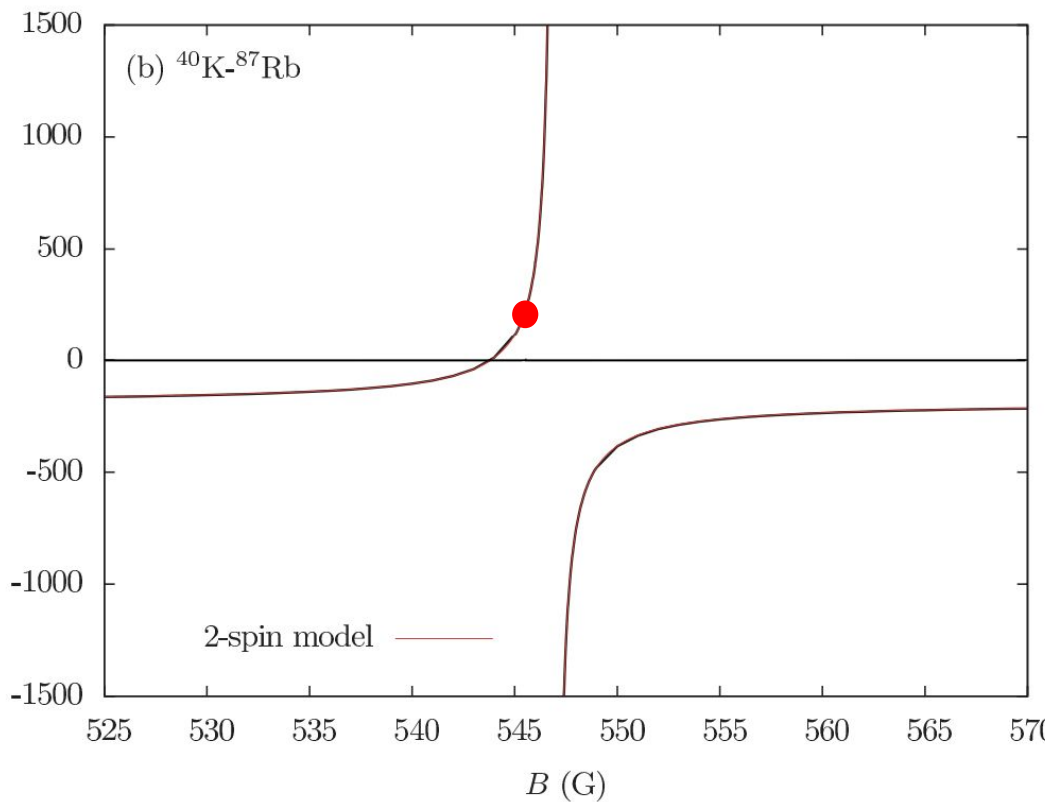
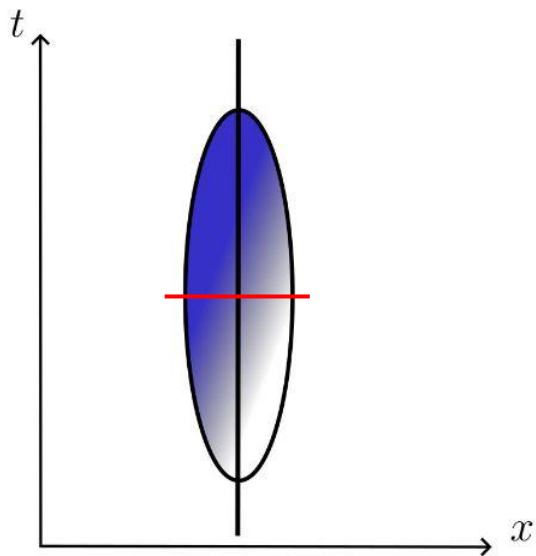


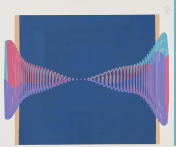
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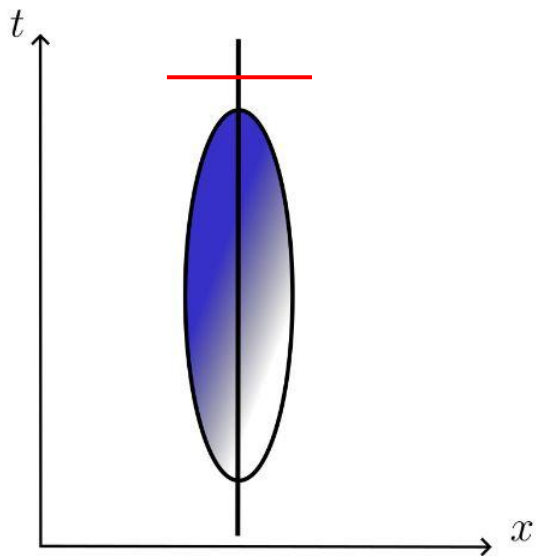


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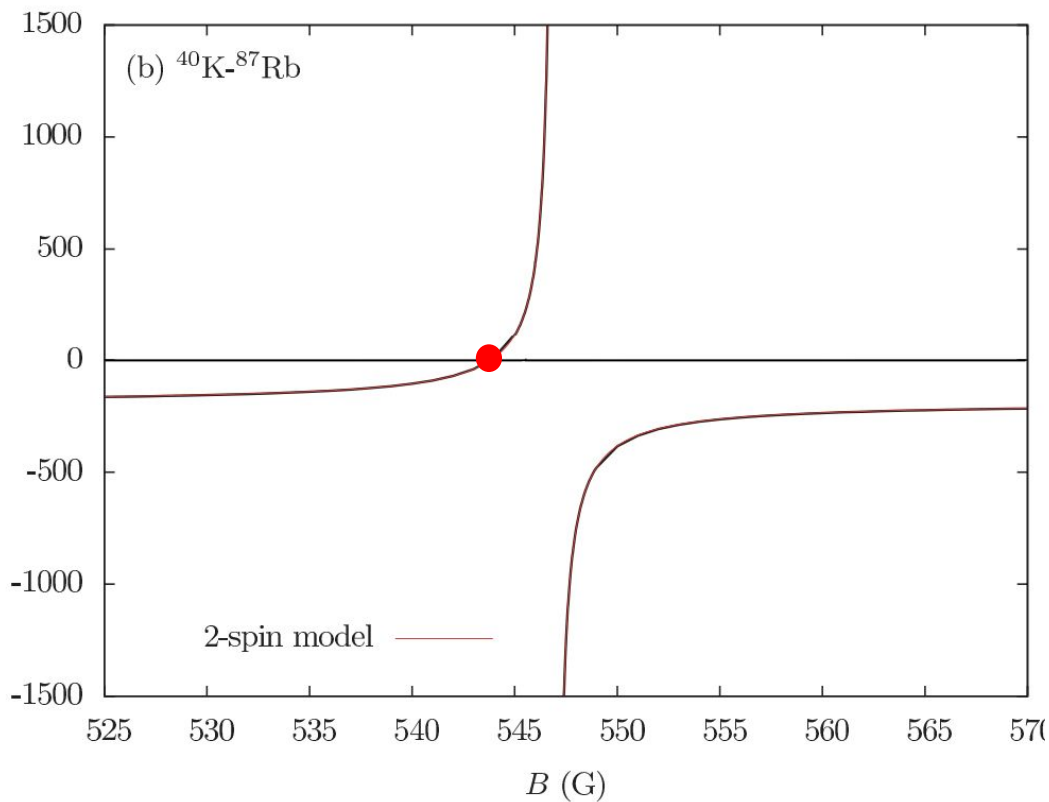


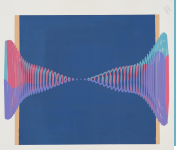


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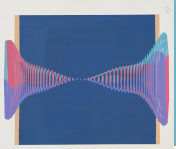


544



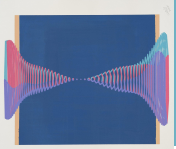


EH in BECs: Summary



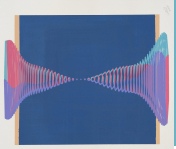
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————→ We can approximate a real scalar quantum field via a BEC



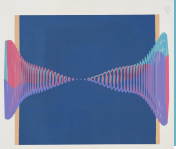
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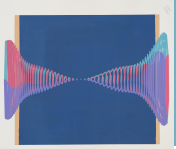
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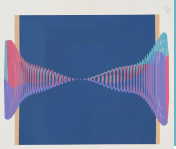
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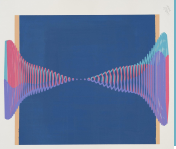
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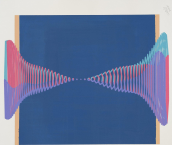
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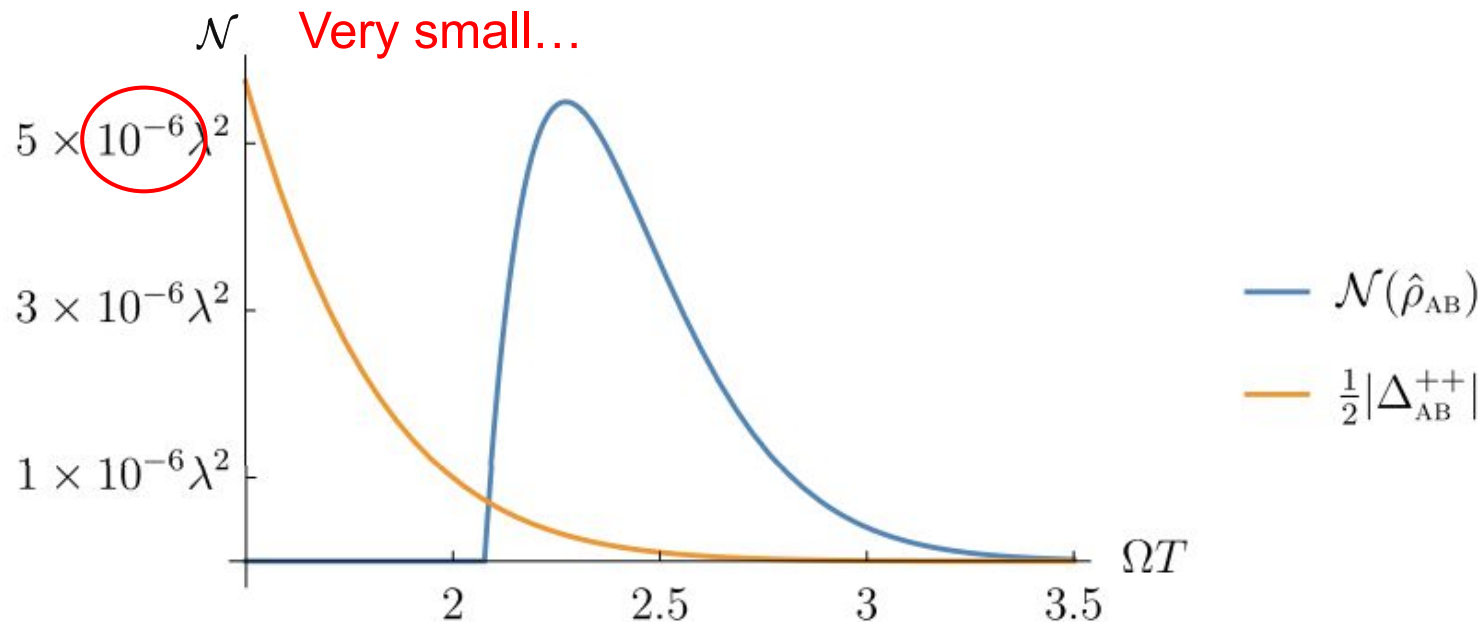


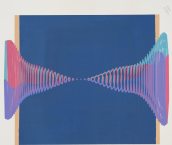
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- It is experimentally challenging to observe negativity of the order of $\mathcal{O}(10^{-6})$

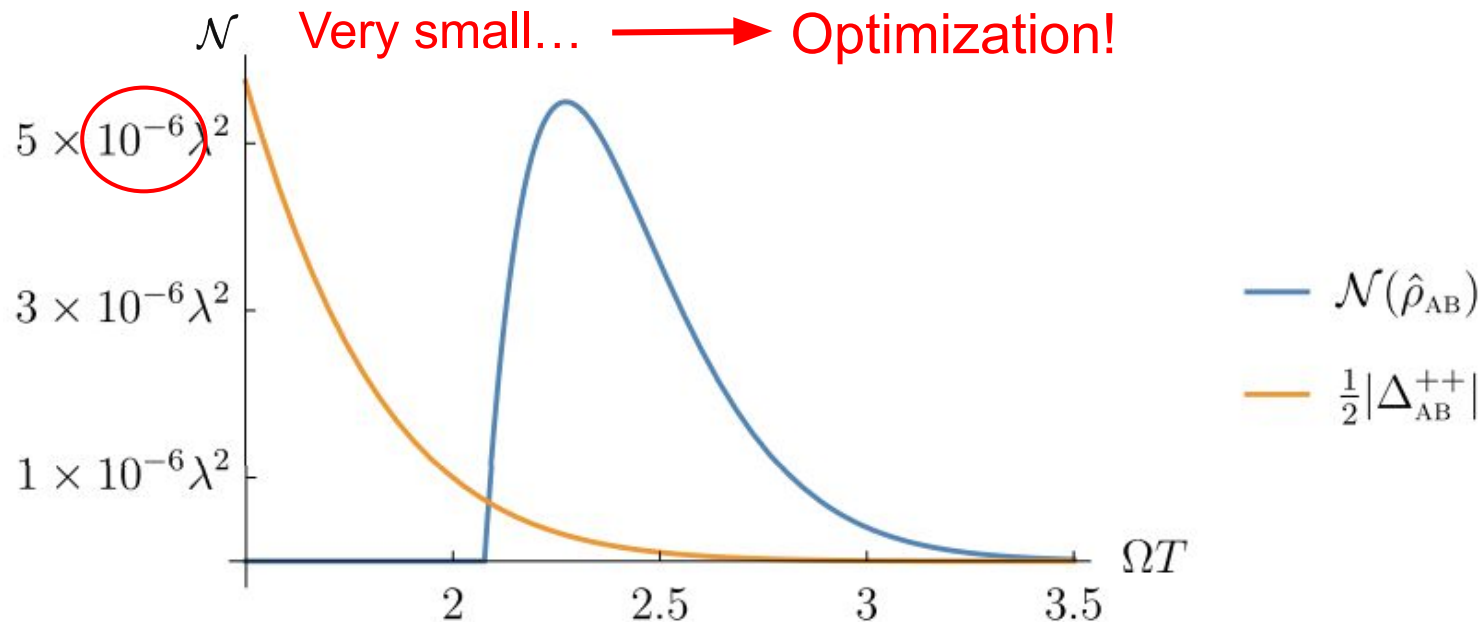


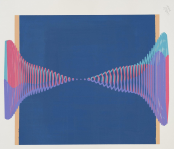
Negativity in practice



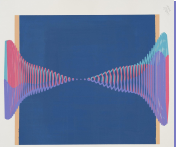


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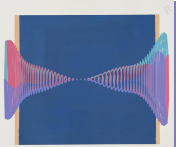


Optimization



Optimization

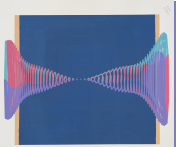
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Optimization

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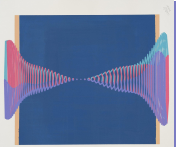


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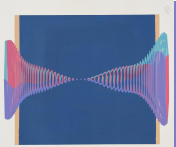
$$\longrightarrow \frac{\Delta}{\mathcal{N}} \leq \varepsilon$$



Optimization

$$\Lambda(\mathbf{x}) \equiv \chi(t)F(\mathbf{x})$$

→ We need to be able to compute the integrals



Optimization

$$\Lambda(\mathbf{x}) \equiv \chi(t)F(\mathbf{x})$$

- We need to be able to compute the integrals
- We need to do so easily so that we can do it many many times



Arbitrary switching: compute negativity

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Arbitrary switching: compute negativity

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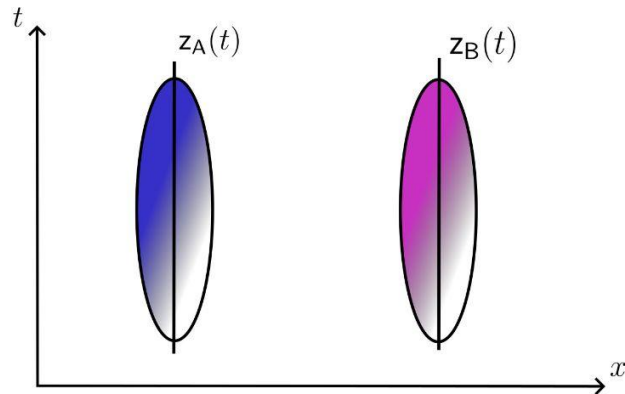
→ Consider $\chi(t)$



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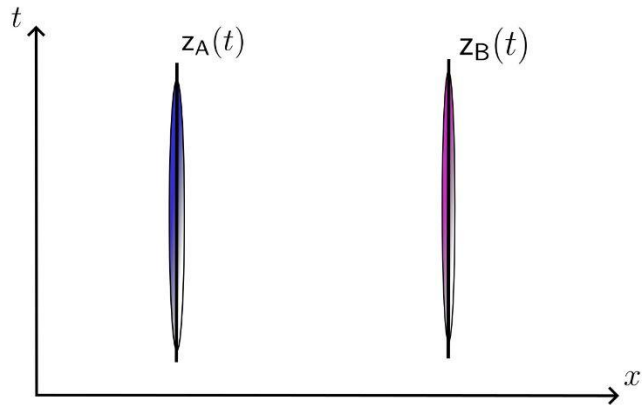




Arbitrary switching: compute negativity

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Arbitrary switching: compute negativity

$$\Lambda(\mathbf{x}) \equiv \chi(t)F(\mathbf{x})$$

→ Consider $\chi(t)$

$$\rightarrow F(\mathbf{x}) = e^{\frac{-|\mathbf{x}|^2}{2\sigma^2}}$$



Arbitrary switching: compute negativity

$$\Lambda(\mathbf{x}) \equiv \chi(t)F(\mathbf{x})$$

→ Consider $\chi(t)$

→ $F(\mathbf{x}) = e^{\frac{-|\mathbf{x}|^2}{2\sigma^2}}$

→ Identical detectors



Arbitrary switching: compute negativity

→ $\chi(t)$



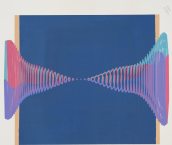
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$$\longrightarrow \chi(t) \approx \sum_{n=0}^N c_n \chi_n(t) \quad \{\chi_n\} \subseteq L^2(\mathbb{R})$$



Arbitrary switching: compute negativity

$$\begin{aligned} \longrightarrow \chi(t) &\approx \sum_{n=0}^N c_n \chi_n(t) & \{\chi_n\} &\subseteq L^2(\mathbb{R}) \\ & & \longrightarrow &\{H_n(t) \mathbf{e}^{-t^2/2T^2}\} \end{aligned}$$

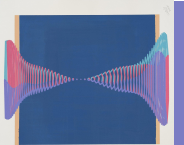


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$$\frac{1}{(2\pi)^3} \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}} \int \frac{d^3\mathbf{k}}{2|\mathbf{k}|} \int dt \int d^3\mathbf{x} \int dt' \int d^3\mathbf{x}' F(\mathbf{x}) F(\mathbf{x}') e^{-i\Omega t} e^{i\Omega t'} e^{i|\mathbf{k}|(t-t') + \mathbf{k} \cdot (\mathbf{x} - \mathbf{x}')} \chi(t) \chi(t') W(\mathbf{x}, \mathbf{x}')$$



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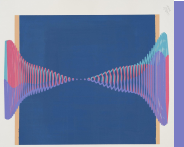
$$\sum_{n=0}^N \sum_{m=0}^N c_n c_m \int dV dV' H_n(t) H_m(t') e^{-t^2/2T^2} F(\mathbf{x}) e^{-t'^2/2T^2} F(\mathbf{x}') W(\mathbf{x}, \mathbf{x}')$$



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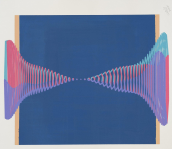
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$$\int dV dV' H_n(t) H_m(t') e^{-t^2/2T^2} e^{-\frac{|\mathbf{x}|^2}{2\sigma^2}} e^{-t'^2/2T^2} e^{-\frac{|\mathbf{x}'|^2}{2\sigma^2}} W(\mathbf{x}, \mathbf{x}')$$



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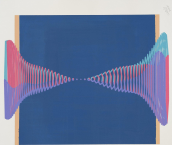


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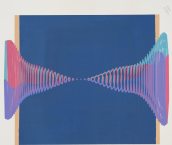


Arbitrary switching: compute negativity

$$\Lambda(\mathbf{x}) = \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}} e^{-\frac{t^2}{2T^2}} e^{-\frac{|\mathbf{x}|^2}{2\sigma^2}}$$

$$G_{AB}^{++} = \frac{\lambda^2 T^2 e^{-\Omega^2 T^2} e^{i\Omega t_0}}{8\sqrt{\pi}|\mathbf{L}|\sqrt{T^2 + \sigma^2}} \left(e^{-\frac{(|\mathbf{L}|+t_0)^2}{4(T^2 + \sigma^2)}} \operatorname{erfi}\left(\frac{|\mathbf{L}| + t_0}{2\sqrt{T^2 + \sigma^2}}\right) + e^{-\frac{(|\mathbf{L}|-t_0)^2}{4(T^2 + \sigma^2)}} \operatorname{erfi}\left(\frac{|\mathbf{L}| - t_0}{2\sqrt{T^2 + \sigma^2}}\right) \right. \\ \left. - i \left(e^{-\frac{(|\mathbf{L}|+t_0)^2}{4(T^2 + \sigma^2)}} \operatorname{erf}\left(\frac{|\mathbf{L}|T^2 - t_0\sigma^2}{2T\sigma\sqrt{T^2 + \sigma^2}}\right) + e^{-\frac{(|\mathbf{L}|-t_0)^2}{4(T^2 + \sigma^2)}} \operatorname{erf}\left(\frac{|\mathbf{L}|T^2 + t_0\sigma^2}{2T\sigma\sqrt{T^2 + \sigma^2}}\right) \right) \right).$$

$$\mathcal{N}(\hat{\rho}_{AB}) = \max \left(0, \frac{\lambda^2 e^{-\Omega^2 T^2}}{4\pi\alpha^2} \left(\sqrt{\pi\alpha} e^{-\frac{|\mathbf{L}|^2}{4\alpha^2 T^2}} \frac{T}{|\mathbf{L}|} \sqrt{\operatorname{erf}\left(\frac{|\mathbf{L}|}{2\alpha\sigma}\right)^2 + \operatorname{erfi}\left(\frac{|\mathbf{L}|}{2\alpha T}\right)^2} + \frac{\sqrt{\pi}\Omega T}{\alpha} e^{\frac{\Omega^2 T^2}{\alpha^2}} \operatorname{erfc}\left(\frac{\Omega T}{\alpha}\right) - 1 \right) \right),$$



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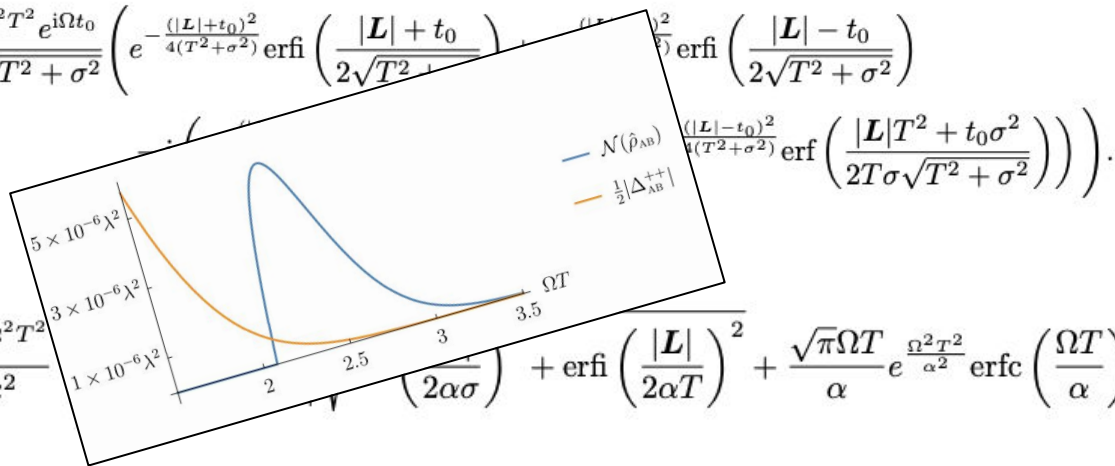
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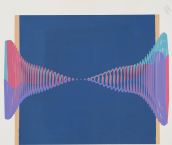
Arbitrary switching: compute negativity

$$\Lambda(\mathbf{x}) = \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}} e^{-\frac{t^2}{2T^2}} e^{-\frac{|\mathbf{x}|^2}{2\sigma^2}}$$

$$G_{AB}^{++} = \frac{\lambda^2 T^2 e^{-\Omega^2 T^2} e^{i\Omega t_0}}{8\sqrt{\pi}|L|\sqrt{T^2 + \sigma^2}} \left(e^{-\frac{(|L|+t_0)^2}{4(T^2 + \sigma^2)}} \operatorname{erfi}\left(\frac{|L|+t_0}{2\sqrt{T^2 + \sigma^2}}\right) + e^{-\frac{(|L|-t_0)^2}{4(T^2 + \sigma^2)}} \operatorname{erfi}\left(\frac{|L|-t_0}{2\sqrt{T^2 + \sigma^2}}\right) \right)$$



$$\mathcal{N}(\hat{\rho}_{AB}) = \max \left(0, \frac{\lambda^2 e^{-\Omega^2 T^2}}{4\pi\alpha^2} \left(\operatorname{erfi}\left(\frac{|L|}{2\alpha T}\right) + \frac{\sqrt{\pi}\Omega T}{\alpha} e^{-\frac{\Omega^2 T^2}{\alpha^2}} \operatorname{erfc}\left(\frac{\Omega T}{\alpha}\right) - 1 \right) \right),$$



Arbitrary switching: compute negativity

$$\int dV dV' H_n(t) H_m(t') e^{-t^2/2T^2} e^{-\frac{|x|^2}{2\sigma^2}} e^{-t'^2/2T^2} e^{-\frac{|x'|^2}{2\sigma^2}} W(\mathbf{x}, \mathbf{x}')$$



$$\int dV dV' e^{-t^2/2T^2} e^{-\frac{|x|^2}{2\sigma^2}} e^{-t'^2/2T^2} e^{-\frac{|x'|^2}{2\sigma^2}} W(\mathbf{x}, \mathbf{x}')$$



Arbitrary switching: compute negativity

$$\int dV dV' H_n(t) H_m(t') e^{-t^2/2T^2} e^{-\frac{|x|^2}{2\sigma^2}} e^{-t'^2/2T^2} e^{-\frac{|x'|^2}{2\sigma^2}} W(x, x')$$



$$H_n\left(\frac{d}{d\alpha}\right) H_m\left(\frac{d}{d\beta}\right) \int dV dV' e^{\alpha t} e^{\beta t'} e^{-t^2/2T^2} e^{-\frac{|x|^2}{2\sigma^2}} e^{-t'^2/2T^2} e^{-\frac{|x'|^2}{2\sigma^2}} W(x, x')$$



Arbitrary switching: compute negativity

$$\int dV dV' H_n(t) H_m(t') e^{-t^2/2T^2} e^{-\frac{|x|^2}{2\sigma^2}} e^{-t'^2/2T^2} e^{-\frac{|x'|^2}{2\sigma^2}} W(x, x')$$



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Arbitrary switching: compute negativity

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$$= W(e^{\alpha t} \Lambda_A^-, e^{\beta t} \Lambda_A^+)$$



Arbitrary switching: compute negativity

$$\int dV dV' H_n(t) H_m(t') e^{-t^2/2T^2} e^{-\frac{|x|^2}{2\sigma^2}} e^{-t'^2/2T^2} e^{-\frac{|x'|^2}{2\sigma^2}} W(x, x')$$



$$H_n\left(\frac{d}{d\alpha}\right) H_m\left(\frac{d}{d\beta}\right) \int dV dV' e^{\alpha t} e^{\beta t'} e^{-t^2/2T^2} e^{-\frac{|x|^2}{2\sigma^2}} e^{-t'^2/2T^2} e^{-\frac{|x'|^2}{2\sigma^2}} W(x, x')$$

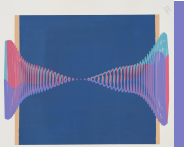
$$= W(e^{\alpha t} \Lambda_A^-, e^{\beta t} \Lambda_A^+) \longrightarrow \text{Closed forms!}$$



Arbitrary switching: compute negativity

$$\longrightarrow \chi(t) \approx \sum_{n=0}^N c_n \chi_n(t) \quad \{\chi_n\} \subseteq L^2(\mathbb{R})$$
$$\longrightarrow \{H_n(t) e^{-t^2/2T^2}\}$$

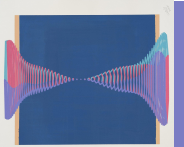
$$\sum_{n=0}^N \sum_{m=0}^N c_n c_m \underbrace{\int dV dV' H_n(t) H_m(t') e^{-t^2/2T^2} F(\mathbf{x}) e^{-t'^2/2T^2} F(\mathbf{x}') W(\mathbf{x}, \mathbf{x}')}_{\longrightarrow \text{closed-forms}}$$



Arbitrary switching: compute negativity

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$$\sum_{n=0}^N \sum_{m=0}^N c_n c_m \underbrace{\int dV dV' H_n(t) H_m(t') e^{-t^2/2T^2} F(\mathbf{x}) e^{-t'^2/2T^2} F(\mathbf{x}') W(\mathbf{x}, \mathbf{x}')}_{W_{nm}} \longrightarrow \text{closed-forms}$$



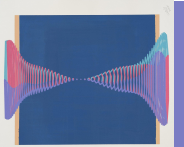
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$$\sum_{n=0}^N \sum_{m=0}^N c_n c_m \underbrace{\int dV dV' H_n(t) H_m(t') e^{-t^2/2T^2} F(\mathbf{x}) e^{-t'^2/2T^2} F(\mathbf{x}') W(\mathbf{x}, \mathbf{x}')}_{W_{nm}}$$

$$\equiv \sum_{n=0}^N \sum_{m=0}^N c_n c_m W_{nm}$$

$W_{nm} \longrightarrow$ closed-forms



Arbitrary switching: compute negativity

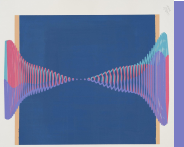
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$$\sum_{n=0}^N \sum_{m=0}^N c_n c_m \underbrace{\int dV dV' H_n(t) H_m(t') e^{-t^2/2T^2} F(\mathbf{x}) e^{-t'^2/2T^2} F(\mathbf{x}') W(\mathbf{x}, \mathbf{x}')}_{W_{nm}}$$

$$\equiv \sum_{n=0}^N \sum_{m=0}^N c_n c_m W_{nm} \quad W_{nm} \longrightarrow \text{closed-forms}$$

$$\equiv \mathbf{c}^\top \mathbf{W} \mathbf{c}$$



Arbitrary switching: compute negativity

$$\longrightarrow \chi(t) \approx \sum_{n=0}^N c_n \chi_n(t) \quad \{\chi_n\} \subseteq L^2(\mathbb{R})$$
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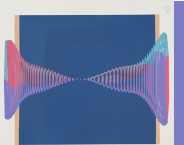
$$\sum_{n=0}^N \sum_{m=0}^N c_n c_m \underbrace{\int dV dV' H_n(t) H_m(t') e^{-t^2/2T^2} F(\mathbf{x}) e^{-t'^2/2T^2} F(\mathbf{x}') W(\mathbf{x}, \mathbf{x}')}_{W_{nm}}$$

$$\equiv \sum_{n=0}^N \sum_{m=0}^N c_n c_m W_{nm}$$

$W_{nm} \longrightarrow$ closed-forms

$\searrow$ Same for other propagators

$$\equiv c^\top W c$$



Arbitrary switching: compute negativity

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$$\equiv \sum_{n=0}^N \sum_{m=0}^N c_n c_m W_{nm}$$

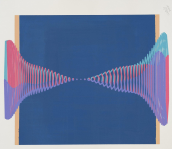
$$\equiv c^\top W c$$

$W_{nm} \longrightarrow$ closed-forms

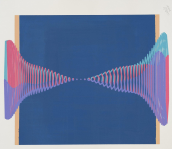
$\searrow$ Same for other propagators

G_{nm}

Δ_{nm}

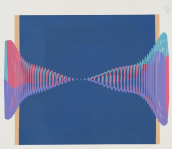


Arbitrary switching: compute negativity



Arbitrary switching: compute negativity

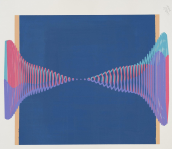
Guide (old)



Arbitrary switching: compute negativity

Guide (old)

————→ For **each** smearing function we want to test: we compute integrals of the form

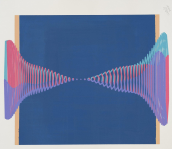


Arbitrary switching: compute negativity

Guide (old)

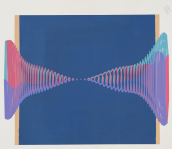
————→ For **each** smearing function we want to test: we compute integrals of the form

$$W(\Lambda_A^-, \Lambda_A^+) = \frac{1}{(2\pi)^3} \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}} \int \frac{d^3\mathbf{k}}{2|\mathbf{k}|} \int dt \int d^3\mathbf{x} \int dt' \int d^3\mathbf{x}' e^{\frac{-|\mathbf{x}|^2}{2\sigma^2}} e^{\frac{-|\mathbf{x}'|^2}{2\sigma^2}} e^{-i\Omega t} e^{i\Omega t'} e^{i|\mathbf{k}|(t-t') + \mathbf{k} \cdot (\mathbf{x} - \mathbf{x}')} \chi(t) \chi(t')$$



Arbitrary switching: compute negativity

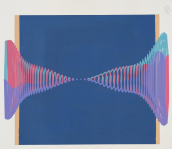
Guide (new)



Arbitrary switching: compute negativity

Guide (new)

————→ We take the closed-form expression of W_{nm}

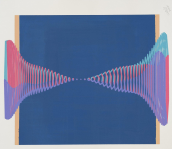


Arbitrary switching: compute negativity

Guide (new)

————→ We take the closed-form expression of W_{nm}

————→ We expand in the Hermite basis the smearing function we want to test



Arbitrary switching: compute negativity

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————→ We get the coefficients c_n



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————→ We take the closed-form expression of W_{nm}

————→ We expand in the Hermite basis the smearing function we want to test

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————→ We calculate $c^\top W c$



Arbitrary switching: compute negativity

Guide (new)

————→ We take the closed-form expression of W_{nm}

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hard integral



Arbitrary switching: compute negativity

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————→ We take the closed-form expression of W_{nm}

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hard integral (not analytical)



Arbitrary switching: compute negativity

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————→ We take the closed-form expression of W_{nm}

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hard integral (not analytical) ————→ Matrix product



Arbitrary switching: compute negativity

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————→ We take the closed-form expression of W_{nm}

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hard integral (not analytical) → Matrix product



Arbitrary switching: compute negativity

Guide (new)

————→ We take the closed-form expression of W_{nm}

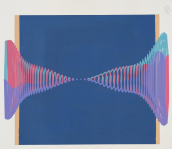
————→ We expand in the Hermite basis the smearing function we want to test

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————→ We calculate $c^T W c$

hard integral (not analytical) —————→ Matrix product





Arbitrary switching: maximize negativity



Arbitrary switching: maximize negativity

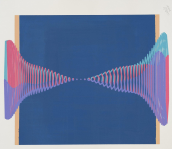
→ find c



Arbitrary switching: maximize negativity

→ find c

→ maximize $\mathcal{N}(c) = \lambda^2 \left(|c^\top G c| - c^\top W c \right)$



Arbitrary switching: maximize negativity



Arbitrary switching: maximize negativity

$$\longrightarrow \mathcal{N}(c) = \lambda^2 \left(|c^\top G c| - c^\top W c \right)$$



Arbitrary switching: maximize negativity

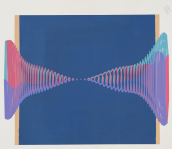
$$\longrightarrow \mathcal{N}(c) = \lambda^2 \left(|c^\top G c| - c^\top W c \right)$$



Arbitrary switching: maximize negativity

$$\longrightarrow \mathcal{N}(c) = \lambda^2 \left(|c^\top G c| - c^\top W c \right)$$

$$\longrightarrow ||\chi(t)||_2 = \int_{-\infty}^{\infty} |\chi(t)|^2 dt$$



Arbitrary switching: maximize negativity

$$\longrightarrow \mathcal{N}(c) = \lambda^2 \left(|c^\top G c| - c^\top W c \right)$$

$$\longrightarrow ||\chi(t)||_2 = \int_{-\infty}^{\infty} |\chi(t)|^2 dt \propto ||c||^2$$



Arbitrary switching: maximize negativity

$$\longrightarrow \mathcal{N}(c) = \lambda^2 \left(|c^\top G c| - c^\top W c \right)$$

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$$= ||e^{-t^2/2}||_2 = \sqrt{\pi}$$



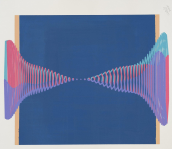
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$$= ||e^{-t^2/2}||_2 = \sqrt{\pi}$$

$$\longrightarrow \frac{\sqrt{\pi}}{||c||^2}$$



Arbitrary switching: maximize negativity

$$\longrightarrow \tilde{\mathcal{N}}(c) = \lambda^2 \sqrt{\pi} \frac{|c^\top G c| - c^\top W c}{||c||^2}$$



Arbitrary switching: maximize negativity

$$\longrightarrow \tilde{\mathcal{N}}(c) = \lambda^2 \sqrt{\pi} \frac{|c^\top G c| - c^\top W c}{||c||^2}$$
$$\longrightarrow \max_{c \in \mathbb{R}^N} \left(\sqrt{\pi} \frac{|c^\top G c| - c^\top W c}{||c||^2} \right)$$



Arbitrary switching: maximize negativity

$$\begin{aligned}\longrightarrow \tilde{\mathcal{N}}(c) &= \lambda^2 \sqrt{\pi} \frac{|c^\top G c| - c^\top W c}{||c||^2} \\ &\longrightarrow \max_{c \in \mathbb{R}^N} \left(\sqrt{\pi} \frac{|c^\top G c| - c^\top W c}{||c||^2} \right) \\ &= \max_{||c||=1} \sqrt{\pi} \left(|c^\top G c| - c^\top W c \right)\end{aligned}$$



Arbitrary switching: maximize negativity

$$\longrightarrow \tilde{\mathcal{N}}(c) = \lambda^2 \sqrt{\pi} \frac{|c^\top G c| - c^\top W c}{||c||^2}$$

$$\longrightarrow \max_{c \in \mathbb{R}^N} \left(\sqrt{\pi} \frac{|c^\top G c| - c^\top W c}{||c||^2} \right)$$

$$= \max_{||c||=1} \sqrt{\pi} \left(|c^\top G c| - c^\top W c \right) \longrightarrow \text{eigenvalue problem}$$



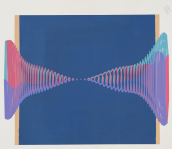
Arbitrary switching: maximize negativity

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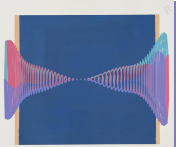
$$= \max_{\theta \in [0, 2\pi)} \lambda_{\max} \left(\cos(\theta) \text{Re}\{G\} + \sin(\theta) \text{Im}\{G\} - \text{Re}\{W\} \right)$$



Arbitrary switching: maximize negativity

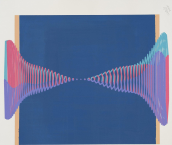
$$= \max_{\theta \in [0, 2\pi)} \lambda_{\max} \left(\cos(\theta) \operatorname{Re}\{G\} + \sin(\theta) \operatorname{Im}\{G\} - \operatorname{Re}\{W\} \right)$$

—————→ will this method give better results?



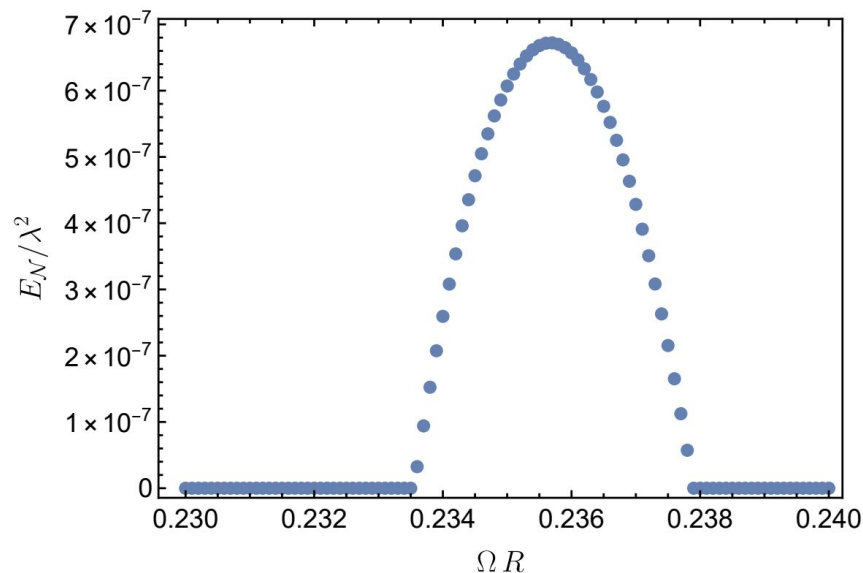
Example: compact support

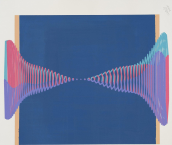
$$\chi(t) = \begin{cases} \left(1 - \frac{4t^2}{T^2}\right)^{5/2}, & t \in [-T/2, T/2] \\ 0 & , \text{ otherwise} \end{cases}$$



Example: compact support

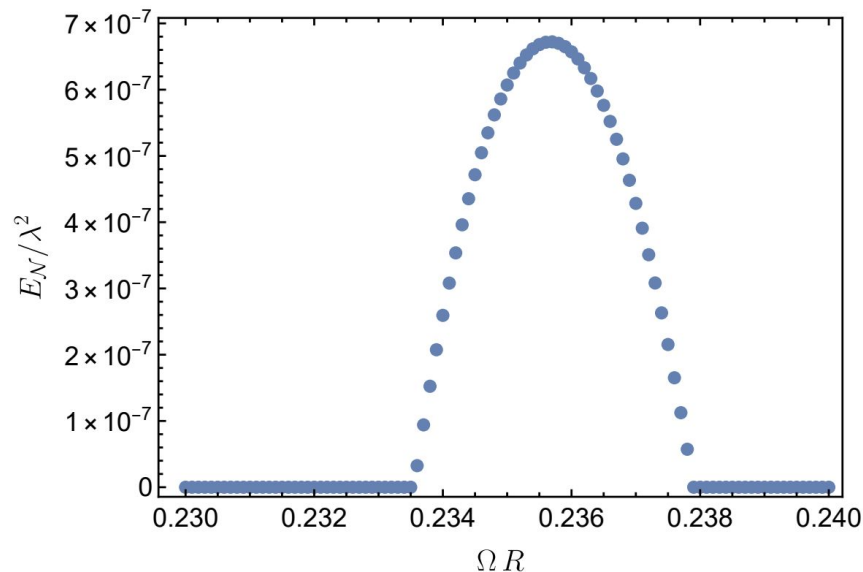
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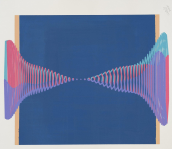




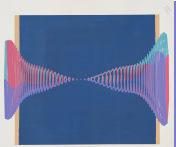
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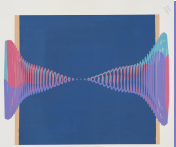


Example: compact support



Example: compact support

$$\longrightarrow \chi(t) \approx \sum_{n=0}^N c_n \chi_n(t)$$



Example: compact support

$$\longrightarrow \chi(t) \approx \sum_{n=0}^N c_n \chi_n(t) \equiv \sum_{n=0}^N c_n H_n(t) \mathbf{e}^{-t^2/2T^2}$$



Example: compact support

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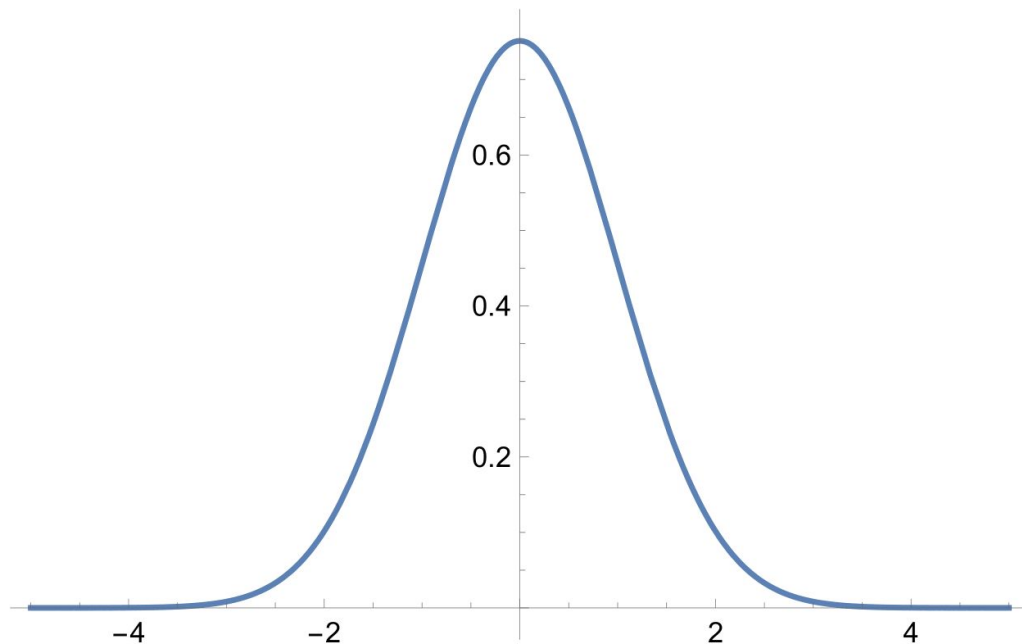
Example: compact support

$$\longrightarrow \chi(t) \approx \sum_{n=0}^N c_n \chi_n(t) \equiv \sum_{n=0}^{\circledast N} c_n \circledast H_n(t) \mathbf{e}^{-t^2/2T^2}$$



Example: compact support

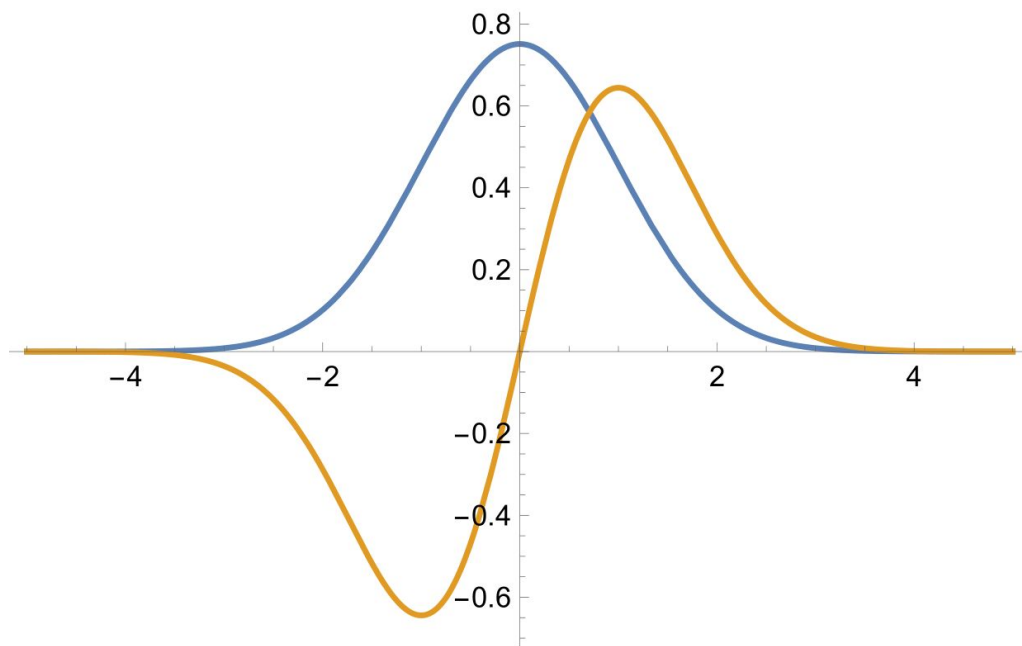
$$\longrightarrow \chi(t) \approx \sum_{n=0}^N c_n \chi_n(t) \equiv \sum_{n=0}^{\overset{N}{\circlearrowleft}} c_n \overset{\circlearrowleft}{H}_n(t) \mathbf{e}^{-t^2/2T^2}$$





Example: compact support

$$\longrightarrow \chi(t) \approx \sum_{n=0}^N c_n \chi_n(t) \equiv \sum_{n=0}^N c_n H_n(t) e^{-t^2/2T^2}$$

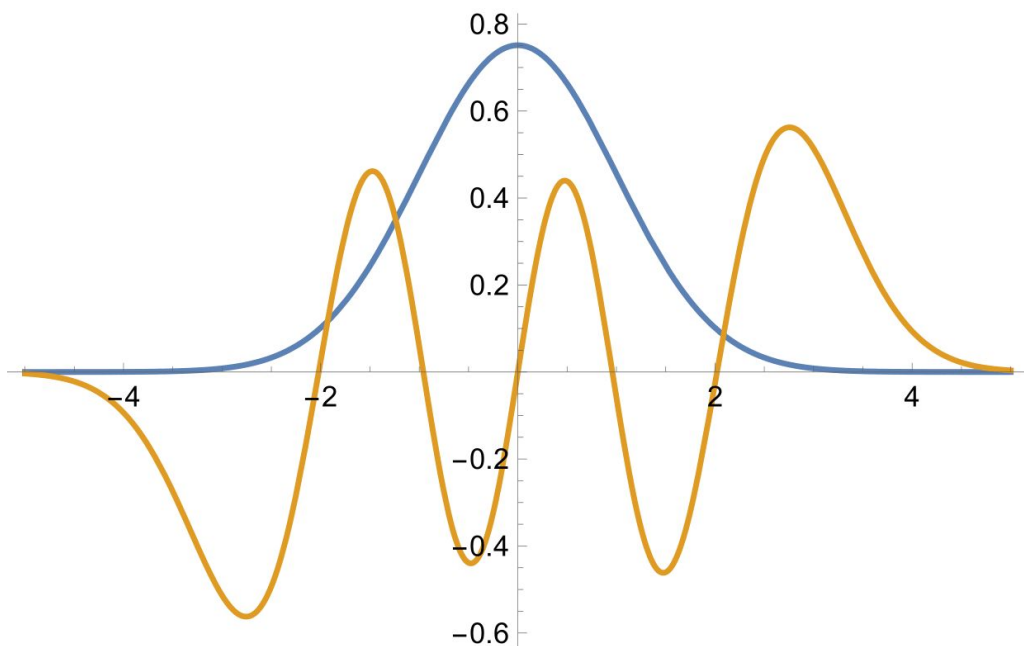


N=1



Example: compact support

$$\longrightarrow \chi(t) \approx \sum_{n=0}^N c_n \chi_n(t) \equiv \sum_{n=0}^N c_n H_n(t) e^{-t^2/2T^2}$$

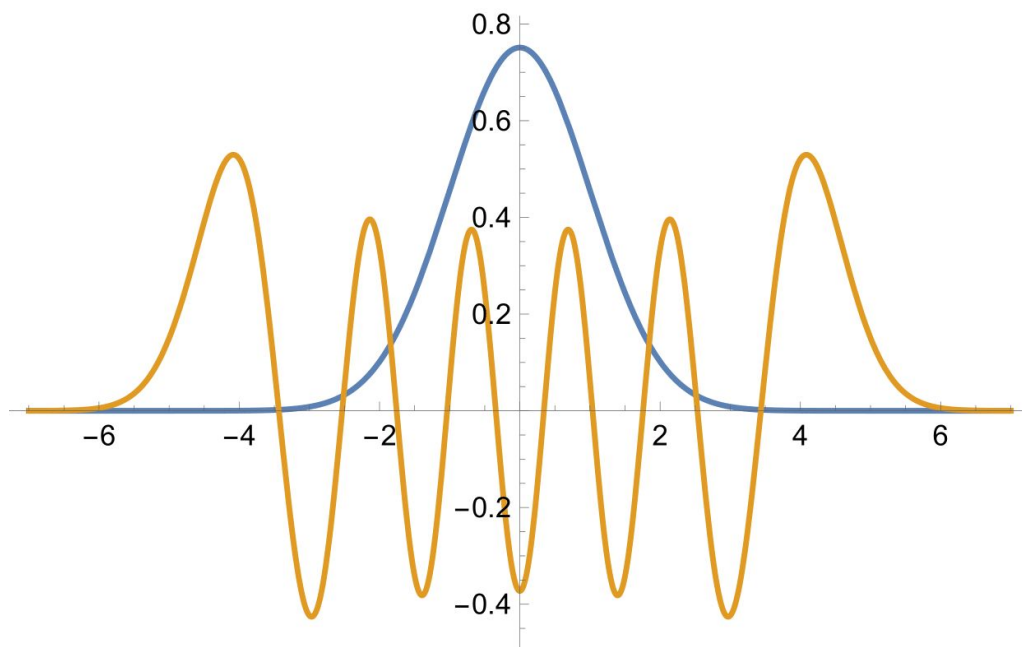


N=5



Example: compact support

$$\longrightarrow \chi(t) \approx \sum_{n=0}^N c_n \chi_n(t) \equiv \sum_{n=0}^{\overset{N}{\circ}} c_n \overset{\circ}{H}_n(t) e^{-t^2/2T^2}$$

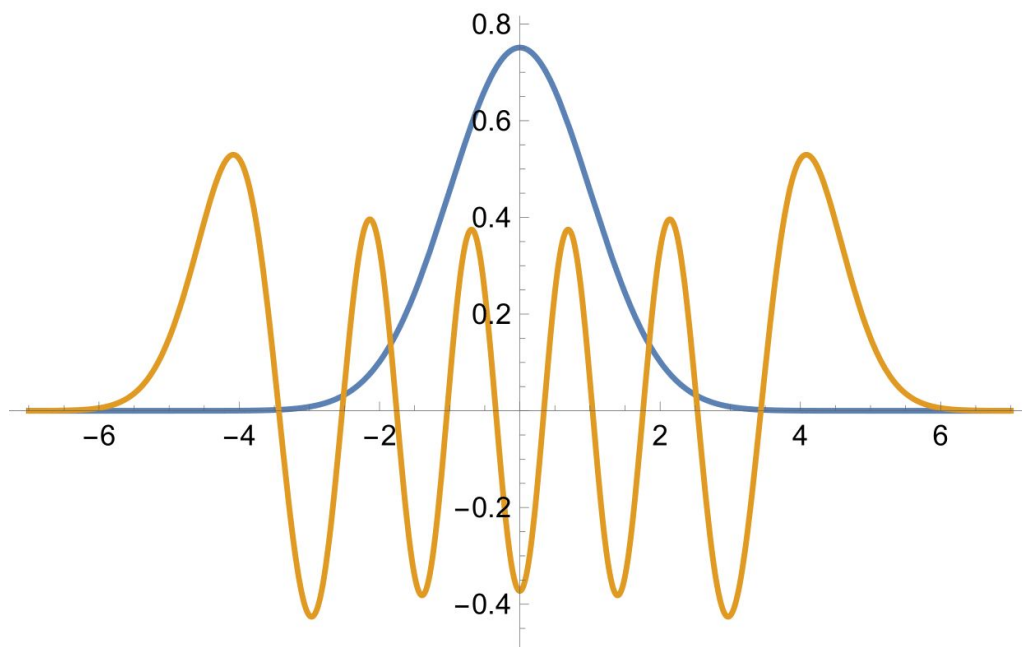


N=10

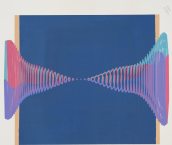


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N=10



Example: compact support

→ Rick's theorem

Any function $f \in L^2([-1, 1])$ can be decomposed as

$$f(t) = \lim_{n \rightarrow \infty} \sum_{k=0}^n f_{k,n} h_{k,n}(t)$$

where

$$h_{k,n}(t) = \frac{1}{\sqrt{2^n n!} \sqrt{\pi} \sigma} H_k \left(\frac{t}{T_n} \right) e^{-\frac{t^2}{2T_n^2}}$$

given that

$$T_n := \frac{g(n)}{\sqrt{2n+1}}, \quad \lim_{n \rightarrow \infty} g(n) = 1$$



Example: compact support

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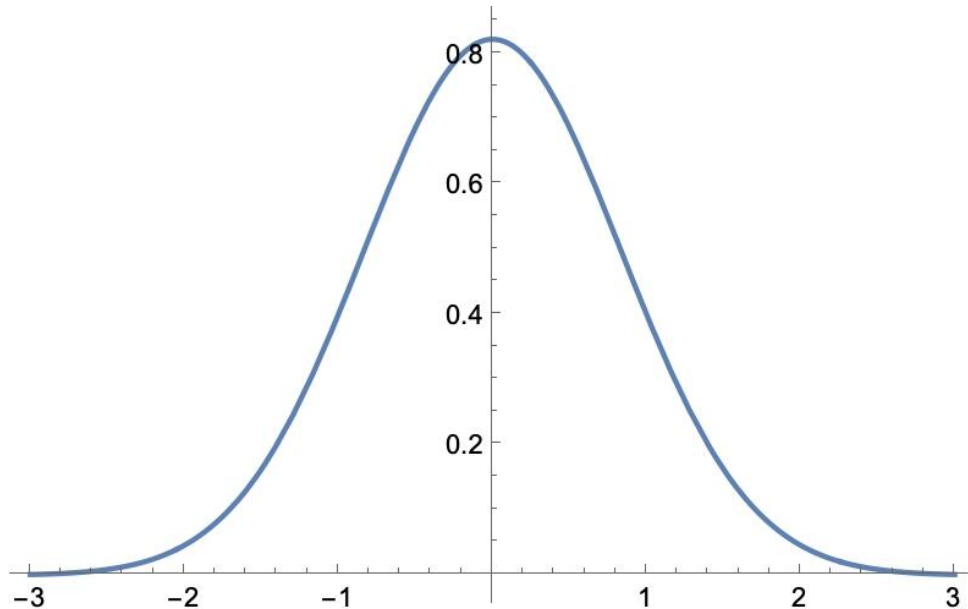
$$T_N = \frac{2.5 T_0}{\sqrt{2N + 1} + 2}$$



Example: compact support

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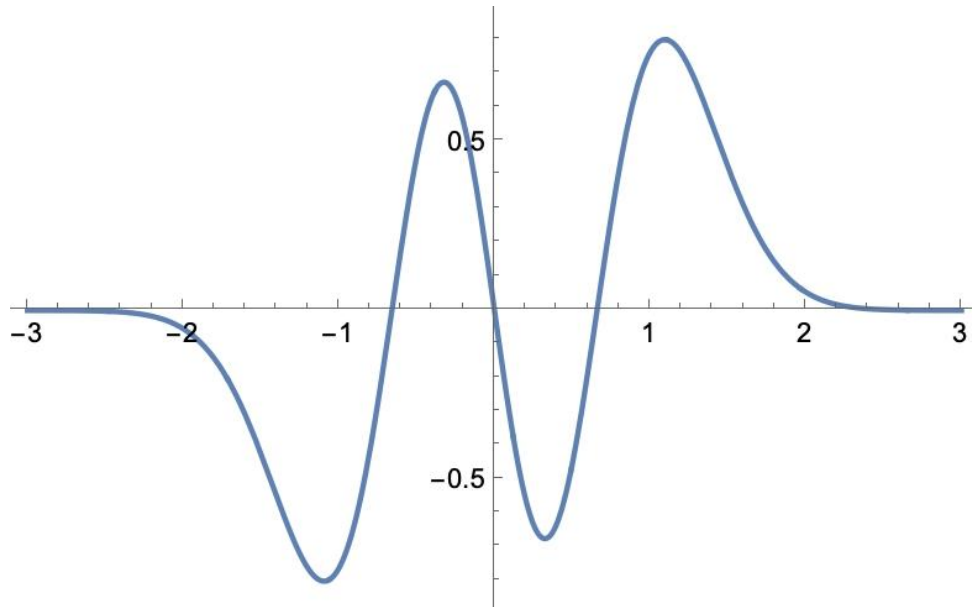
$N=1$





Example: compact support

$$T_N = \frac{2.5 T_0}{\sqrt{2N + 1} + 2}$$

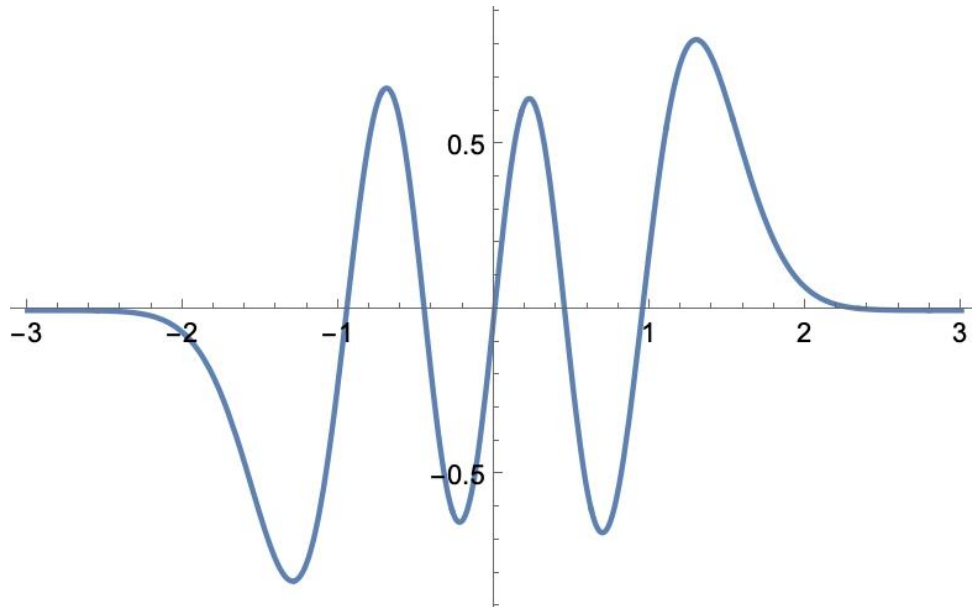


N=3



Example: compact support

$$T_N = \frac{2.5 T_0}{\sqrt{2N + 1} + 2}$$

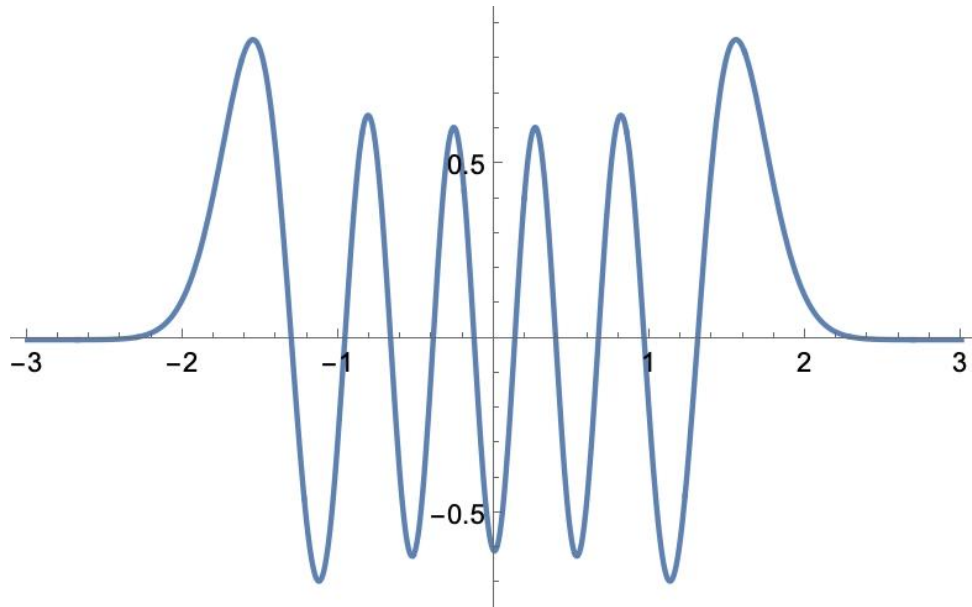


$N=5$



Example: compact support

$$T_N = \frac{2.5 T_0}{\sqrt{2N + 1} + 2}$$

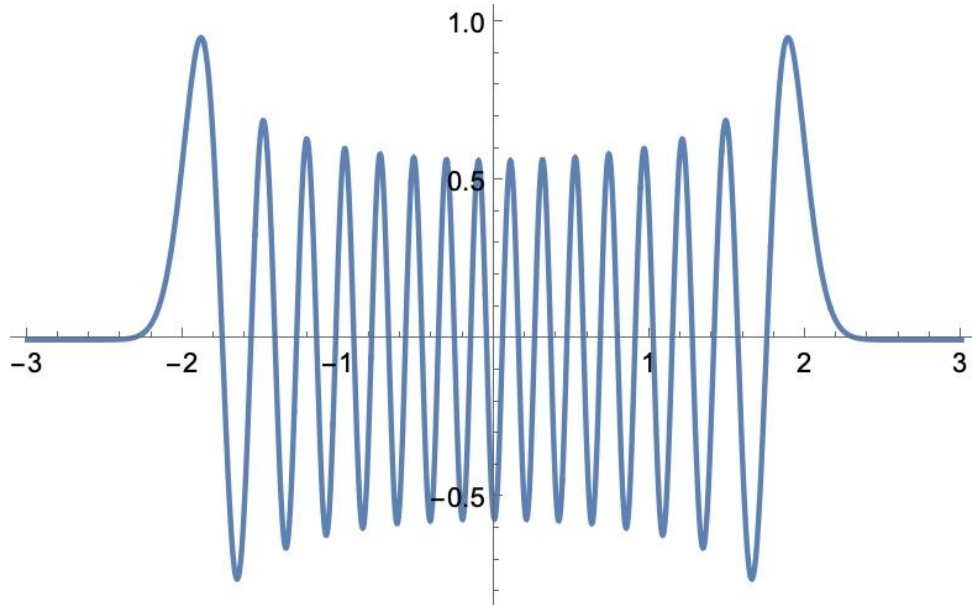


$N=10$

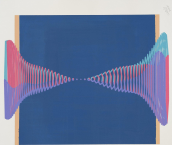


Example: compact support

$$T_N = \frac{2.5 T_0}{\sqrt{2N + 1} + 2}$$



$N=30$



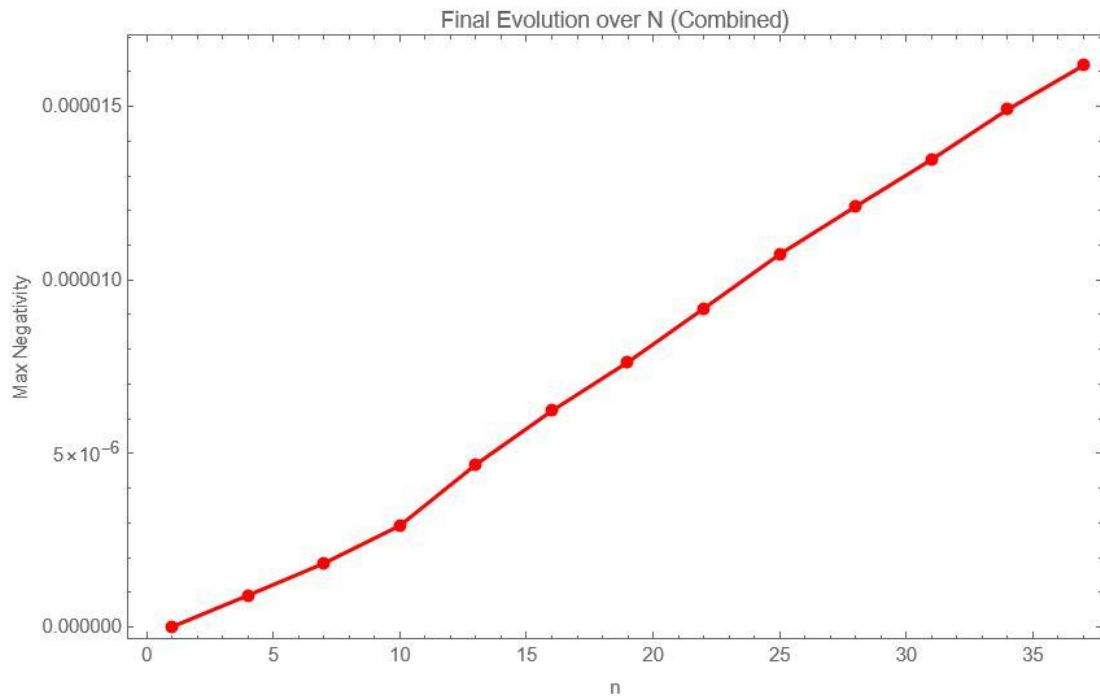
Example: compact support

$$\max_{\theta \in [0, 2\pi)} \lambda_{\max} \left(\cos(\theta) \operatorname{Re}\{G\} + \sin(\theta) \operatorname{Im}\{G\} - \operatorname{Re}\{W\} \right)$$



Example: compact support

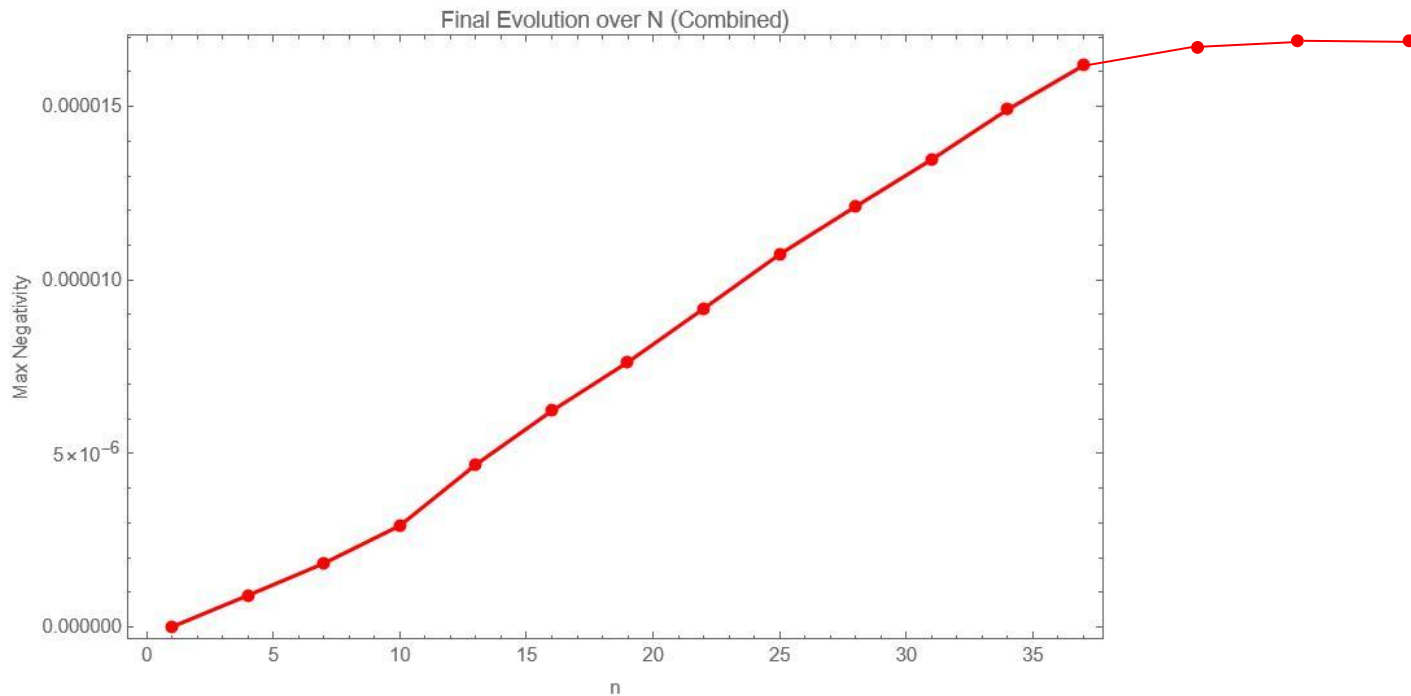
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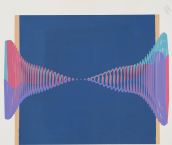




Example: compact support

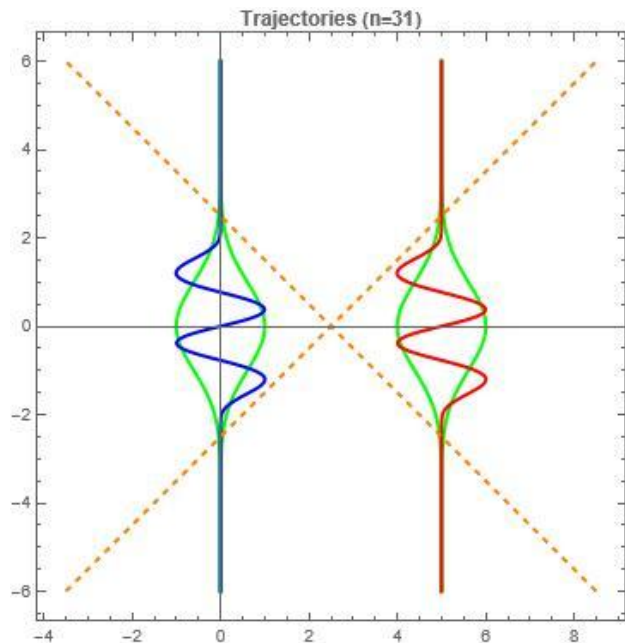
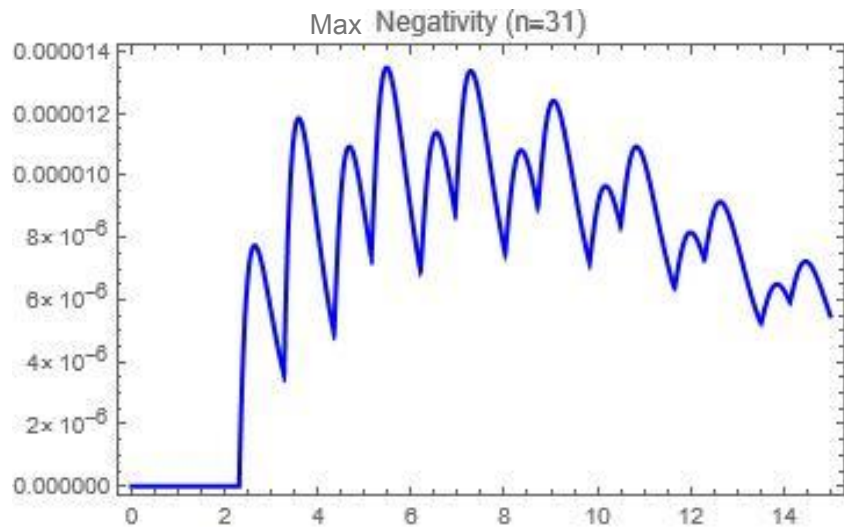
$$\max_{\theta \in [0, 2\pi)} \lambda_{\max} \left(\cos(\theta) \operatorname{Re}\{G\} + \sin(\theta) \operatorname{Im}\{G\} - \operatorname{Re}\{W\} \right)$$





Example: compact support

N=31 $\max_{\theta \in [0, 2\pi)} \lambda_{\max} \left(\cos(\theta) \text{Re}\{G\} + \sin(\theta) \text{Im}\{G\} - \text{Re}\{W\} \right)$

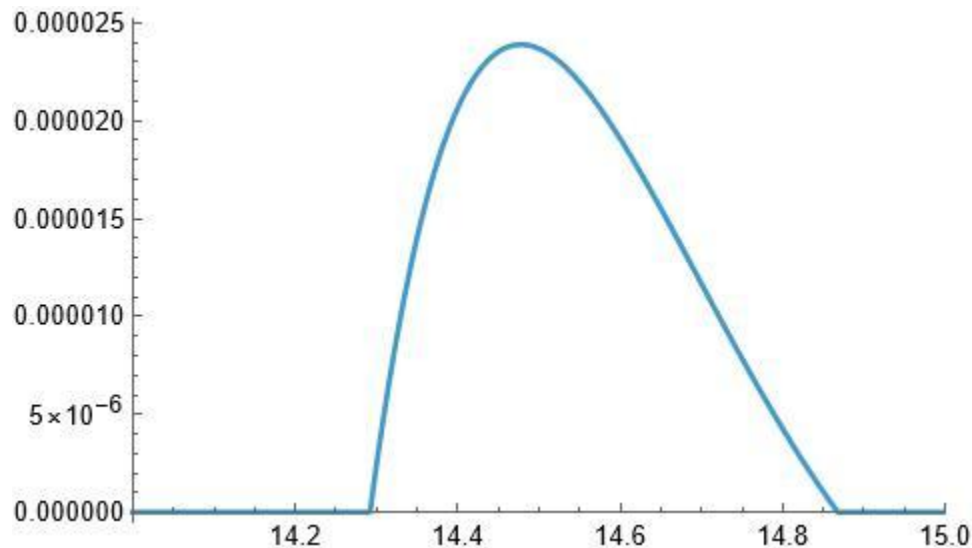




Example: compact support

N=31

$$\max_{\theta \in [0, 2\pi)} \lambda_{\max} \left(\cos(\theta) \operatorname{Re}\{G\} + \sin(\theta) \operatorname{Im}\{G\} - \operatorname{Re}\{W\} \right)$$





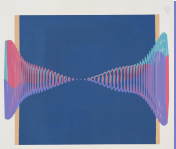
Optimization: summary

————→ We are looking for a compactly supported switching function that maximizes negativity



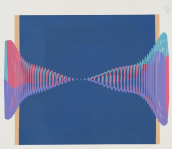
Optimization: summary

- We are looking for a compactly supported switching function that maximizes negativity
 - Computations are very hard for general functions



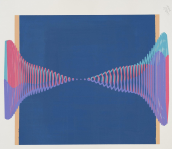
Optimization: summary

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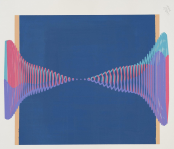
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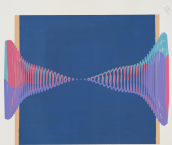


Optimization: summary

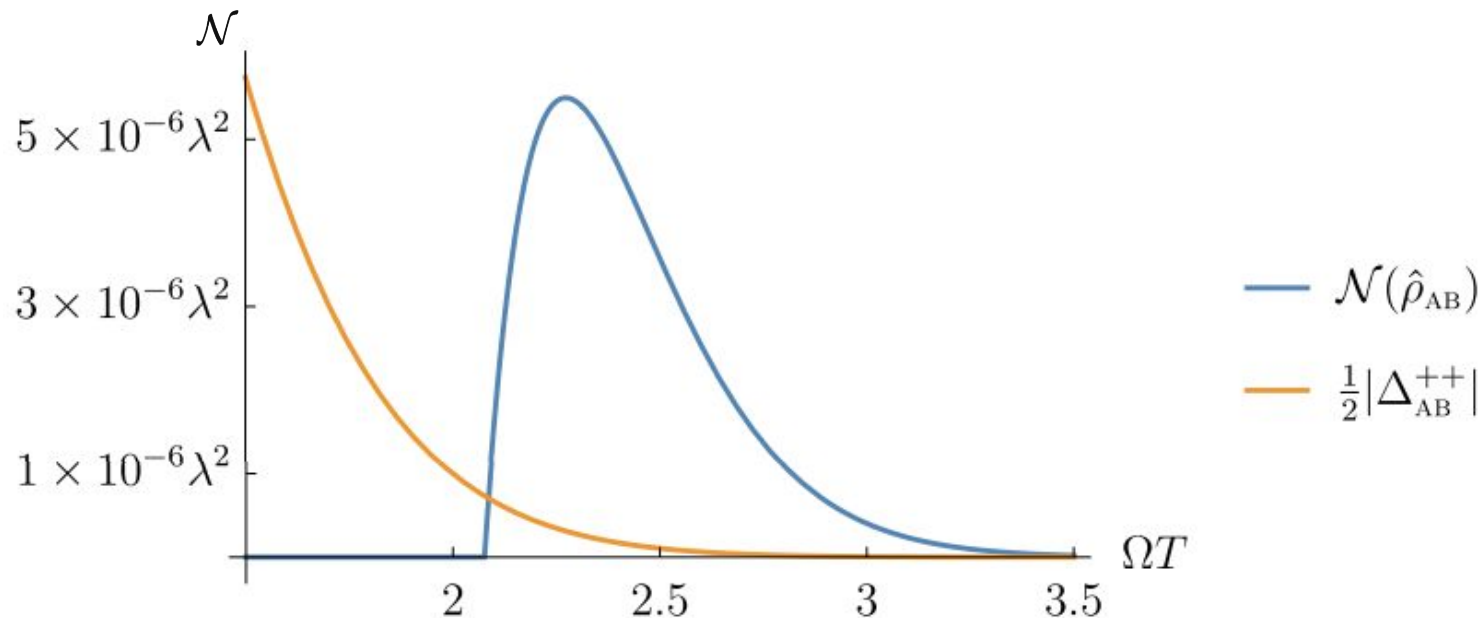
- We are looking for a compactly supported switching function that maximizes negativity
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- We expand switching function in Hermite Basis of L^2 and re-arrange
 - The hard computations become trivial matrix products
- We solve the maximization problem as a simple eigenvalue problem
 - We find an optimal compactly supported function that maximizes the negativity that can be harvested



Further work: non compact support



Further work: non compact support



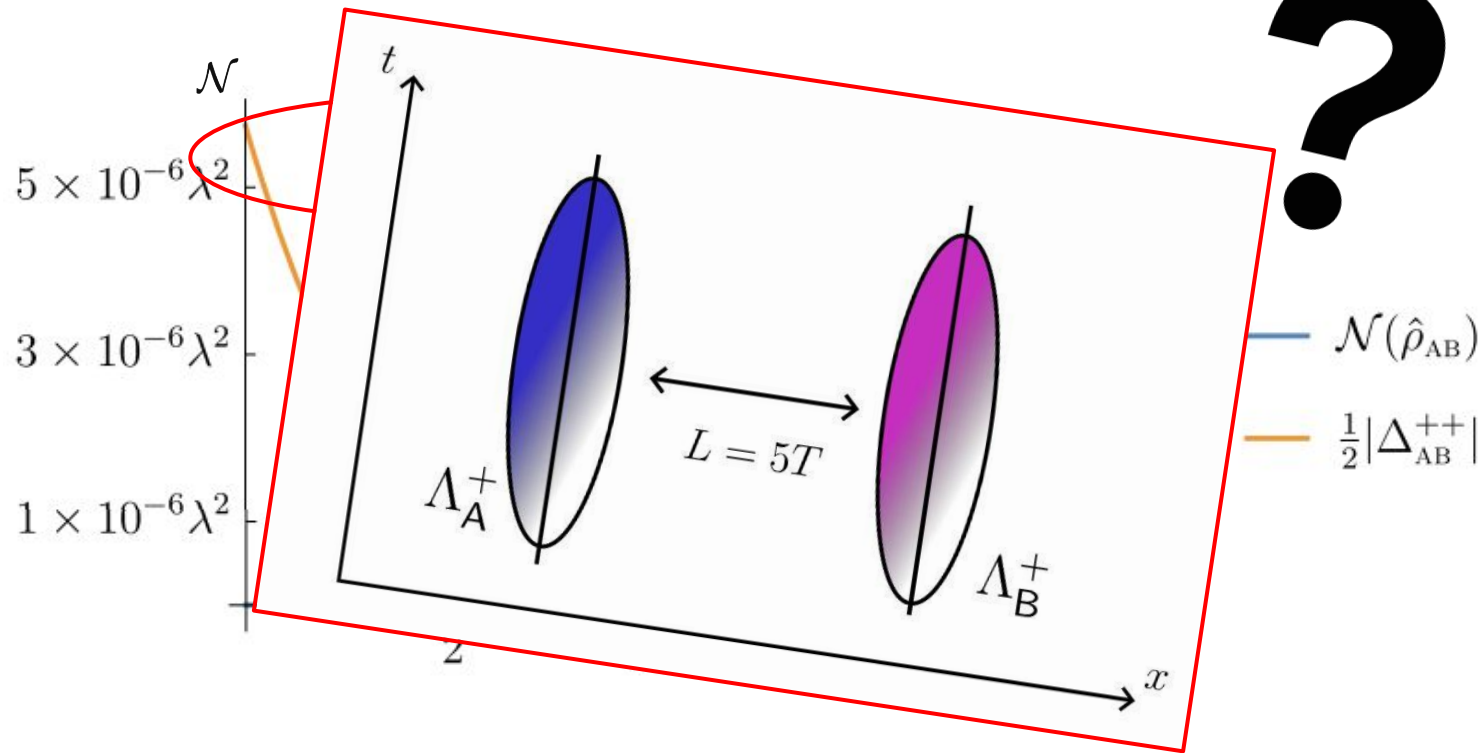


Further work: non compact support





Further work: non compact support





Further work: non compact support

$$\Lambda_A^+ = \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}} e^{\frac{-t^2}{2T^2}} e^{\frac{-|\mathbf{x}|^2}{2\sigma^2}}$$

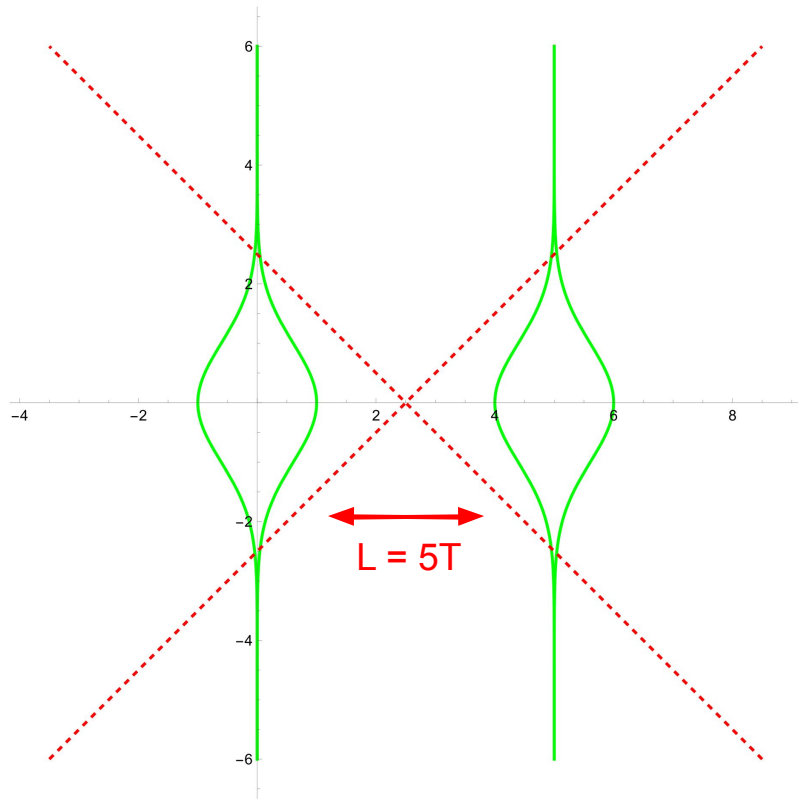
$$\Lambda_B = \frac{1}{(2\pi\sigma^2)^{\frac{3}{2}}} e^{\frac{-t^2}{2T^2}} e^{\frac{-|\mathbf{x}-\mathbf{L}|^2}{2\sigma^2}}$$



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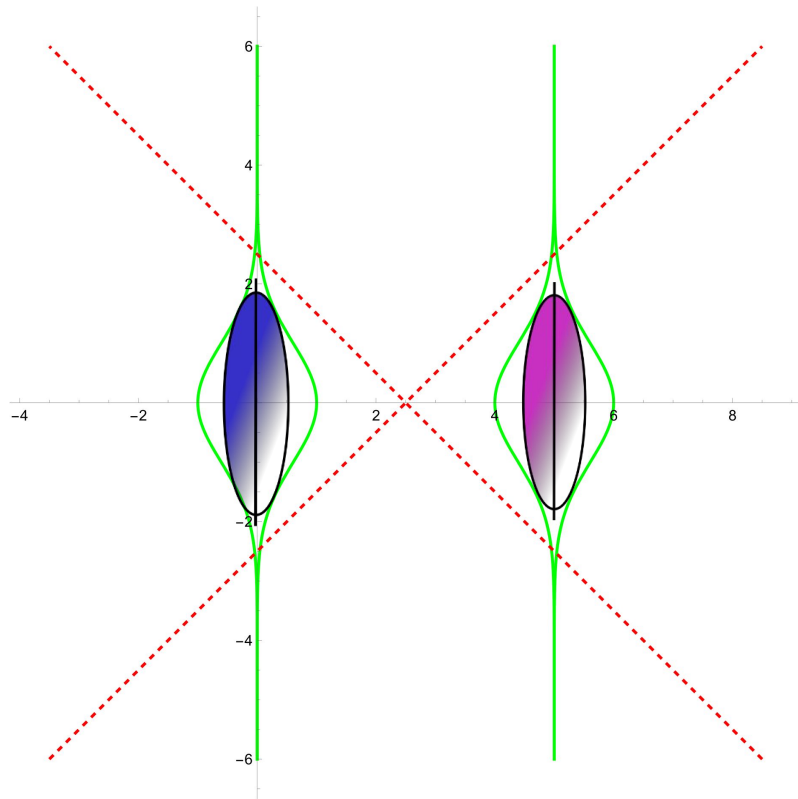




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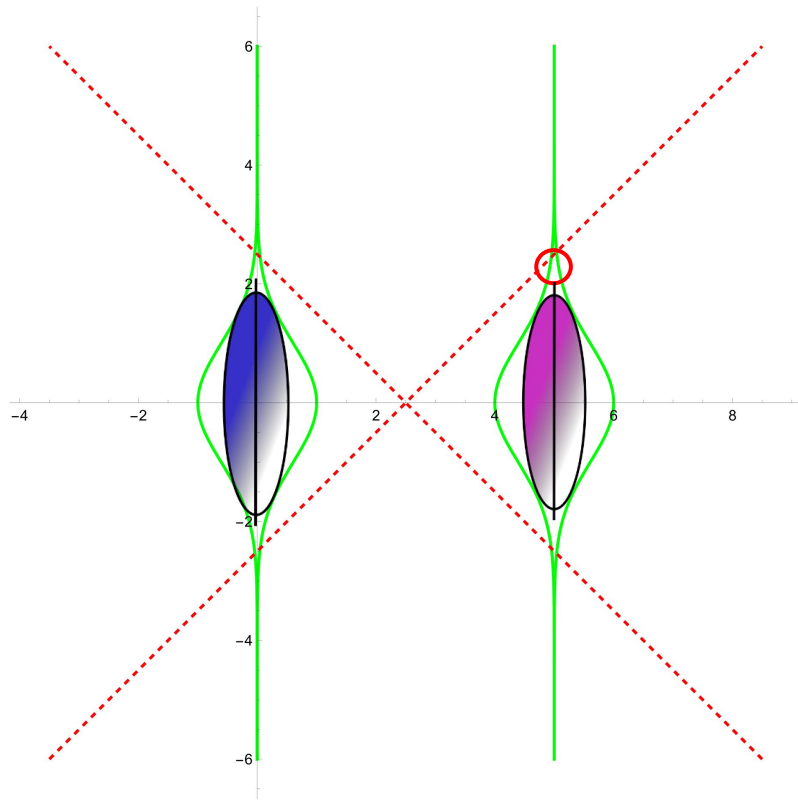




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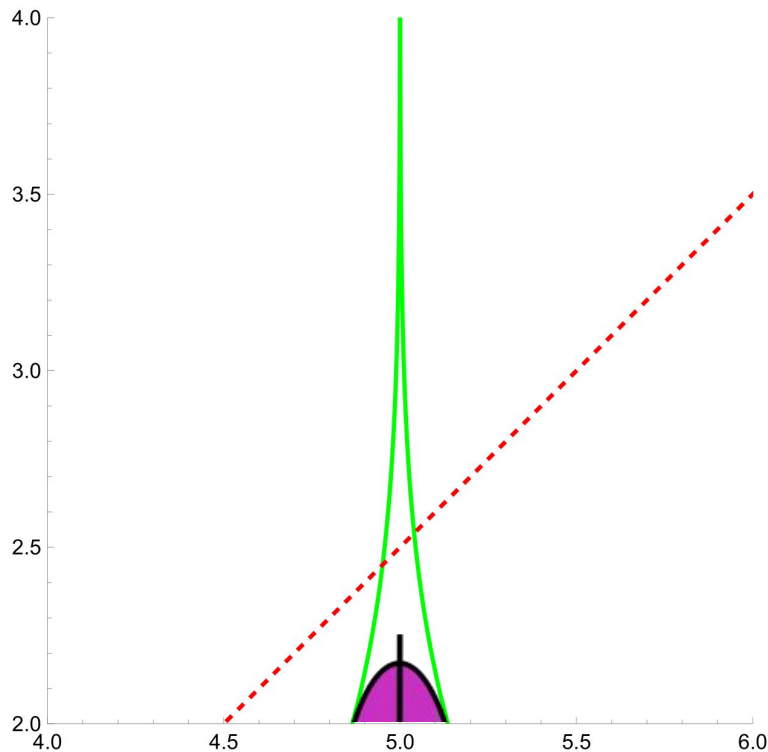




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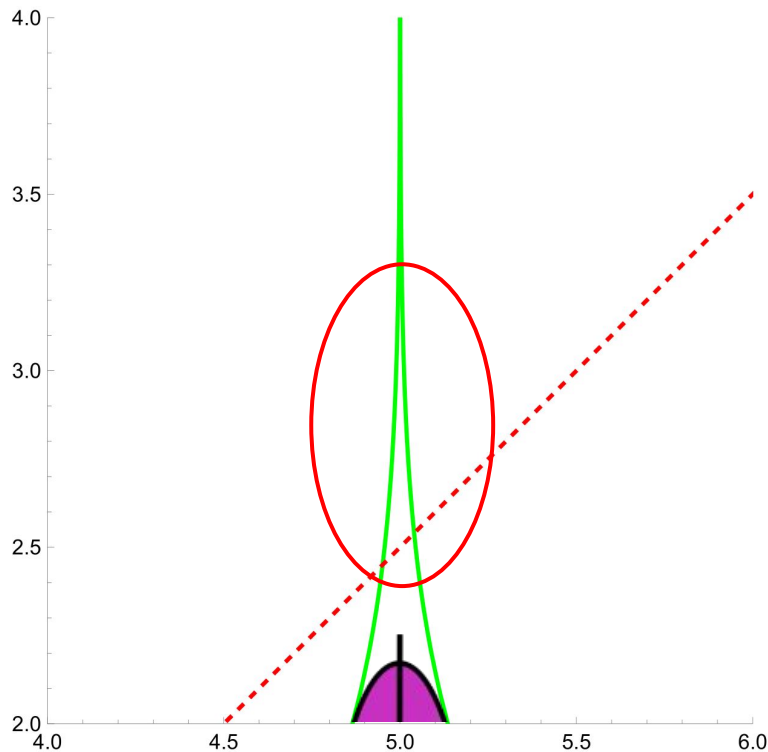




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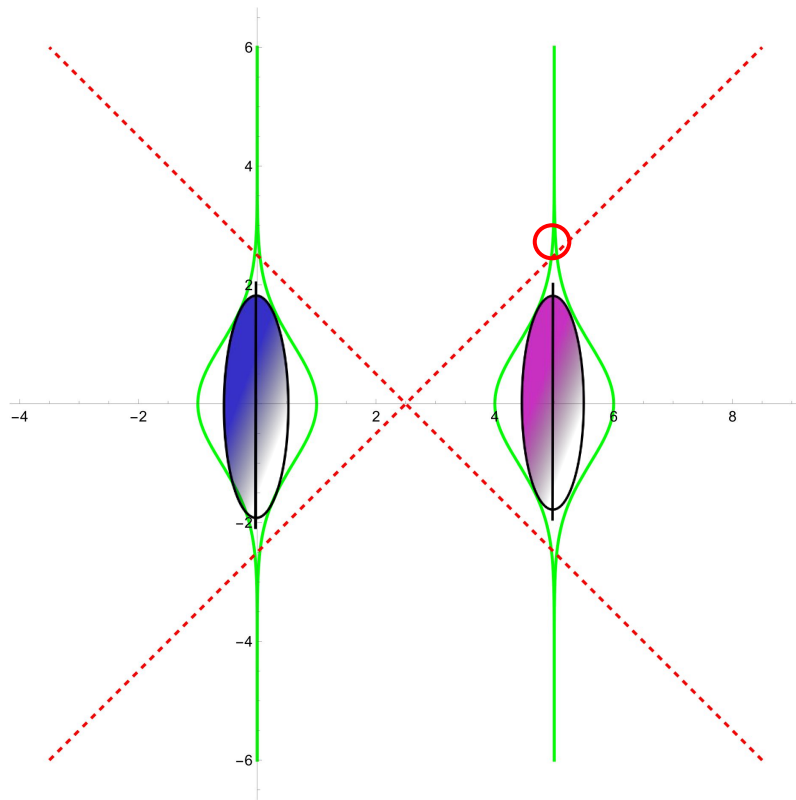




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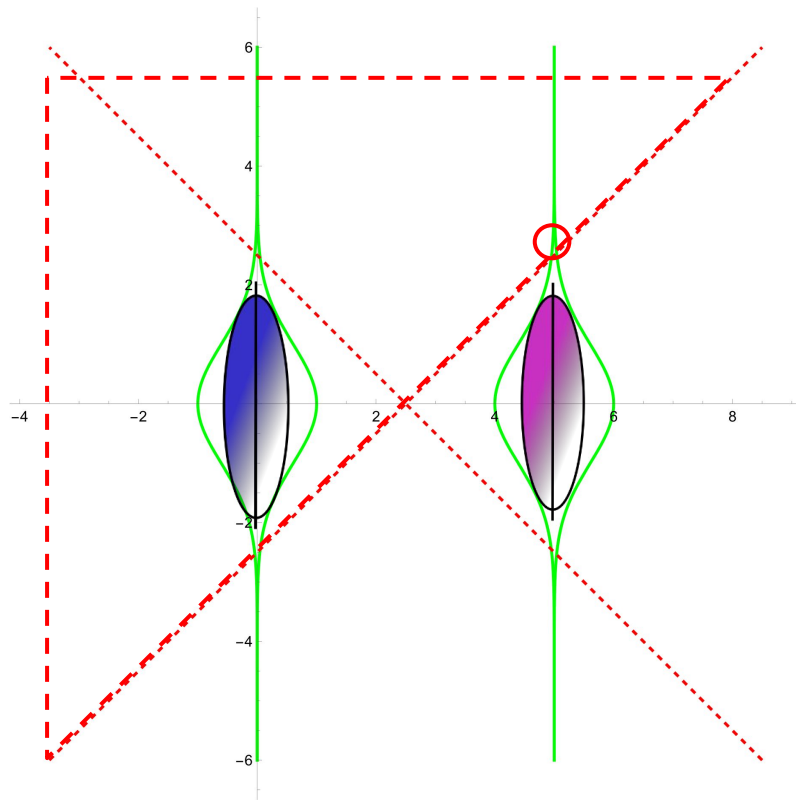


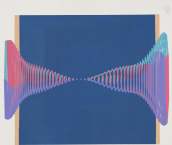


Further work: non compact support

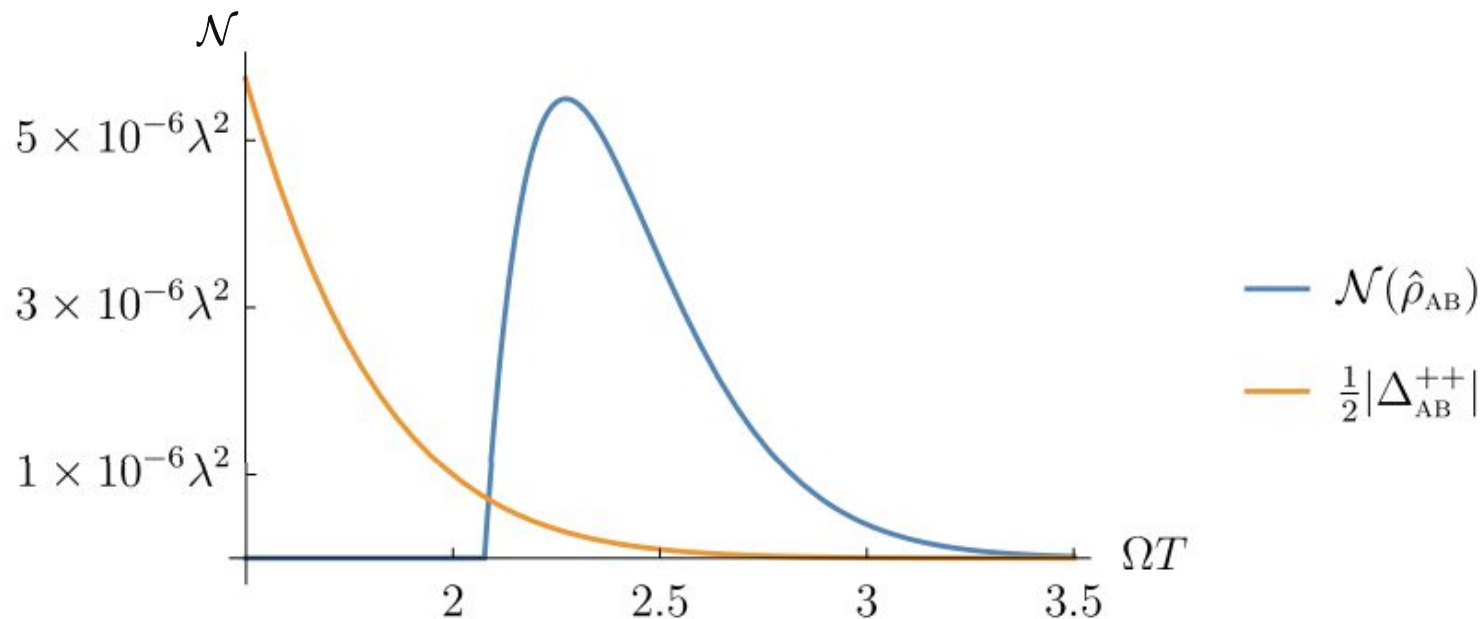
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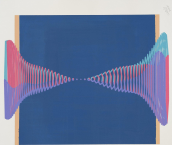
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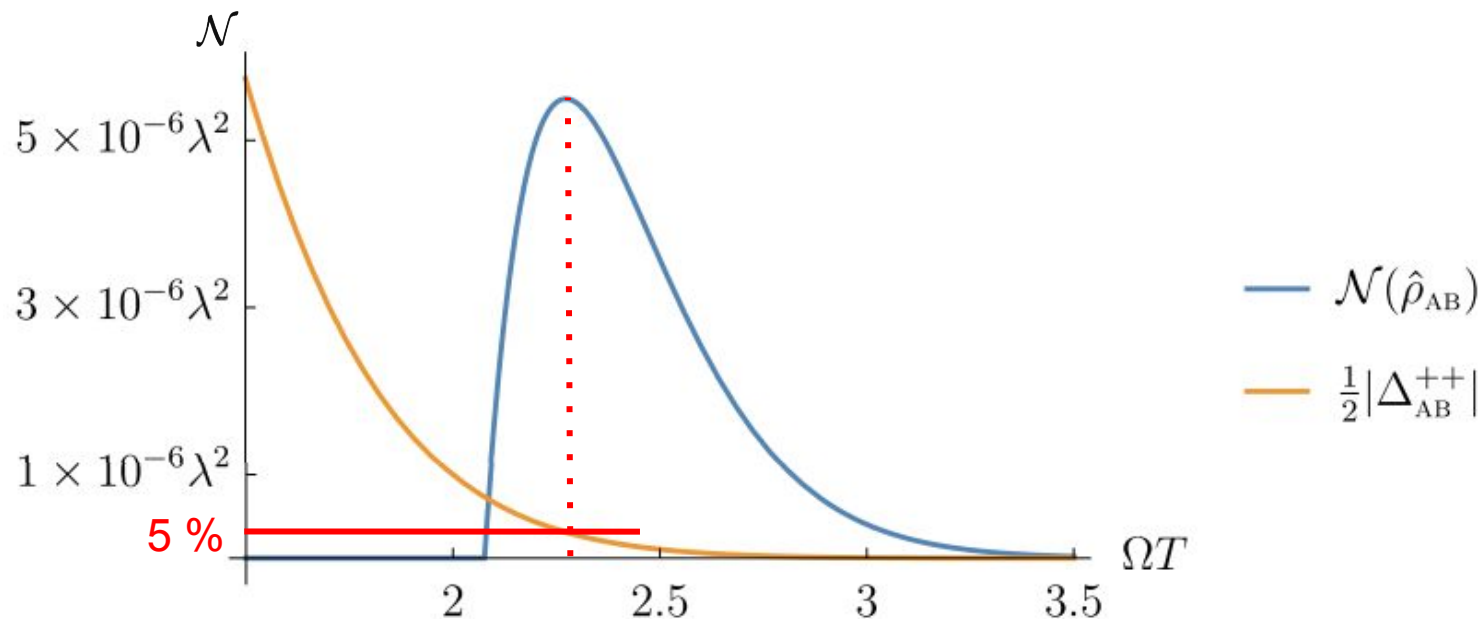


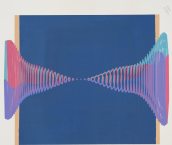
Further work: non compact support





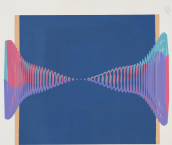
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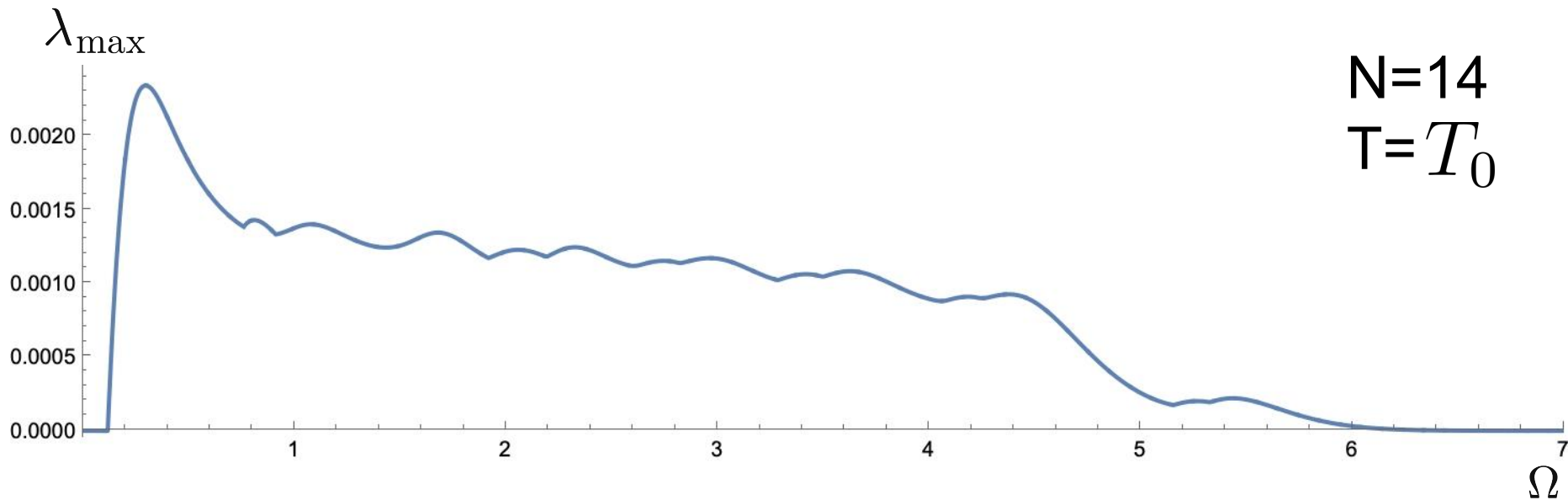
Further work: non compact support

$$\max_{\theta \in [0, 2\pi)} \lambda_{\max} \left(\cos(\theta) \operatorname{Re}\{G\} + \sin(\theta) \operatorname{Im}\{G\} - \operatorname{Re}\{W\} \right)$$



Further work: non compact support

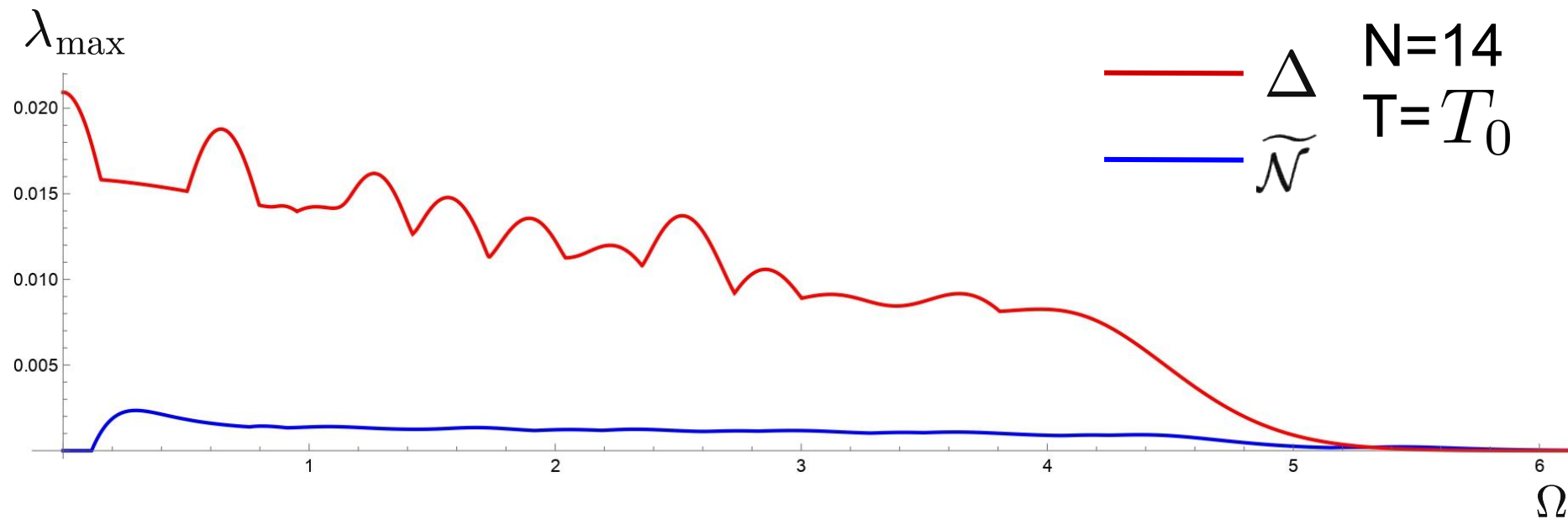
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Further work: non compact support

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Further work: non compact support



Further work: non compact support

→ find a good T for which we can perform the optimization

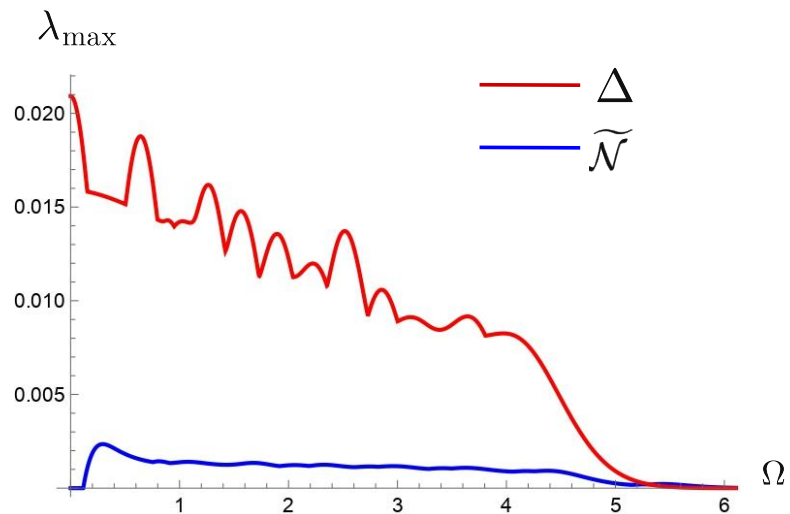


Further work: non compact support

- find a good T for which we can perform the optimization
 - small enough to keep signalling at most 5 %



Further work: non compact support



$$T = 1 \times T_0$$

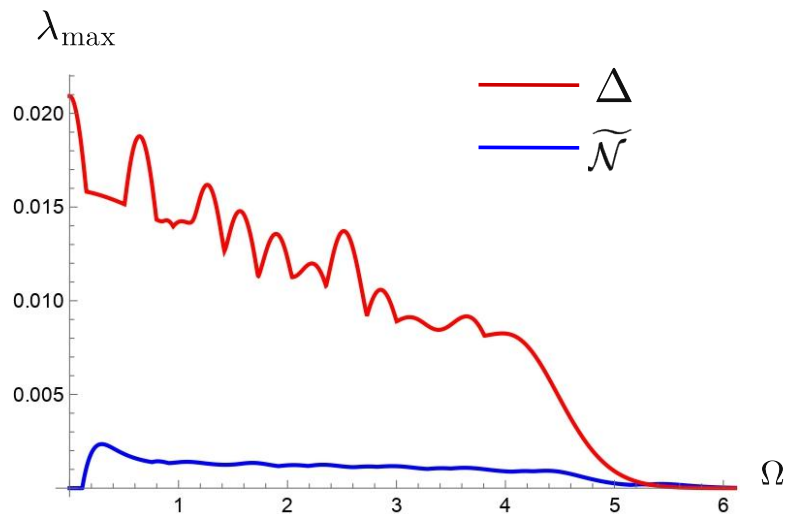


Further work: non compact support

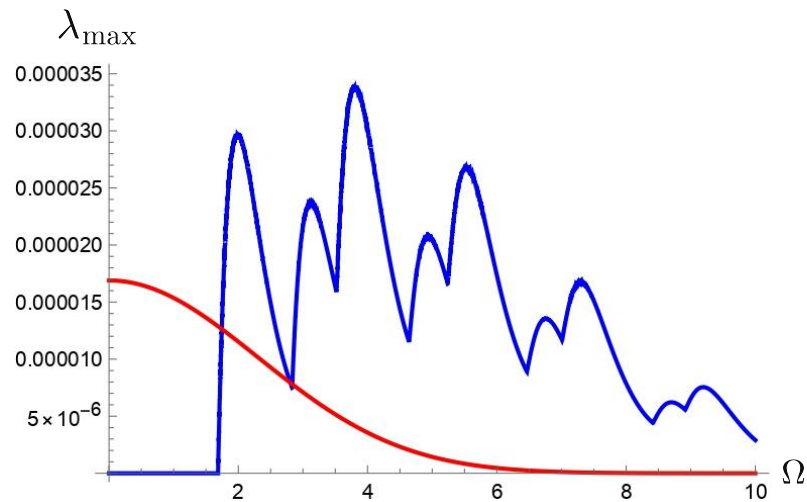
- find a good T for which we can perform the optimization
 - small enough to keep signalling at most 5 %
 - large enough to allow more entanglement



Further work: non compact support



$$T = 1 \times T_0$$

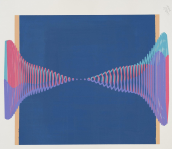


$$T = 0.4 \times T_0$$



Further work: non compact support

- find a good T for which we can perform the optimization
 - small enough to keep signalling at most 5 %
 - large enough to allow more entanglement
 - constrained optimization



Further work: non compact support

→ Point-like identical detectors, $N=14$



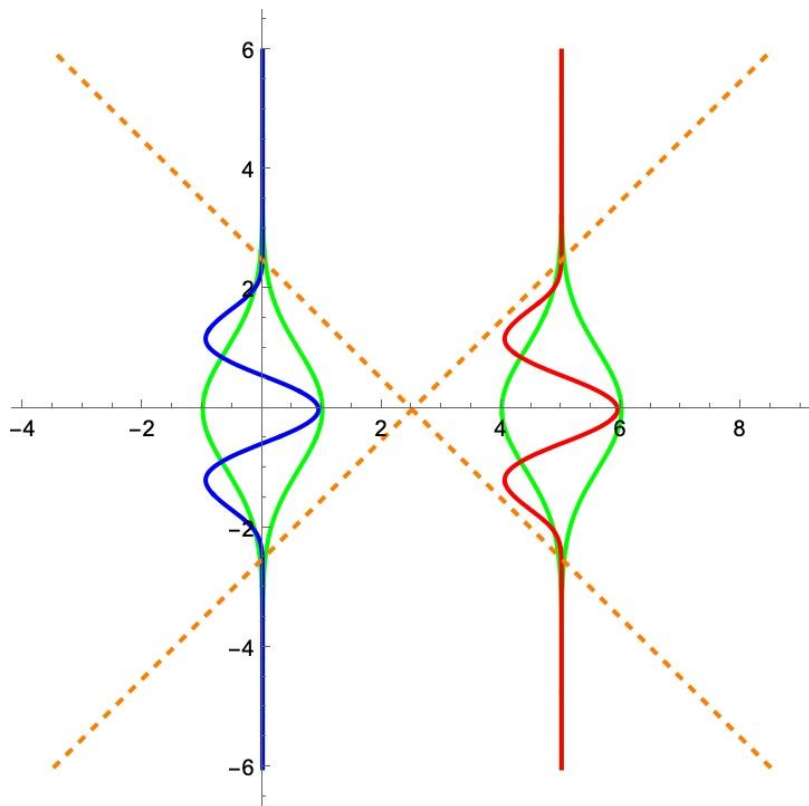
Further work: non compact support

→ Point-like identical detectors, $N=14$, $T=0.53 \cdot T_0$



Further work: non compact support

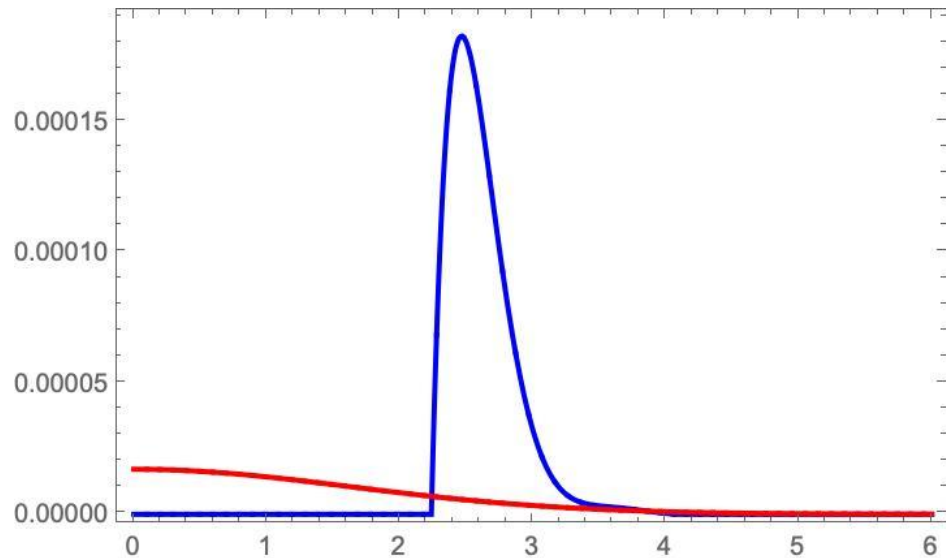
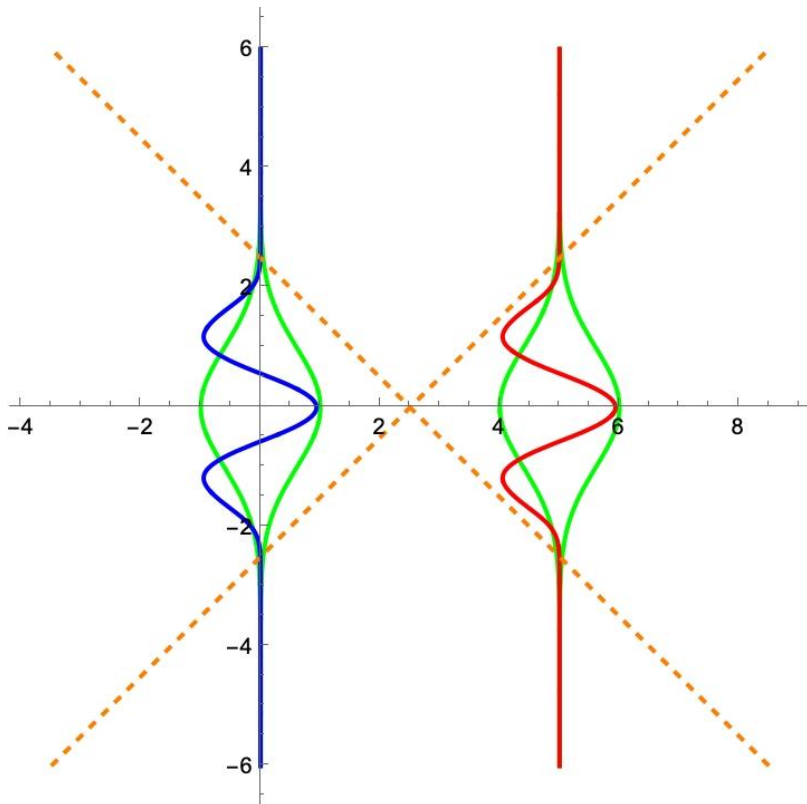
→ Point-like identical detectors, $N=14$, $T=0.53 \cdot T_0$





Further work: non compact support

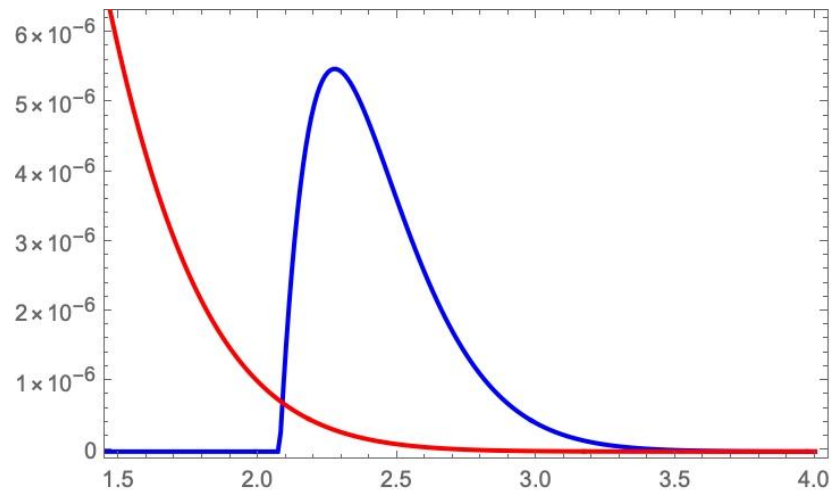
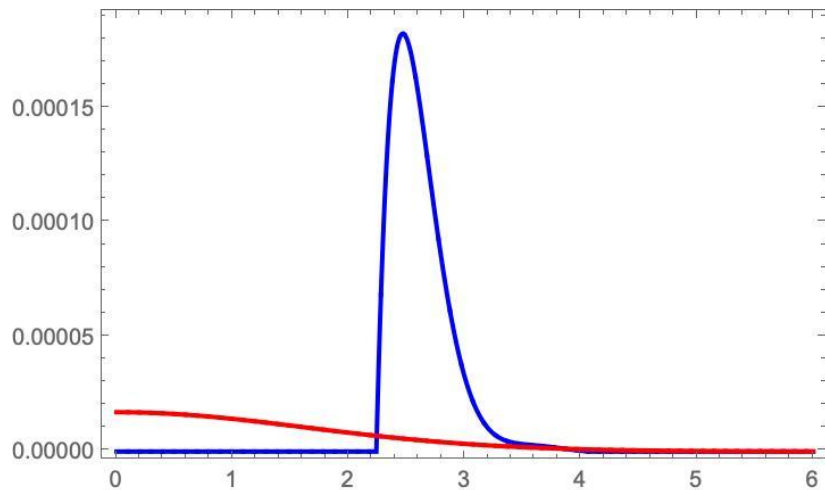
→ Point-like identical detectors, $N=14$, $T=0.53 \cdot T_0$





Further work: non compact support

→ Point-like identical detectors, $N=14$, $T=0.53 \cdot T_0$

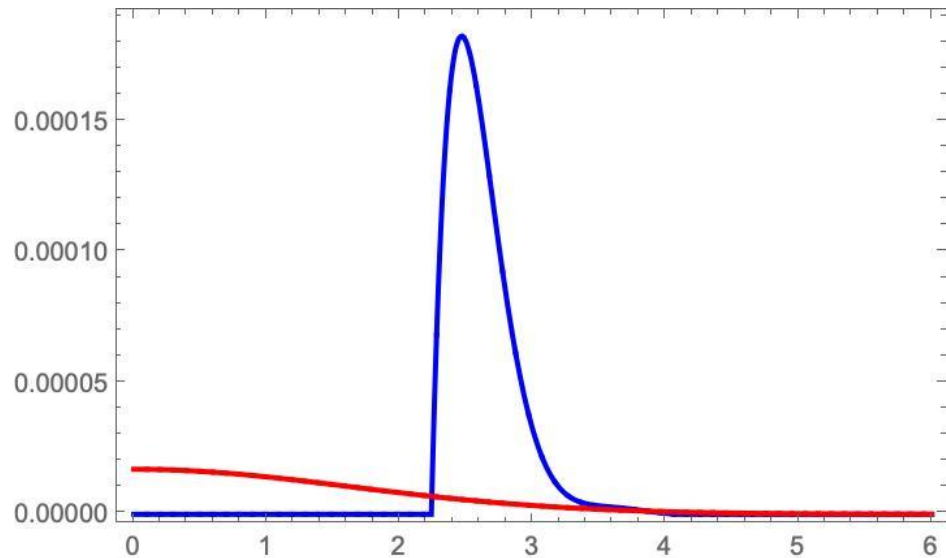
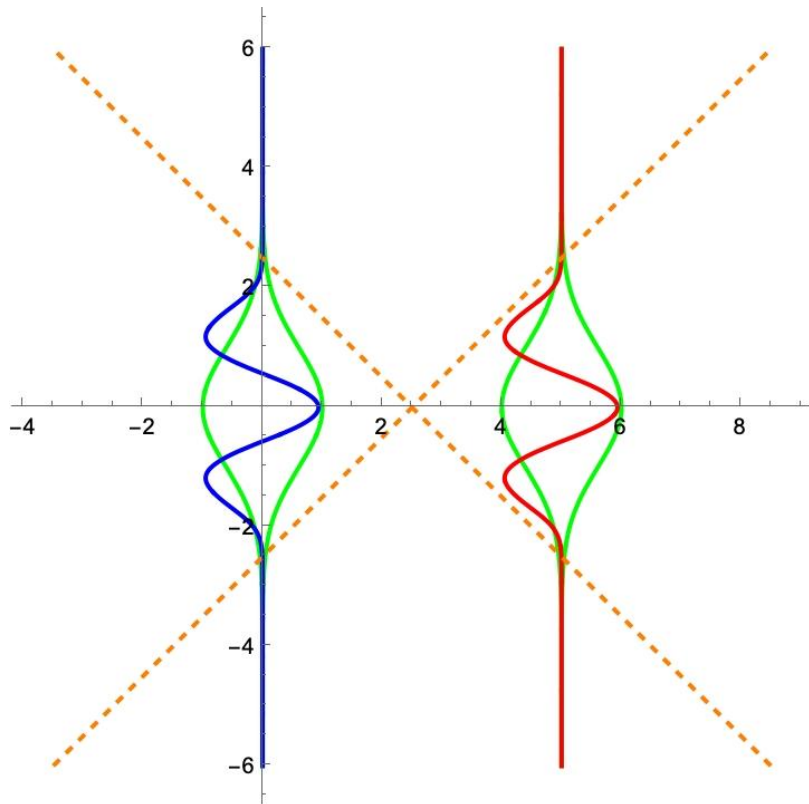


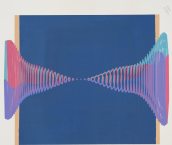
Standard gaussian



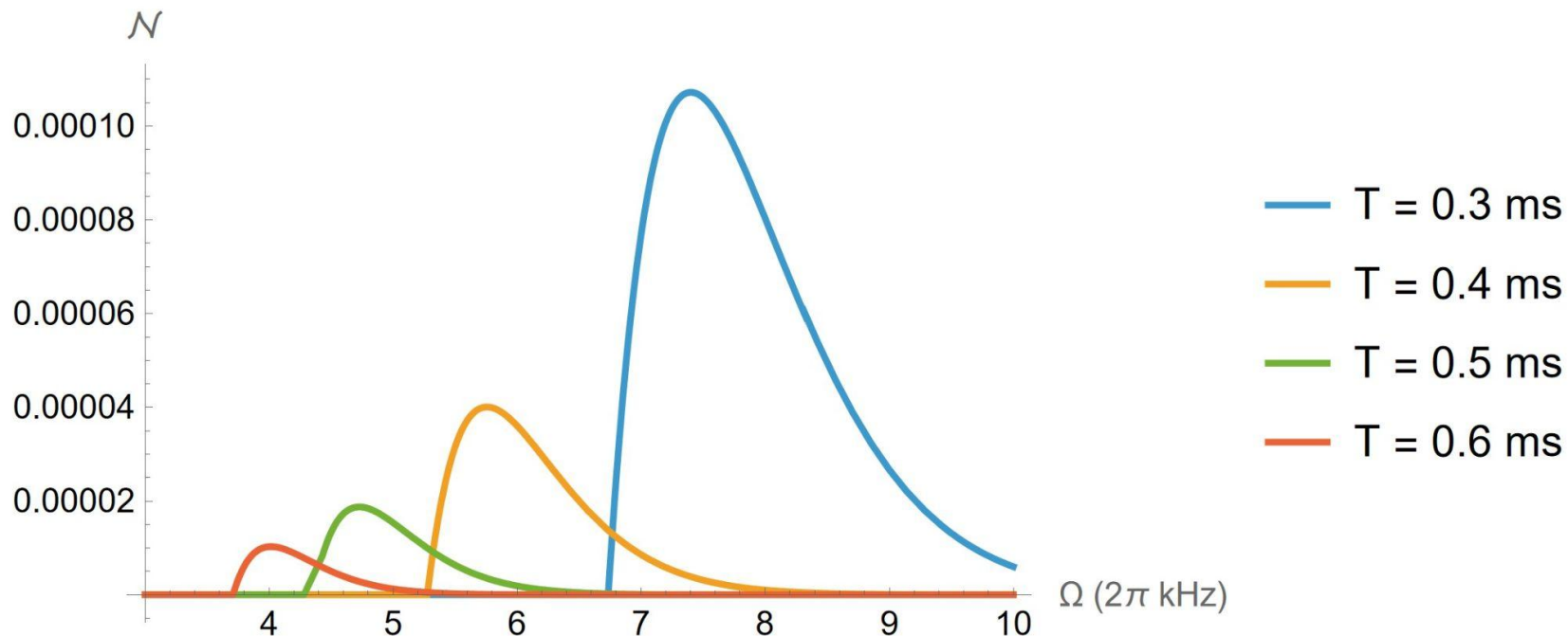
Further work: non compact support

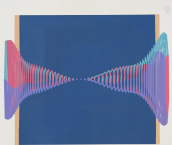
→ Point-like identical detectors, $N=14$, $T=0.53 \cdot T_0$





Extra further work: Heidelberg





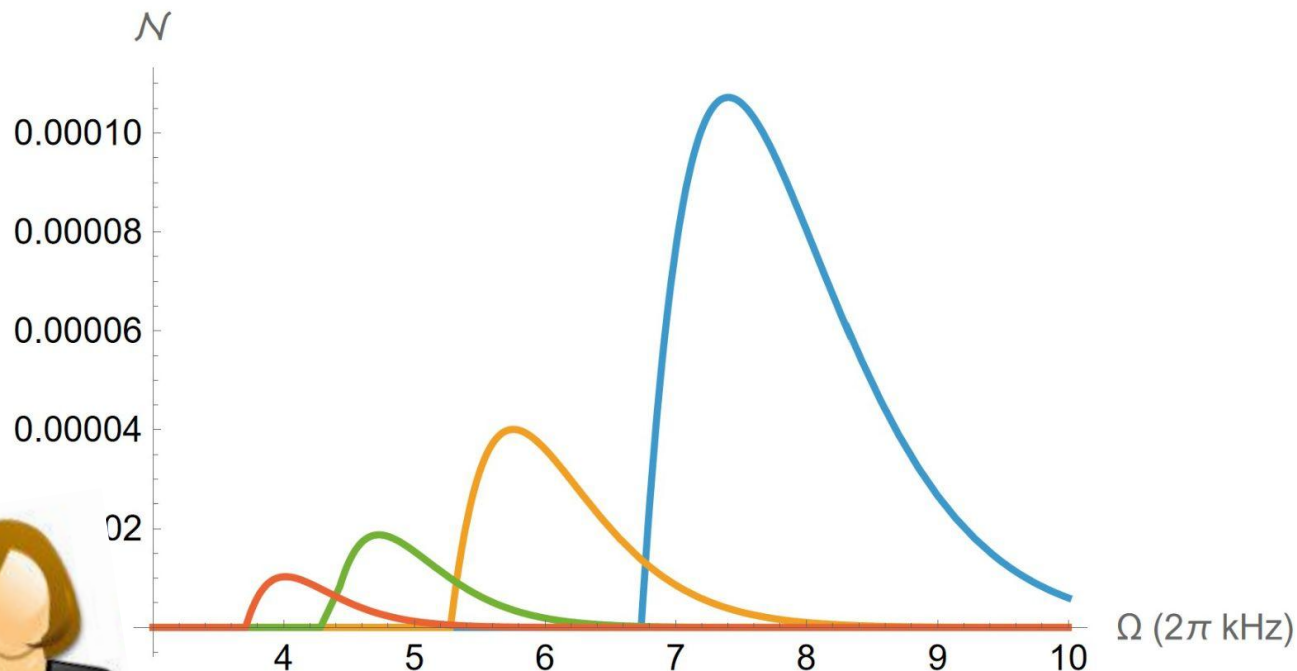
Extra further work: Heidelberg



Bob



Alice



- $T = 0.3$ ms
- $T = 0.4$ ms
- $T = 0.5$ ms
- $T = 0.6$ ms



Thank you!

